STRUCTURE OF ATOM

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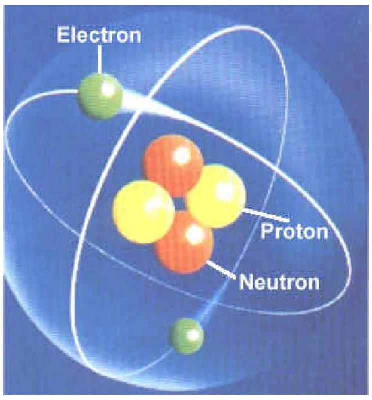


IMPORTANT INFORMATION

S.No.	Common Name	Chemical Name	Formulae
1.	Alum	Hydrated double sulphate of potassium and aluminium	$K_2SO_4.Al_2(SO_4)_3.24\ H_2O$
2.	Alumina	Aluminium oxide	Al_2O_3
3.	Ammonia water	Ammonium hydroxide	NH_4OH
4.	Angelsite	Lead sulphate	$PbSO_4$
5.	Aqua fortis	Concentrated nitric acid	HNO_3
6.	Aqua regia	....	$3HCl(Conc) + HNO_3(Conc)$
7.	Azote gas	Nitrogen	N_2
8.	Azurite blue	Basic copper carbonate	$2CuCO_3.Cu(OH)_2$
9.	Baking Soda	Sodium hydrogen carbonate	$NaHCO_3$
10.	Baryta	Barium Hydroxide	$Ba(OH)_2$
11.	Basic lead acetate	....	$Pb(OH)_2Pb(CH_3COO)_2$
12.	Bauxite	Dihydrate Aluminium oxide	$Al_2O_3.2H_2O$
13.	Blue Vitriol	Pentahydrate of cupric sulphate	$CuSO_4.5H_2O$
14.	Bleaching powder	Calcium oxychloride or calcium (hypochlorite) chloride	$CaOCl_2$ or $Ca(OCl)Cl$
15.	Borax	Sodium tetraborate	$Na_2B_4O_7.10H_2O$
16.	Brimstone	Sulphur	S
17.	Brine or common salt or rock salt	Sodium Chloride	$NaCl$
18.	Cerusite	Lead carbonate	$PbCO_3$
19.	Chalk or limestone	Calcium Carbonate	$CaCO_3$
20.	Chile saltpetre of Caliche	Sodium nitrate	$NaNO_3$
21.	Cuprite or ruby copper	Cuprous oxide	Cu_2O
22.	Copper glance	Copper sulphide	Cu_2S
23.	Carnalite	Potassium magnesium chloride	$KCl.MgCl_2.6H_2O$
24.	Calomel	Mercurous chloride	Hg_2Cl_2
25.	Cane sugar or Beet sugar	Sucrose	$C_{12}H_{22}O_{11}$
26.	Calgon	Sodium hexa-metaphosphate	$Na_2[Na_4(PO_3)_6]$
27.	Caustic soda	Sodium hydroxide	$NaOH$
28.	Caustic potash	Potassium hydroxide	KOH
29.	Calamine	Zinc carbonate	$ZnCO_3$
30.	Corundum	Aluminium oxide	Al_2O_3
31.	Dead burnt gypsum	Anhydrous calcium sulphate	$CaSO_4$
32.	Diaspore	Monohydrate aluminium oxide	$Al_2O_3.H_2O$
33.	Dry ice	Solid carbon dioxide	CO_2 (solid)
34.	Epsom salt	Heptahydrate of magnesium sulphate	$MgSO_4.7H_2O$
35.	Fluorspar	Calcium fluoride	CaF_2
36.	Galena	Lead sulphide	PbS
37.	Glauber's salt	Hydrated sodium sulphate	$Na_2SO_4.10H_2O$
38.	Green Vitriol	Ferrous sulphate	$FeSO_4.7H_2O$
39.	Gypsum	Calcium sulphate dihydrate	$CaSO_4.2H_2O$
40.	Hydrolith	Calcium hydride	CaH_2
41.	Hypo	Sodium thiosulphate	$Na_2S_2O_3.5H_2O$
42.	Iron pyrites	Iron disulphide	FeS_2
43.	Laughing gas	Dinitrogen oxide	N_2O
44.	Limonite	Hydrated ferric oxide	$Fe_2O_3.3H_2O$
45.	Litharge	Lead monoxide	PbO
46.	Lime	Calcium oxide	CaO

47.	Lime water or slaked lime	Calcium hydroxide	$Ca(OH)_2$
48.	Lime stone or chalk or marble	Calcium carbonate	$CaCO_3$
49.	Lunar caustic	Silver nitrate	$AgNO_3$
50.	Magnetite	Ferrosoferric oxide	Fe_3O_4
51.	Milk of lime	Calcium hydroxide	$Ca(OH)_2$
52.	Milk of magnesia	Magnesium hydroxide	$Mg(OH)_2$
53.	Marsh gas	Methane	CH_4
54.	Malachite (green)	Basic copper carbonate	$CuCO_3.Cu(OH)_2$
55.	Magnesia	Magnesium oxide	MgO
56.	Magnesite	Magnesium carbonate	$MgCO_3$
57.	Mohr's salt	Ferrous ammonium sulphate hexahydrate	$(NH_4)_2SO_4\ FeSO_4.6H_2O$
58.	Muriatic acid	Hydrochloric acid	HCl
59.	Nessler's reagent	Potassium tetra-idomercurate	$K_2[Hg\,I_4]$
60.	Nitre	Potassium nitrate	KNO_3
61.	Nitre cake	Sodium hydrogen sulphate	$NaHSO_4$
62.	Nitrolim	Calcium cyanamide plus carbon	$CaCN_2 + C$
63.	Nitrate of lime (Nitrolime)	Basic calcium nitrate	$Ca(NO_3)_2.CaO$
64.	Oil of Vitriol	Concentrated Sulphuric acid	H_2SO_4
65.	Oleum	Pyrosulphuric acid	$H_2S_2O_7$
66.	Pearl ash	Potassium carbonate	K_2CO_3
67.	Philosopher's wool	Zinc oxide	ZnO
68.	Phosgene	Carbonyl chloride	$COCl_2$
69.	Potash	Potassium carbonate	K_2CO_3
70.	Potash alum	Potassium aluminium sulphate	$K_2SO_4.Al_2(SO_4)_3.24H_2O$
71.	Plaster of Paris	Hemihydrate of calcium sulphate	$CaSO_4.\frac{1}{2}H_2O$
72.	Pyrogallol	Benzene-1, 2, 3-triol	$C_6H_3(OH)_3$
73.	Pyrolusite	Manganese dioxide	MnO_2
74.	Quartz or silica	Silicon dioxide	SiO_2
75.	Quick lime	Calcium oxide	CaO
76.	Quick silver	Mercury	Hg
77.	Red lead	Triplumbic tetroxide	Pb_3O_4
78.	Rock salt	Sodium chloride	$NaCl$
79.	Rough	Ferric oxide	Fe_2O_3
80.	Sal ammoniac	Solid ammonium chloride	NH_4Cl
81.	Sal volatile	Solid ammonium carbonate	$(NH_4)_2CO_3$
82.	Salt cake	Anhydrous sodium sulphate	Na_2SO_4
83.	Salt petre	Potassium nitrate	KNO_3
84.	Siderite	Ferrous carbonate	$FeCO_3$
85.	Silica	Silicon dioxide	SiO_2
86.	Silica gel	Hydrated silica	SiO_2, xH_2O
87.	Slaked lime	Calcium hydroxide	$Ca(OH)_2$
88.	Soda ash	Anhydrous sodium carbonate	Na_2CO_3
89.	Water glass	Sodium dioxide containing excess of silica	$Na_2SiO_3.SiO_2$
90.	Super phosphate of lime	Mixture of calcium dihydrogen phosphate and calcium sulphate	$Ca(H_2PO_4)_2.H_2O + 2CaSO_4.2H_2O$
91.	Verdigris	Basic copper acetate	$Cu(CH_3COO)_2$
92.	White lead	Basic lead carbonate	$2PbCO_3.Pb(OH)_2$
93.	White Vitriol	Heptahydrate of zinc sulphate	$ZnSO_4.7H_2O$
94.	Washing soda	Hydrated sodium carbonate	$Na_2CO_3.10H_2O$
95.	Zincite	Zinc oxide	ZnO
96.	Zinc blende	Zinc sulphide	ZnS

VALENCY OF SOME COMMON BASIC AND ACIDIC RADICALS

Electrovalent positive ions (basic radicals)

Monovalent

1.	Ammonium	NH_4^+
2.	Aurous [Gold (I)]	Au^+
3.	Cuprous [Copper (I)]	Cu^+
4.	Hydrogen	H^+
5.	Mercurous [Mercury (I)]	Hg^+
6.	Potassium	K^+
7.	Sodium	Na^+
8.	Argentous [Silver (I)]	Ag^+
9.	Lithium	Li^+

Divalent

1.	Argentic [Silver(II)]	Ag^{2+}
2.	Barium	Ba^{2+}
3.	Calcium	Ca^{2+}
4.	Cupric [Copper(II)]	Cu^{2+}
5.	Ferrous [Iron (II)]	Fe^{2+}
6.	Magnesium	Mg^{2+}
7.	Mercuric [Mercury (II)]	Hg^{2+}
8.	Plumbous [Lead (II)]	Pb^{2+}
9.	Stannous [Tin (II)]	Sn^{2+}
10.	Zinc	Zn^{2+}
11.	Platinous [Platinum (II)]	Pt^{2+}
12.	Nickel	Ni^{2+}

Trivalent

1.	Aluminium	Al^{3+}
2.	Ferric [Iron (III)]	Fe^{3+}
3.	Chromium	Cr^{3+}
4.	Auric [Gold (III)]	Au^{3+}
5.	Bismuth	Bi^{3+}
6.	Arsenic	As^{3+}

Tetravalent

1.	Plumbic [Lead (IV)]	Pb^{4+}
2.	Stannic [Tin (IV)]	Sn^{4+}
3.	Platinic [Platinium (IV)]	Pt^{4+}

ELECTROVALENT NEGATIVE IONS (ACIDIC RADICALS)

Monovalent

1.	Acetate	CH_3COO^-
2.	Bicarbonate or Hydrogen carbonate	HCO_3^-
3.	Bisulphide or Hydrogen sulphide	HS^-
4.	Bisulphate or Hydrogen sulphate	HSO_4^-
5.	Bisulphite or Hydrogen sulphite	HSO_3^-
6.	Bromide	Br^-
7.	Chloride	Cl^-
8.	Permanganate	MnO_4^-
9.	Fluoride	F^-
10.	Hydride	H^-
11.	Hydroxide	OH^-
12.	Iodide	I^-
13.	Cyanide	CN^-
14.	Nitrate	NO_3^-
15.	Nitrite	NO_2^-
16.	Chlorite	ClO_2^-
17.	Hypochlorite	ClO^-
18.	Chlorate	ClO_3^-
19.	Perchlorate	ClO_4^-
20.	Meta aluminate	AlO_2^-

Divalent

1.	Carbonate	CO_3^{2-}
2.	Dichromate	$Cr_2O_7^{2-}$
3.	Oxide	O^{2-}
4.	Peroxide	O_2^{2-}
5.	Sulphate	SO_4^{2-}
6.	Sulphite	SO_3^{2-}
7.	Sulphide	S^{2-}
8.	Silicate	SiO_3^{2-}
9.	Thiosulphate	$S_2O_3^{2-}$
10.	Zincate	ZnO_2^{2-}
11.	Plumbite	PbO_2^{2-}
12.	Stannate	SnO_3^{2-}
13.	Manganate	MnO_4^{2-}
14.	Chromate	CrO_4^{2-}
15.	Oxalate	$(COO)_2^{2-}$

Trivalent

1.	Arsenate	AsO_4^{3-}
2.	Nitride	N^{3-}
3.	Arsenite	AsO_3^{3-}
4.	Phosphide	P^{3-}
5.	Phosphite	PO_3^{3-}
6.	Phosphate	PO_4^{3-}
7.	Borate	BO_3^{3-}
8.	Aluminate	AlO_3^{3-}

Tetravalent

1.	Carbide	C^{4-}
2.	Ferrocyanide	$Fe(CN)_6^{4-}$

THEORY

1. INTRODUCTION

In this chapter, we explore the inside world of atoms which is full of mystery and surprises. Whole chemistry is based on atoms and their structures. We will also study the behaviour exhibited by the electrons and their consequences.

1.1 Discovery of fundamental particles

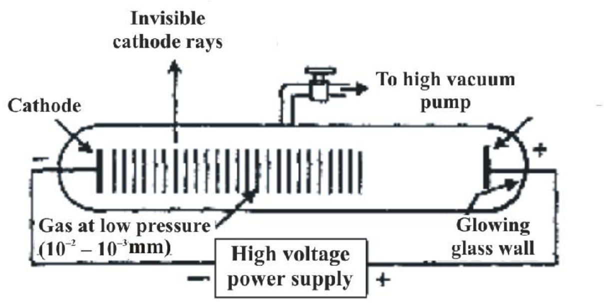
Dalton's atomic theory was able to explain the law of conservation of mass, law of constant composition and law of multiple proportion very successfully but it failed to explain the results of many experiments like it was known that substances like glass or ebonite when rubbed with silk or fur generate electricity

1.1.1 Discovery of electron

William Crookes in 1879 studied the electrical discharge in partially evacuated tubes known as **cathode ray discharge tubes.**

A discharge tube is made of glass, about 60cm long containing two thin pieces of metals called electrodes, sealed in it. This is known as crooke's tube. The negative electrode is known as **cathode** and positive electrode is known **anode.**

When a gas enclosed at low pressure(~ 10^{-4} atm) in discharge tube is subjected to a high voltage (~ 10,000V), invisible rays originating from the cathode and producing a greenish glow behind the perforated anode on the glass wall coated with phosphorescent material ZnS is observed. These rays were called **cathode rays.**

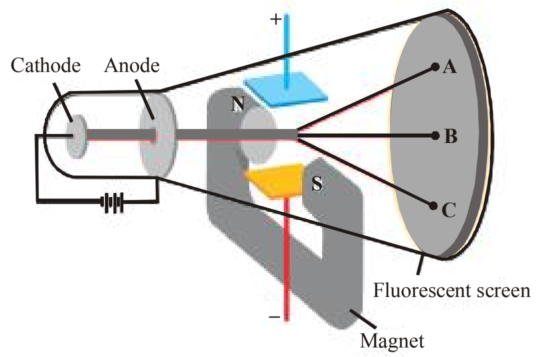


1.1.2 Properties

1. They produce sharp shadow of the solid object in their path suggesting that they travel in straight line.
2. They are deflected towards the positive plate in an electric field suggesting that they are negatively charged. They were named as electrons by Stoney.
3. They can make a light paddle wheel to rotate placed in their path. This means they possess kinetic energy and are material particles.
4. They have a charge to mass ratio = 1.75882×10^{11}C/kg
5. They ionise gases through which they travel.
6. They produce X-rays when they strike a metallic target.
7. The characterstics of cathode rays (electrons) do not depend on the material of electrodes and nature of the gas present in the cathode ray tube.

Thus, we can conclude that electrons are basic constituents of all matter.

1.1.3 Charge to mass ratio of electron

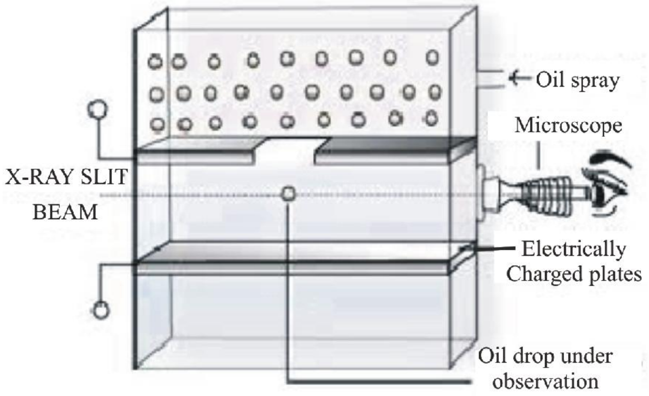


In 1897 J.J. Thomson measured e/m ratio of electron by using cathode ray tube and applying electric and magnetic field perpendicular to each other as well as to the path of electrons. The extent of deviation of electrons from their path in the presence of electric and magnetic field depends on:

(a) Charge on the electron
(b) Mass of the particle
(c) The strength of electric or magnetic field

When only electric field is applied, the electrons are deflected to the point A. When only magnetic field is applied the electrons are deflected to the point C. By balancing the strengths of electric or magnetic fields, the electrons are allowed to hit the screen at point B i.e. the point where electrons hit in the absence of electric and magnetic field. By measuring the amount of deflections Thomson was able to calculate the value of e/m as 1.758820×10^{11}C/kg.

1.1.4 Charge on the electron



R.A Millikan devised a method known as oil drop experiment to determine the charge on the electrons.

In this method, oil droplets in the form of mist, produced by the atomiser, were allowed to enter through a tiny hole in the upper plate of electrical condenser. The downward motion of these droplets was viewed through the telescope, equipped with a micrometer eye piece. By measuring the rate of fall of these droplets, Millikan was able to measure the mass of oil droplets. The air inside the chamber was ionized by passing a beam of X-rays through it. The electrical charge on these oil droplets was acquired by collisions with gaseous ions. The fall of these charged oil droplets can be retarded, accelerated or made stationary depending upon the charge on the droplets and the polarity and strength of the voltage applied to the plate. By carefully measuring the effects of electrical field strength on the motion of oil droplets, Millikan concluded that the magnitude of electrical charge, q, on the droplets is always an integral multiple of the electrical charge e, that is, $q = n$ e, where n = 1, 2, 3... .

Charge on the electron is found to be -1.6022×10^{-19}C.

The mass of electron was thus calculated as

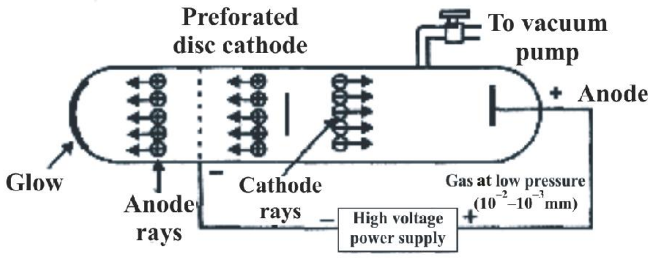
$$m = \frac{e}{e/m} = \frac{1.6022 \times 10^{-19}\,C}{1.758820 \times 10^{11}\,C/kg} = 9.1094 \times 10^{-31}\,kg.$$

1.1.5 Origin of cathode rays

The cathode rays are first produced from the material of the cathode and then from the gas inside the discharge tube due to bombardment of the gas molecules by the high speed electrons emitted first from the cathode.

1.2.1 Discovery of proton

Since the atom as a whole is electrically neutral and the presence of negatively charged particles in it was established, therefore it was thought that some positively charged particles must also be present in the atom. So, during the experiments with cathode rays, the scientist Goldstein designed a special type of discharge tube. He discovered new rays called **Canal rays.** The name canal rays is derived from the fact that the rays travelled in straight line through a vacuum tube in the opposite direction to cathode rays, pass through and emerge from a canal or hole in the cathode. They are also known as **anode rays.**



1.2.2 Properties:

1. They travel in straight lines.
2. They carry a positive charge.
3. They are made up of material particles.
4. The value of the charge on the particles constituting the anode rays is found to depend on the nature of gas taken.
5. The mass of the particles constituting the anode rays is found to depend on the nature of gas taken.
6. The charge to mass ratio(e/m) of the particles is also found to depend on the gas taken.
7. Their behaviour in electric and magnetic field is opposite to that observed for electron.

1.2.3 Origin of anode rays:

These rays are believed to be produced as a result of the knock out of the electrons from the gaseous atoms by the bombardment of high speed electrons of the cathode rays on them. These anode rays are not emitted from the anode but are produced in the space between the anode and the cathode.

The lightest charged particles were obtained when the gas taken in the discharge tube was hydrogen. The e/m value of these particles were maximum. They had minimum mass and unit positive charge. The particle was called a proton.

Charge on a proton = + 1.6022×10^{-19}C

Mass of a proton = 1.672×10^{-27}kg

1.3.1 Discovery of neutron

The theoretical requirement for the existence of a neutron particle in the atomic nucleus was put forward by Rutherford in 1920.It was proposed to be a particle with no charge and having mass almost equal to that of a proton. He named it as neutron. In 1932 Chadwick proved its existence. He observed that, when a beam of α particles $({}^{2}_{4}He)$ is incident on Beryllium (Be), a new type of particle was ejected. It had mass almost equal to that of a **proton** (1.674×10^{-27}kg) and carried no charge.

$$^{11}_{5}B + {}^{4}_{2}He \rightarrow {}^{14}_{7}N + {}^{1}_{0}n$$

$$^{9}_{4}Be + {}^{4}_{2}He \rightarrow {}^{12}_{6}C + {}^{1}_{0}n$$

SUMMARY :

ATOMIC BUILDING BLOCKS

Name	Discoverer	Symbol	Charge	Relative Charge	Mass
Electron	J.J. Thomson	e	-1.6022×10^{-19} C	–1	9.1094×10^{-31} Kg
Proton	Goldstein	p	$+1.6022 \times 10^{-19}$C	+1	1.6726×10^{-27} kg
Neutron	Chadwick	n	0	0	1.6749×10^{-27} kg

2. DEFINATIONS

2.1 Electron

A fundamental particle which carries one unit negative charge and has a mass nearly equal to 1/1837th of that of hydrogen atom.

2.2 Proton

A fundamental particle which carries one unit positive charge and has a mass nearly equal to that of hydrogen atom.

2.3 Neutron

A fundamental particle which carries no charge but has a mass nearly equal to that of hydrogen atom.

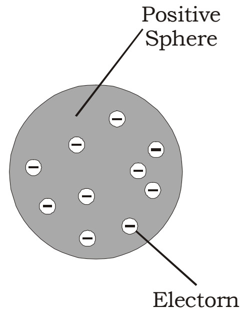
3. THOMSON MODEL

Sir J. J. Thomson, who discovered the electron, was the first to suggest a model of atomic structure.

(i) All atoms contain electrons.

(ii) The atom as a whole is neutral. The total positive charge and total negative charge must be equal.

He visualised all the positive charge of the atom as being spread out uniformly throughout a sphere of atomic dimensions (i.e. approx. 10^{-10} m in diameter). The electrons were smaller particles together carrying a negative charge, equal to the positive charge in the atom. They were studded in the atom like plums in a pudding. The charge distribution was such, that it gave the most stable arrangement. This model of the atom was often called the plum – pudding model. Also the raisin pudding model or watermelon model.



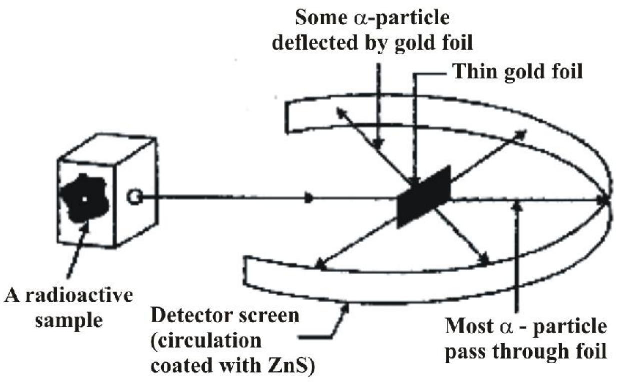
Thomson's proposed model of atom.

3.1 Drawbacks

Though the model was able to explain the overall neutrality of the atom, it could not satisfactorily explain the results of scattering experiments carried out by Rutherford.

4. RUTHERFORD'S α-SCATTERING EXPERIMENT

Rutherford conducted α- particles scattering experiments in 1909. In this experiment, a very thin foil of gold (0.004nm) is bombarded by a fine stream of alpha particles. A fluorescent screen (ZnS) is placed behind the gold foil, where points were recorded which were emerging from α-particles. Polonium was used as the source of α-particles.



RUTHERFORD'S SCATTERING EXPERIMENT

4.1 Observations

Rutherford carried out a number of experiments, involving the scattering of α – particles by a very thin foil of gold. Observations were:

(i) Most of the α – particles (99%) passes through it, without any deviation or deflection.

(ii) Some of the α – particles were deflected through small angles.

(iii) Very few α – particles were deflected by large angles and occasionally an α – particle got deflected by 180°

4.2 Conclusions

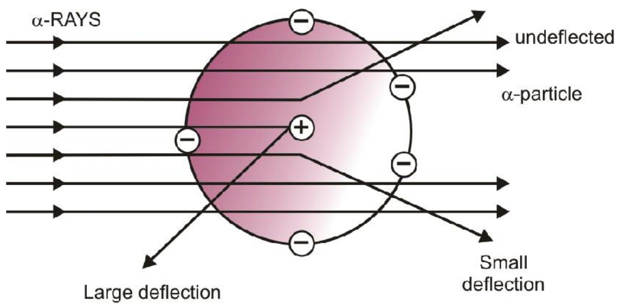
(i) An atom must be extremely hollow and must consist of mostly empty space because most of the particles passed through it without any deflection.

(ii) Very few particles were deflected to a large extent. This indicates that:

(a) Electrons because of their negative charge and very low mass cannot deflect heavy and positively charged α particles

(b) There must be a very heavy and positively charged body in the atom i.e. nucleus which does not permit the passage of positively charged α particles.

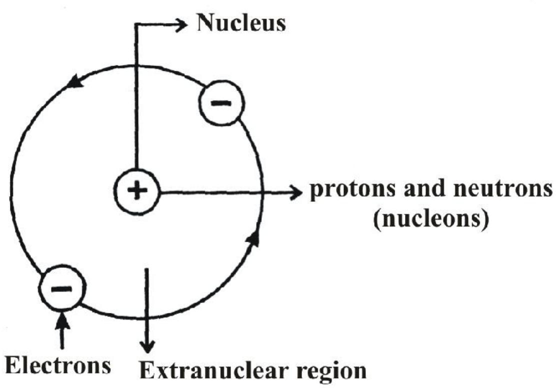
(c) Because, the number of α particles which undergo deflection of 180°, is very small, therefore the volume of positively charged body must be extremely small fraction of the total volume of the atom. This positively charged body must be at the centre of the atom which is called **nucleus.**



It has been found that radius of atom is of the order of 10^{-10}m while the radius of the nucleus is of the order of 10^{-15}m.

Thus if a cricket ball represents a nucleus, then the radius of atom would be about 5 km.

4.3 Rutherford's nuclear atomic model



(i) An atom consists of tiny positively charged nucleus at the centre and it is surrounded by hollow portion called extra nuclear part.

(ii) The positive charge of the nucleus is due to **nucleons** which consist of protons and neutrons while the electrons, present in extra nuclear portion has negligible mass and carry a negative charge.

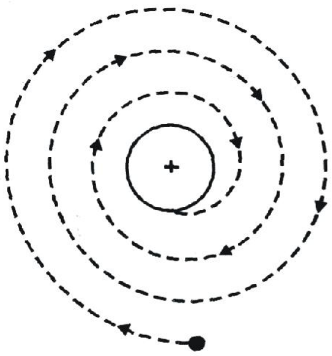
(iii) The atom is electrically neutral, as the number of electrons is equal to number of protons in it. Thus, total positive charge of the nucleus is balanced by the total negative charge of electrons.

(iv) The electrons in the extra nuclear part are revolving around the nucleus in circular paths called **orbits.** Thus, an atom resembles the solar system in which the sun plays the role of nucleus and the planets that of revolving electrons and the model is known as **planetary model.**

(v) Electrons and nucleus are held together by the electrostatic force of attraction.

(vi) Forces of attraction operating on the electron are exactly balanced by centrifugal forces.

4.4 Drawbacks



ELECTRON FALLING INTO THE NUCLEUS

(i) According to classical mechanics, any charged body in motion under the influence of attractive forces should radiate energy continuously. If this is so, the electron will follow a spiral path and finally fall into the nucleus and the structure would collapse. This behaviour is never observed.

(ii) It says nothing about the electronic structure of atoms i.e. how the electrons are distributed around the nucleus and what are the energies of these electrons.

5. ATOMIC NUMBER AND MASS NUMBER

5.1 Atomic number (Z)

Atomic number of an element is equal to the number of unit positive charges or number of protons present in the nucleus of the atom of the element. It also represents the number of electrons in the neutral atom. Eg. Number of protons in Na = 11 , thus atomic number of Na=11

5.2 Mass number (A)

The elementary particles (protons and neutrons) present in the nucleus of an atom are collectively known as nucleons.

The mass number (A) of an atom is equal to the sum of protons and neutrons. It is always a whole number. Thus,

Mass number (A) = Number of protons(Z) + Number of neutrons(n)

Therefore, **number of neutrons (n) = Mass Number (A) – Number of protons (Z)**

n = A – Z

Note..

The general notation that is used to represent the mass number and atomic number of a given atoms is ${}^{A}_{Z}X$

Where, X – symbol of element

A – Mass number

Z – atomic number

5.3 Isotopes, Isobars, isotones and Isoelectronic

5.3.1 Isotopes:

Isotopes are the atoms of the same element having identical atomic number but different mass number. The difference is due to the difference in number of neutrons.

The chemical properties of atoms are controlled by the number of electrons. Thus, isotopes of an element show same chemical behaviour.

Isotopes of Hydrogen

Isotope	Formula	Mass number	No. of protons	No. of neutrons
Protium	${}^{1}_{1}H$ (H)	1	1	0
Deuterium	${}^{2}_{1}H$ (D)	2	1	1
Tritium	${}^{3}_{1}H$ (T)	3	1	2

Isotopes of Oxygen

Isotope number	Mass number	No. of protons	No. of neutrons
${}^{16}_{8}O$	16	8	8
${}^{17}_{8}O$	17	8	9
${}^{18}_{8}O$	18	8	10

Isotopes of some common elements

Element	Isotopes
Carbon (C)	$^{12}_{6}C$, $^{13}_{6}C$, $^{14}_{6}C$
Nitrogen (N)	$^{14}_{7}N$, $^{15}_{7}N$
Uranium	$^{233}_{92}U$, $^{235}_{92}U$, $^{238}_{92}U$
Sulphur	$^{32}_{16}S$, $^{33}_{16}S$, $^{34}_{16}S$, $^{36}_{16}S$

5.3.2 Relative Abundance:

Isotopes of an element occur in different percentages in nature, which is termed as relative abundance.

Using this relative abundance the average atomic mass of the element can be calculated. For Example,

the average atomic mass of Cl is 35.5 due to existence of two isotopes ^{35}Cl and ^{37}Cl in 75% and 25% abundance respectively

5.3.3 Isobars:

Atoms of different elements having different atomic numbers but same mass numbers are called isobars. Eg

Isobar	Atomic number	Mass number	No. of elctrons	No. of protons	No. of neutrons
$^{40}_{18}Ar$	18	40	18	18	22
$^{40}_{19}K$	19	40	19	19	21
$^{40}_{20}Ca$	20	40	20	20	20

5.3.4 Isotones:

Atoms of different elements which contain the same number of neutrons are called isotones. Eg

Isotones	Atomic	Mass number	No. of neutros
$^{36}_{16}S$	16	36	20
$^{37}_{17}Cl$	17	37	20
$^{38}_{18}Ar$	18	38	20
$^{39}_{19}K$	19	39	20
$^{40}_{20}Ca$	20	40	20

5.3.5 Isoelectronic:

The species (atoms or ions) containing the same number of electrons are called isoelectronic. Eg.

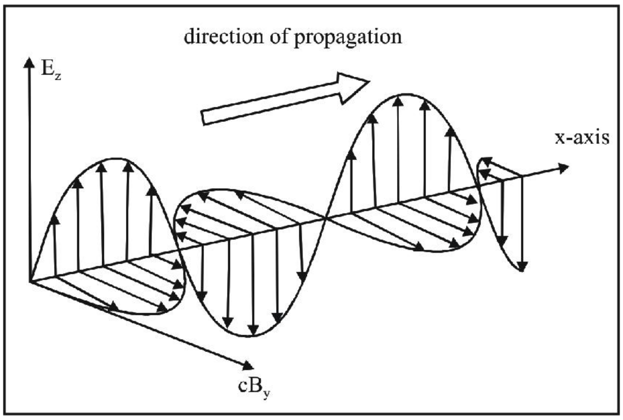
O^{2-}, F^{-}, Na^{+}, Mg^{+2}, Al^{+3}, Ne etc

To go further into the atomic mysteries, we will have to understand the nature of electromagnetic radiations and study **Maxwell's Electromagnetic Wave theory".**

James Maxwell was the first to give a comprehensive explanation about the interaction between the charged bodies and the behaviour of electric and magnetic fields.

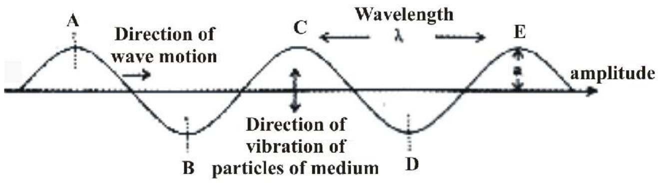
6. ELECTROMAGNETIC RADIATIONS

Electromagnetic Radiations are waves which are formed as a result of oscillating magnetic and electric fields which are perpendicular to each other and both are perpendicular to direction of motion.



They do not require any medium and can move in vacuum unlike sound waves.

Light is a form of radiation and has wave characterstics. The various characterstics of a wave are:



PROPAGATION OF A WAVE

1) **Amplitude :** It is height of the crest or trough (depth) of a wave. Units : metre (m)

2) **Frequency (ν) :** The number of waves passing through a point in one second. **Units :** Hertz (Hz) or s^{-1}

3) **Time Period :** The time taken by a wave to complete one vibration is called time period. **Units :** sec

4) Velocity : The distance travelled by a wave in one second is called velocity. **Units :** m/s

In vacuum, all types of electromagnetic radiations travel at the same speed i.e. **3×10^8 m/s.** This is called speed of light.

5) Wavelength(λ) : The distance between two adjacent crests or troughs is called wavelength. **Units :** Angstrom(Å)

[1 Å=10^{-10}m]

6) Wave Number $(\bar{\nu})$: It is the number of wavelengths per centimetre of length. **Units :** m^{-1}

$$\bar{\nu} = 1/\lambda$$

6.1 Relationship between velocity, frequency & wavelength

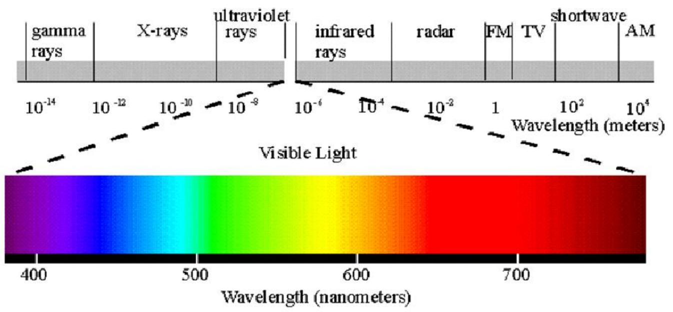
$$c = \nu\lambda$$

where **c :** speed of light i.e. 3×10^8 m/s in vaccum

v : frequency; **λ :** wavelength

7. ELECTROMAGNETIC SPECTRUM

When all the electromagnetic radiations are arranged in increasing order of wavelength or decreasing frequency the band of radiations obtained is termed as electromagnetic spectrum.



The visible spectrum is a subset of this spectrum (VIBGYOR) whose range of wavelength is 380-760nm.

The wavelengths increase in the order:

Gamma Rays < X-rays < Ultra-violet rays < Visible< Infrared < Micro-waves <Radio waves.

7.1 Electromagnetic Wave Theory

The main points of this theory are:

(1) A source (like the heated rod) emits energy **continuously** in the form of radiations (i.e. no change in wavelength or frequency of the emitted radiations even on increasing the energy radiated).

(2) These radiations are Electromagnetic in nature.

7.2 Failure of EM wave theory

The theory failed because of 2 experiments:

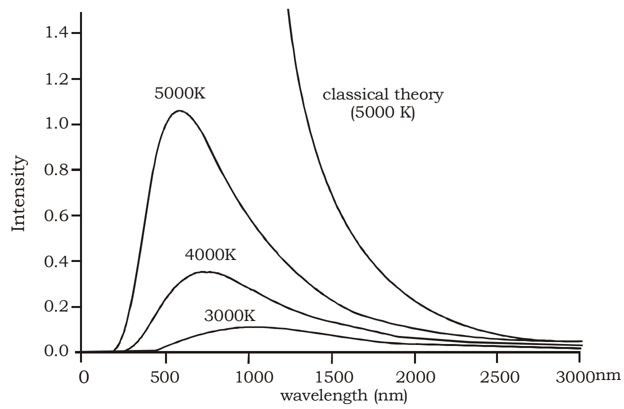
(1) Black Body Radiation :

According to Maxwell's theory on heating a body the intensity should increase, that is, energy radiated per unit area should increase without having any effect on the wavelength or frequency.

But we observe that when we heat an iron rod, it first turns to red then white and then becomes blue at very high temperatures. This means that frequency of emitted radiations is changing.

An ideal body, which emits and absorbs radiations of all frequencies is called **black body** and radiation emitted by a black body is called **black body radiation**

The variation of intensity with wavelength at different temperatures for a black body is shown below:

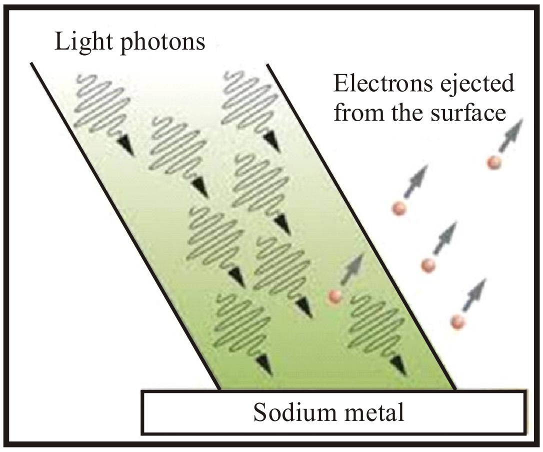


So it is observed that with increasing temperature the dominant wavelength in the emitted radiations decreases and the frequency increases.

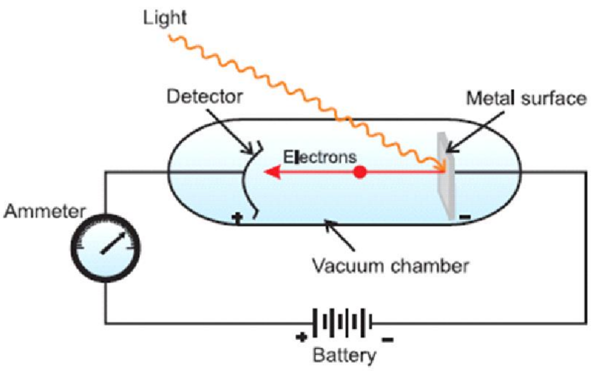
That is at higher temperatures, though the intensity rises as predicted by Maxwell's theory but the wavelength decreases. If **$T_1>T_2>T_3$ then $\lambda_1<\lambda_2<\lambda_3$.**

(2) Photoelectric Effect

When radiations with certain minimum frequency (ν_0) strike the surface of a metal, the electrons are ejected from the surface of the metal. This phenomena is called **photoelectric effect.** The electrons emitted are called **photoelectrons.**



According to Maxwell's Theory the ejection of electrons should depend on intensity of radiation that is if electrons are not being ejected, then on increasing the intensity they can be ejected.



The following **observations** are made:

(i) The electrons are ejected from the metal surface as soon as the beam of light strikes the surface, i.e., there is no time lag between the striking of light beam and the ejection of electrons from the metal surface.

(ii) The number of electrons ejected is proportional to the intensity or brightness of light.

(iii) For each metal, there is a characteristic minimum frequency, ν_0 (also known as **threshold frequency**) below which photoelectric effect is not observed. At a frequency $\nu > \nu_0$, the ejected electrons come out with certain kinetic energy. The kinetic energies of these electrons increase with the increase of frequency of the light used.

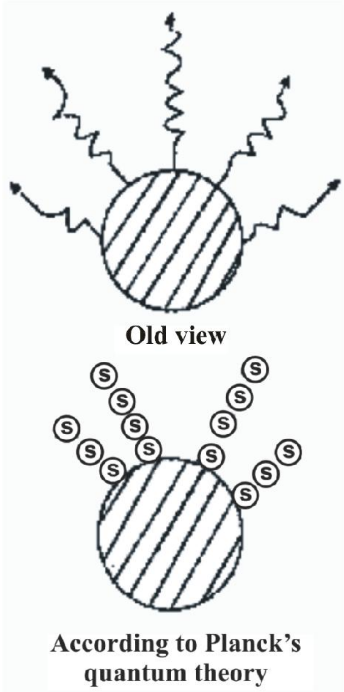
Thus, these findings were contradictory to the Maxwell's theory. The number of electrons ejected and kinetic energy associated with them should depend on the intensity of light. It has been observed that though the number of electrons ejected does depend upon the brightness of light, the kinetic energy of the ejected electrons does not.

To justify these findings Max Von Planc gave his Quantum theory.

8. PLANCK'S QUANTUM THEORY

The main points of this theory are:

(i) The energy is emitted or absorbed not continuously but **discontinuously** in the form of small discrete packets of energy. Each such packet of energy is called a **'quantum'.** In case of light this quantum of energy is called a **photon.**



(ii) At a time only one photon can be supplied to one electron or any other particle.

(iii) One quantum cannot be divided or distributed.

(iv) The energy of each quantum is directly proportional to the frequency of radiation.

$$E \propto \nu \text{ or } E = h\nu = \frac{hc}{\lambda}$$

h = Planck's constant = 6.626×10^{-34}Js

(v) The total energy emitted or absorbed by a body will be in whole number quanta.

Hence $\mathbf{E} = nh\nu = \dfrac{nhc}{\lambda}$

This is also called **"Quantisation of energy".**

8.1 Explanation of Black body radiation

As the temperature is increased the energy emitted increases thereby increasing the frequency of the emitted radiations. As the frequency increases the wavelength shifts to lower values.

8.2 Explanation of Photoelectric effect

(i) When light of some particular frequency falls on the surface of metal, the photon gives its entire energy to the electron of the metal atom. The electron will be ejected from the metal only if the energy of the photon is sufficient to overcome the force of attraction of the electron by the nucleus. So, photoelectrons are ejected only when the incident light has a certain minimum frequency (threshold frequency ν_0). The Threshold energy required for emission is called **"Work Function"** that is "$h\nu_0$".

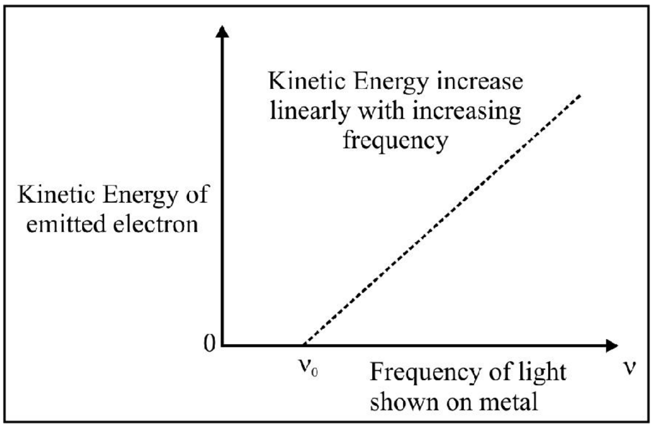
(ii) If the frequency of the incident light (ν) is more than the threshold frequency (ν_0), the excess energy is imparted to the electron as kinetic energy. Hence,

Energy of one quantum = Threshold Energy + Kinetic Energy

$$h\nu = h\nu_0 + (1/2)\, m_e v^2$$

(iii) When $\nu > \nu_0$, then on increasing the intensity the number of quanta incident increases thereby increasing the number of photoelectrons ejected.

(iv) When $\nu > \nu_0$, then on further increasing the frequency, the energy of each photon increases and thus kinetic energy of each ejected electron increases.



Note...

Energy can also be expressed in Electron Volt(eV).

The energy acquired by an electron when it is accelerated through a potential difference of one Volt.

$1 eV = 1.602 \times 10^{-19} J$

CONCLUSION :

Light has both the **Wave nature** (shows the phenomena of diffraction and interference) **and Particle nature** (could explain the black body radiation and photoelectric effect) . Thus, light has dual nature.

Bohr's Model is based on "Atomic Spectra", therefore before moving further we will study:

9. WHAT IS SPECTRUM?

A spectrum is a group or band of wavelengths/colours and the study of emission or absorption spectra is known as spectroscopy.

9.1 Types of spectrum

There are two types of spectrum:

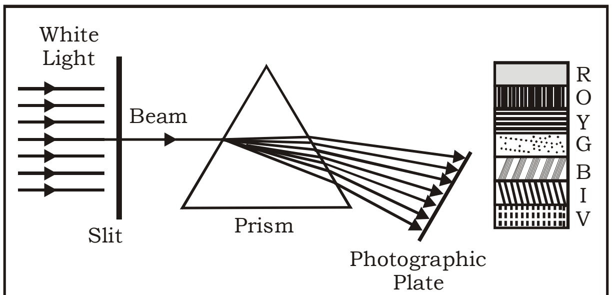
1) Emission Spectrum
2) Absorption Spectrum

9.2 Emission Spectrum

When radiations emitted from a source are incident on a prism and are separated into different wavelengths and obtained on a photographic plate.

(a) Continuous Emission Spectra:

There are no gaps between various wavelengths, one wavelength merges into another.

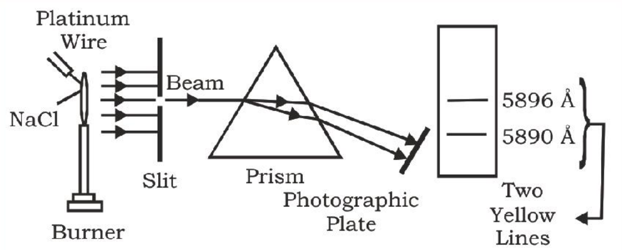


Continuous spectrum of white light

(b) Discontinuous Emission Spectra:

It is also known as **Line Spectra** or **atomic spectra.**

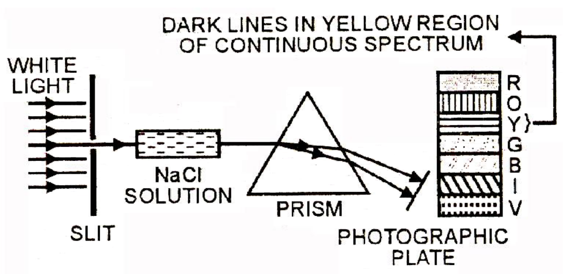
In this, certain wavelengths go missing from a group and that leaves dark spaces in between giving discontinuity to the spectrum. It is also known **as fingerprint of an element.**



Line Spectrum produced from a volatile salt placed in a flame.

9.3 Absorption Spectra

When light from any source is first passed through the solution of a chemical substance and then analysed, it is observed that there are some dark lines in the otherwise continuous spectra.



Production of absorption spectrum

Bohr studied the atomic spectra of hydrogen and based on that he proposed his model.

10. BOHR'S MODEL

Note: This model is applicable to H-atom or H-like species like He^+,Li^{2+},Be^{3+}**.**

10.1 Postulates

1) An atom consists of a small, heavy, positively charged nucleus in the centre and the electrons revolve around it in circular orbits.

2) Electrons revolve only in those orbits which have a fixed value of energy. Hence, these orbits are called energy levels or stationary states.

They are numbered as 1,2,3,...... These numbers are known as **Principal quantum Numbers**.

(a) **Energy** of an electron is given by:

$E_n = -R_H(Z^2/n^2)$ n = 1,2,3.......

where R_H is Rydberg's constant and its value is 2.18×10^{-18} J.

Z = atomic number

$$E_n = -2.18\times10^{-18}\frac{Z^2}{n^2}\,J/atom$$

$$E_n = -13.6\frac{Z^2}{n^2}\,eV/atom$$

$$E_n = -1312\frac{Z^2}{n^2}\,kJ/mol$$

Thus, energies of various levels are in the order:

K < L < M < N...... and so on.

Energy of the lowest state(n=1) is called **ground state.**

(b) **Radii** of the stationary states:

$$r_n = \frac{52.9n^2}{Z}\,pm$$

For H-atom (Z = 1), the radius of first stationary state is called **Bohr orbit** (52.9 pm)

(c) **Velocities** of the electron in different orbits:

$$v_n = \frac{2.188\times10^6 Z}{n}\,m/s$$

3) Since the electrons revolve only in those orbits which have fixed values of energy, hence electrons in an atom can have only certain definite values of energy and not any of their own. Thus, **energy of an electron is quantised.**

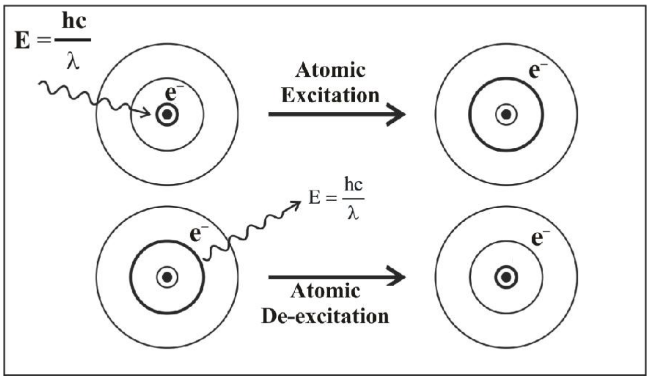
4) Like energy, the angular momentum of an electron in an atom can have certain definite values and not any value of their own.

$$mvr = \frac{nh}{2\pi}$$

Where n=1,2,3...... and so on.

5) An electron does not lose or gain energy when it is present in the same shell.

6) When an electron gains energy, it gets excited to higher energy levels and when it de-excites, it loses energy in the form of electromagnetic radiations and comes to lower energy values.



Excitation by absorption of light
and de-excitation by emission of light

10.2 What does negative energy for Hydrogen atom means?

This negative sign means that the energy of the electron in the atom is lower than the energy of a free electron at rest. A free electron at rest is an electron that is infinitely far away from the nucleus $(n = \infty)$ and is assigned the energy value of zero. As the electron gets closer to the nucleus (as *n* decreases), *En* becomes more and more negative. The most negative energy value is given by *n*=1 which corresponds to the most stable orbit.

10.3 Transition of Electron

We know that energy is absorbed or emitted when electron excites or de-excites respectively. The energy gap between the two orbits is

$$\Delta E = E_f - E_i$$

$$\Delta E = \left(-\frac{R_H}{n_f^2}\right) - \left(-\frac{R_H}{n_i^2}\right)$$

$$\Delta E = R_H\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right) = 2.18\times10^{-18}\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right) J/atom$$

The wavelength associated with the absorption or emission of the photon is:

$$\frac{1}{\lambda} = \frac{\Delta E}{hc} = \frac{R_H}{hc}\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right) = 1.09677\times10^7\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right) m^{-1}$$

This is known as Rydberg's formula.

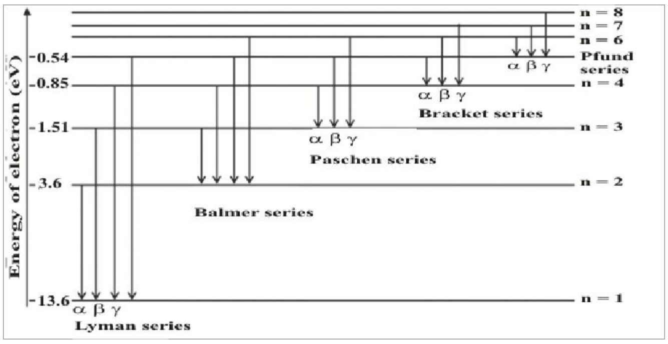
Note..

$1.09677 \times 10^7\ m^{-1}$ is also known as Rydberg's constant.

10.4 Line Spectrum of Hydrogen

When an electric discharge is passed through gaseous hydrogen, the H_2 molecules dissociate and the energetically excited hydrogen atoms produced emit electromagnetic radiations of discrete frequency. The hydrogen spectra consists of several lines named after their discoverer.

We get discrete lines and not a continuous spectra because the energy of an electron cannot change continuously but can have only definite values. Thus we can say that energy of an electron is quantised.



LYMAN SERIES :

When an electron jumps from any of the higher states to the ground state or first state (n = 1) ,the series of spectral lines emitted lies in the **ultra violet** region and are called as Lyman series.

Therefore , in Rydberg's formula $n_1 = 1$, $n_2 = 2,3,4,5...$

BALMER SERIES :

When an electron jumps from any of the higher states to the state with n=2,the series of spectral lines emitted lies in the **visible** region and are called as Balmer series.

Therefore , in Rydberg's formula $n_1 = 2$, $n_2 = 3,4,5,6....$

PASCHEN SERIES :

When an electron jumps from any of the higher states to the state with n=3 ,the series of spectral lines emitted lies in the **infrared** region and are called as Paschen series.

Therefore , in Rydberg's formula $n_1 = 3$, $n_2 = 4,5,6...$

BRACKETT SERIES :

When an electron jumps from any of the higher states to the state with n = 4,the series of spectral lines emitted lies in the **infrared** region and are called as Brackett series.

Therefore , in Rydberg's formula $n_1 = 4$, $n_2 = 5,6,7...$

PFUND SERIES :

When an electron jumps from any of the higher states to the state with n = 4,the series of spectral lines emitted lies in the **infrared** region and are called as Pfund series.

Therefore , in Rydberg's formula $n_1 = 5$, $n_2 = 6,7...$

10.5 Ionisation Energy

It is the energy required to remove the electron completely from the atom so as to convert it into a positive ion.

Thus, $n_1 = 1$ and $n_2 = \infty$

10.6 Limiting Line

It is the line of shortest wavelength i.e. $n_2 = \infty$

Note..

Although a hydrogen atom has only one electron, yet its spectra contains large number of lines. This is because a sample of hydrogen gas contains large number of molecules. When such a sample is heated to a high temperature, the hydrogen molecules split into hydrogen atoms. The electrons in different hydrogen atoms absorb different amounts of energies and are excited to different levels. When these electrons return, different electrons adopt different routes to return to the ground state. Thus, they emit different amounts of energies and thus large number of lines in atomic spectra in hydrogen.

Maximum no. of lines that can be emitted when an electron in an excited state n_2 de-excites to n_1 ($n_2 > n_1$) :

$$\frac{(n_2 - n_1 + 1)(n_2 - n_1)}{2}$$

10.7 Limitations of Bohr's Model

1) Inability to explain line spectra of multi-electron atoms.
2) It fails to account for the finer details (doublet-two closely spaced lines) of the hydrogen spectra.
3) Inability to explain splitting of lines in the magnetic field (Zeeman Effect) and in the electric field (Stark Effect)- If the source emitting the radiation is placed in magnetic or electric field, it is observed that each spectral line splits up into a number of lines. Splitting of spectral lines in magnetic field is known as Zeeman Effect while splitting of spectral lines in electric field is known as Stark Effect.
4) It could not explain the ability of atoms to form molecules by covalent bonds.
5) He ignores dual behaviour of matter and also contradicts Heisenberg uncertainty principle.

11. TOWARDS QUANTUM MECHANICAL MODEL

This model was based on two concepts:

(1) de Broglie Concept of dual nature of matter

(2) Heisenberg uncertainty Principle

11.1 Dual behaviour of matter

de Broglie proposed that matter should also exhibit dual behaviour i.e. both particle and wave like properties.

$$\lambda = \frac{h}{mv} = \frac{h}{p} = \frac{h}{\sqrt{2m(KE)}} = \frac{h}{\sqrt{2mqV}}$$

Where p is linear momentum of a particle.

According to de Broglie, every object in motion has a wave character. The wavelengths associated with ordinary objects are so short (because of their large masses) that their wave properties cannot be detected. The wavelengths associated with electrons and other subatomic particles (with very small mass) can however be detected experimentally.

11.2 Heisenberg's Uncertainty Principle

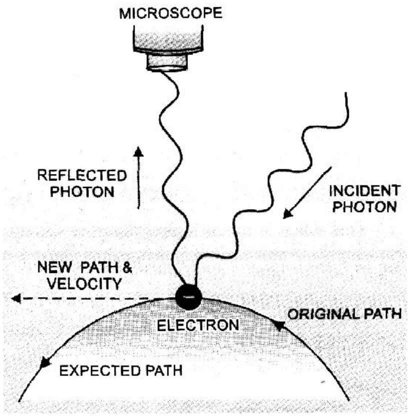
It is impossible to measure simultaneously the position and momentum of a small particle with absolute accuracy. If an attempt is made to measure any of these two quantities with higher accuracy, the other becomes less accurate. The product of the uncertainty in the position (Δx) and the uncertainty in momentum (Δp) is always a constant and is equal to or greater than $h/4\pi$.

$(\Delta x).(\Delta p) \geq h/4\pi$

Or $(\Delta x).(m\Delta v) \geq h/4\pi$

Or $(\Delta x).(\Delta x) \geq h/4\pi m$

11.2.1 Explaination



Change of momentum and position of electron on impact with a photon

Suppose we attempt to measure both the position and momentum of an electron. To pin point the position of the electron we have to use light so that the photon of light strikes the electron and the reflected photon is seen in the microscope. As a result of the hitting, the position as well as the velocity of the electron are disturbed.

11.2.2 Significance of Uncertainity Principle:

It rules out the existence of definite paths or trajectories of electrons as stated in Bohr's Model.

Note..

The effect of Heisenberg Uncertainty Principle is significant only for motion of microscopic objects, and is negligible for that of macroscopic objects.

12. QUANTUM MECHANICAL MODEL OF ATOM

Quantum mechanics is a theoretical science that deals with the study of the motion of microscopic objects which have both particle like and wave like properties. The fundamental equation of quantum mechanics was developed by Schrodinger.

This equation describes a function called **electron wave function (ψ)**. This wave function stores all the information about an electron like energy, position, orbital etc. As such it does not have any physical significance. The information stored in ψ about an electron can be extracted in terms of **Quantum Numbers.**

12.1 Probability Density

$|\psi|^2$ is the probability of finding the electron at a point within an atom.

12.2 Concept of Orbital

It is a three dimensional space around the nucleus within which the probability of finding an electron of given energy is maximum (say upto 90%).

12.3 Quantum Numbers

They may be defined as a set of four numbers with the help of which we can get complete information about all the electrons in an atom i.e. location, energy, type of orbital occupied, shape and orientation of that orbital etc.

The three quantum numbers called as Principal, Azimuthal and Magnetic quantum number are derived from Schrodinger wave equation. The fourth quantum number i.e. the Spin quantum number was proposed later on.

1) Principal Quantum Number (n):

It tells about the **shell** to which an electron belongs.

n = 1,2,3,4,5..... and so on.

This number helps to explain the **main lines of the spectrum** on the basis of electronic jumps between these shells.

(a) It gives the **average distance** of the electron from the nucleus. Larger the value of n, larger is the distance from the nucleus.

(b) It completely determines the **energy** in hydrogen atom or hydrogen like species.

The energy of H-atom or H-like species depends only on the value of n.

Order of energy : $1 < 2 < 3 < 4 < 5$....... and so on.

For multi-electron species, energy depends on both principal and azimuthal quantum number.

The maximum number of electrons present in any shell = $\mathbf{2n^2}$

2) Azimuthal Quantum Number (*l*) :

Also known as **Orbital Angular momentum** or **Subsidiary quantum number.** Within the same shell, there are number of sub-shells, so number of electronic jumps increases and this explains the presence of fine lines in the spectrum. This quantum number tells about :

(a) The number **of subshells** present in a shell.

(b) Angular momentum of an electron present in subshell.

(c) **Shapes** of various subshells present within the same shell.

(d) Relative energies of various subshells.

Value of l varies from 0 to n – 1

For 1st shell (n = 1): l = 0

For 2nd shell (n = 2): l = 0,1

For 3rd shell (n = 3): l = 0,1,2

For 4th shell (n = 4): l = 0,1,2,3

Value of I	Designation of subshell
0	s
1	p
2	d
3	f
4	g
5	h

The notations s,p,d,f represent the initial letters of the word sharp, principal, diffused and fundamental. In continuation $l = 4$ is called g subshell and $l = 5$ is called h subshell and so on.

Principal shell	Subshells
1st shell	$l = 0$ (s-subshell)
2nd shell	$l = 0,1$ (s & p subshell)
3rd shell	$l = 0,1,2$ (s,p & d subshell)
4th shell	$l = 0,1,2,3$ (s,p,d & f subshell)

Note..

The number of subshells present in any principal shell is equal to the number of the principal shell.

Energies of various subshell present within the same shell

is: s < p < d < f

Angular momentum of an electron in orbital :

$$\frac{h}{2\pi}\sqrt{l(l+1)} = \hbar\sqrt{l(l+1)}$$

(3) Magnetic Quantum Number(m):

This quantum number is required to explain the fact that when the source producing the line spectrum is placed in a magnetic field, each spectral line splits up into a number of line (Zeeman effect).

Under the influence of external magnetic field, electrons of a subshell can orient themselves in a certain preferred regions of space around the nucleus called orbitals.

The magnetic quantum number determines the number of preffered orientations of the electrons present in a subshell. Since each orientation corresponds to an orbital, thus magnetic quantum number determines the **number of orbitals** present in any subshell.

Value of m ranges from $-l$ to $+l$ including zero.

Sub-shell	Orbitals (m)	Number of orbitals
s-subshell (l=0)	m = 0	1
p-subshell (l=1)	m = -1, 0, 1	3
d-subshell (l=2)	m = –2, –1, 0, 1 ,2	5
f-subshell (l=3)	m = –3, –2, –1, 0, 1, 2, 3	7

Orbital	Value of m
P_x	m = 0
P_y	m = + 1
P_z	m = – 1

Orbital	Value of m
d_{z^2}	m = 0
d_{xz}	m = + 1
d_{yz}	m = – 1
$d_{x^2-y^2}$	m = + 2
d_{xy}	m = – 2

These orbitals of the same subshell having equal energy are called **degenerate orbitals.** Eg.

The three p-orbitals of a particular principal shell have the same energy in the absence of magnetic field.

Similarly, all five orbitals of d-subshell of a particular shell have the same energy.

Thus, for H-atom order of energy is:

1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f < ....

For multi electron atoms, the energy of the orbitals decreases with increase in effective nuclear charge. Eg

$E_{2s}(H) > E_{2s}(Li) > E_{2s}(Na) > E_{2s}(K)$

The total possible values of m in a given subshell = $2l + 1$

Total no. of orbitals in a given shell = n^2

4) Spin Quantum Number(s):

The electron in an atom not only moves around the nucleus but also spins about its own axis. Since the electron in an orbital can spin either in clockwise or anti-clockwise direction. Thus s can have only two values

$$+\frac{1}{2} \text{ or } -\frac{1}{2}$$

This quantum number helps to explain the magnetic properties of substances.

Note..

An orbital cannot have more than two electrons and these electrons should be of opposite spin.

Thus, maximum number of electrons in s-subshell = 2

Maximum number of electrons in p-subshell = 6

Maximum number of electrons in d-subshell = 10

Maximum number of electrons in f-subshell = 14

12.4 Shapes of atomic orbitals

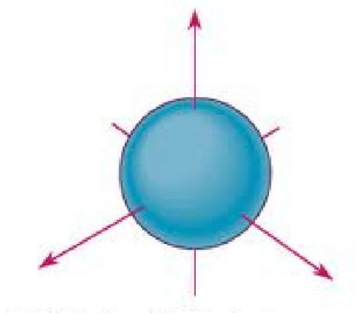
(1) Shape of s-orbitals:

(a) They are non-directional and spherically symmetric i.e. probability of finding the electron at a given distance is equal in all directions.

(b) 1s orbital and 2s orbital have same shape but size of 2s is larger.

(c) There is a spherical shell within 2s orbital where electron density is zero and is called a node.

(d) The value of azimuthal quantum number(l) is zero (l=0) and magnetic quantum number can have only one value i.e. m = 0



(2) Shape of p-orbitals:

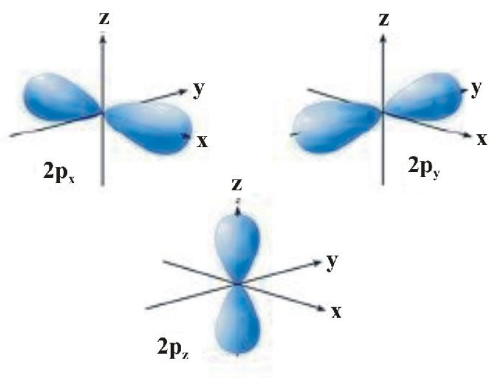
(a) It consists of two lobes present on either side of the plane that passes through the nucleus. The p-orbital is dumb-bell shaped.

(b) There are three possible orientations of electron cloud in p-orbitals. Therefore, the lobes of p-orbital may be considered to be along x,y and z axis. Hence they are designated as p_x, p_y, p_z. The three p-orbitals are oriented at right angles to one another.

(c) First main energy level(Principal quantum number n = 1) does not contain any p-orbital.

(d) The three p-orbitals of a particular energy level have same energy in absence of an external electric and magnetic field and are called degenerate orbitals.

(e) Like s orbitals, p-orbitals increase in size with increase in the energy of main shell of an atom. Thus, value of azimuthal quantum number is one (l=1) and magnetic quantum number has three values (m= –*1*, 0, +1)



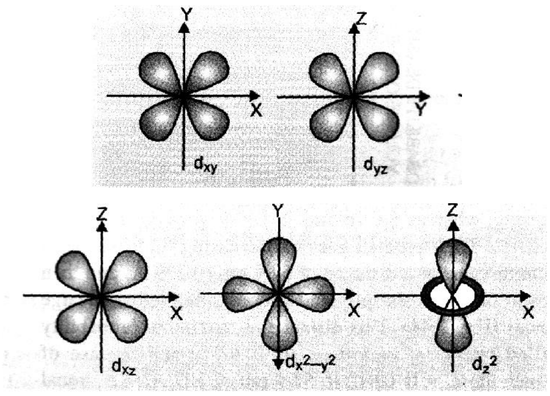
(3) Shapes of d-orbitals:

(a) They are designated as d_{xy}, d_{yz}, d_{zx} and $d_{x^2-y^2}$. They have a shape like a four leaf clover. The fifth d orbital designated as d_{z^2} looks like a doughnut.

(b) All five d orbitals have same energy in the absence of magnetic field.

(c) The d orbitals have azimuthal quantum number $l = 2$ and magnetic quantum number values –2, –1,0,+ 1,+ 2.

(d) For principal shell number 1 and 2, there are no d orbitals.



We have drawn boundary surface diagrams i.e the surface is drawn in space for an orbital on which probability density $(\psi)^2$ is constant. It encloses the region where probability of finding the electron is very high. We do not draw a boundary surface diagram that encloses 100% probability of finding the electron because probability density has some value, howsoever small it may be, at any finite distance from the nucleus. Thus it is not possible to draw a boundary surface diagram of a rigid size in which the probability of finding the electron is 100%.

12.5 Nodes and nodal planes

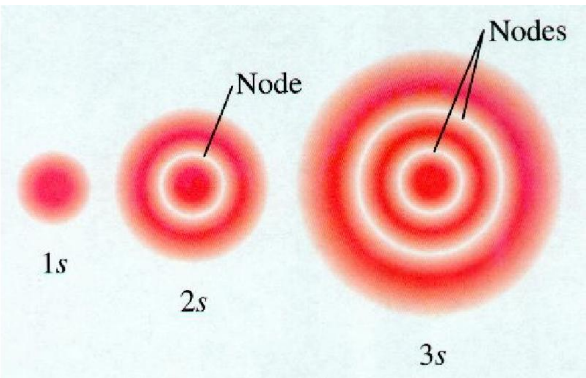
Node:

It is a region of zero probability.

There are two types of nodes:

(1) Radial or Spherical nodes: Three dimensional regions in an orbital where probability of finding the electron becomes zero.

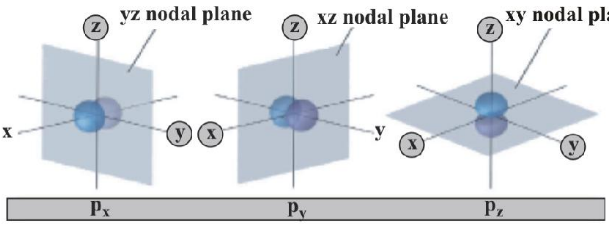
Number of radial/ spherical nodes = $n - l - 1$



(2) Planar or Angular Nodes: They are the planes cutting through the nucleus on which probability of finding the electron is zero.

Number of Planar/Angular Nodes: l

Total Number of nodes: n - 1



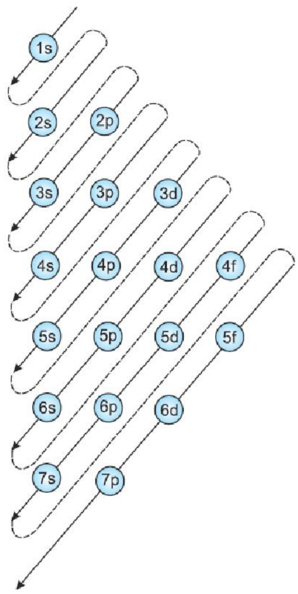
12.6 Filling of orbitals in an atom

(1) Aufbau Principle:

In the ground state of the atoms, the orbitals are filled in order of their increasing energies. In other words, electrons first occupy the lowest energy orbital available to them and enter into higher energy orbitals only when the lower energy orbitals are filled.

Unlike H-atom where energy of orbitals depend only on n, energy of orbitals of multi-electron orbitals depend on both n and l. Their order of energy can be determined using **(n+l) rule.**

According to this rule, lower the value of (n+l) for an orbital, lower is its energy. If two different types of orbitals have the same value of (n+l), the orbital with lower value of n has lower energy.



$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2$......

(2) Pauli Exclusion Principle: An orbital can have maximum two electrons and these must have opposite spin.

Correct : $\boxed{\uparrow\downarrow}$ *(i)* or $\boxed{\downarrow\uparrow}$ *(ii)*

Incorrect : $\boxed{\uparrow\uparrow}$ *(iii)* or $\boxed{\downarrow\downarrow}$ *(iv)*

(3) Hund's rule of maximum multiplicity: Electron pairing in p,d and f orbitals cannot occur until each orbital of a given subshell contains one electron each. Also all the singly occupied orbitals will have **parallel spin.**

Electronic configurations of elements in the ground state

Atomic No.	Element	Electronic Config
1	H	$1s^1$
2	He	$1s^2$
3	Li	[He] $2s^1$
4	Be	[He] $2s^2$
5	B	[He] $2s^2\ 2p^1$
6	C	[He] $2s^2\ 2p^2$
7	N	[He] $2s^2\ 2p^3$
8	O	[He] $2s^2\ 2p^4$
9	F	[He] $2s^2\ 2p^5$
10	Ne	[He] $2s^2\ 2p^6$
11	Na	[Ne] $3s^1$
12	Mg	[Ne] $3s^2$
13	Al	[Ne] $3s^2\ 3p^1$
14	Si	[Ne] $3s^2\ 3p^2$
15	P	[Ne] $3s^2\ 3p^3$
16	S	[Ne] $3s^2\ 3p^4$
17	Cl	[Ne] $3s^2\ 3p^5$
18	Ar	[Ne] $3s^2\ 3p^6$
19	K	[Ar] $4s^1$
20	Ca	[Ar] $4s^2$
21	Sc	[Ar] $3d^1\ 4s^2$
22	Ti	[Ar] $3d^2\ 4s^2$
23	V	[Ar] $3d^3\ 4s^2$

24	Cr	$[Ar]\,3d^5\,4s^1$
25	Mn	$[Ar]\,3d^5\,4s^2$
26	Fe	$[Ar]\,3d^6\,4s^2$
27	Co	$[Ar]\,3d^7\,4s^2$
28	Ni	$[Ar]\,3d^8\,4s^2$
29	Cu	$[Ar]\,3d^{10}\,4s^1$
30	Zn	$[Ar]\,3d^{10}\,4s^2$

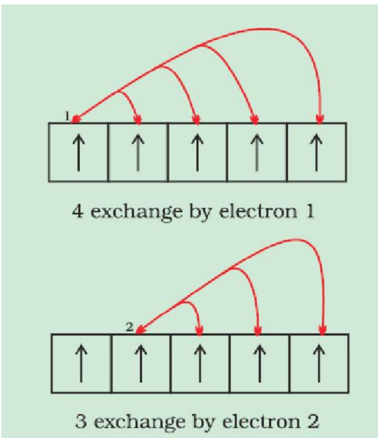
12.7 Exceptional Configuration of Cr & Cu

The completely filled and completely half filled sub-shells are stable due to the following reasons:

1. **Symmetrical distribution of electrons*:*** It is well known that symmetry leads to stability. The completely filled or half filled subshells have symmetrical distribution of electrons in them and are therefore more stable. Electrons in the same subshell (here *3d*) have equal energy but different spatial distribution. Consequently, their shielding of one another is relatively small and the electrons are more strongly attracted by the nucleus.

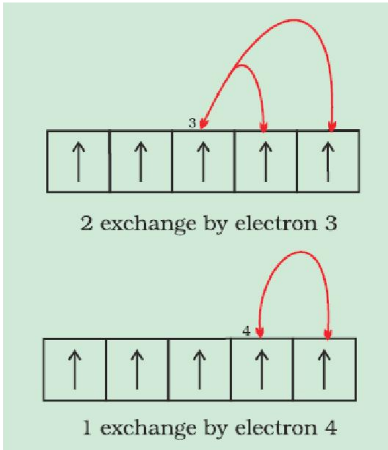
2. **Exchange Energy *:*** The stabilizing effect arises whenever two or more electrons with the same spin are present in the degenerate orbitals of a subshell. These electrons tend to exchange their positions and the energy released due to this exchange is called exchange energy. The number of exchanges that can take place is maximum when the subshell is either half filled or completely filled. As a result the exchange energy is maximum and so is the stability.

eg. for Cr : $[Ar]\,4s^1\,3d^5$



4 exchange by electron 1

3 exchange by electron 2



2 exchange by electron 3

1 exchange by electron 4

Thus, total number of exchanges=10

12.8 Electronic Configuration of Ions

12.8.1 Cations:

They are formed when outermost electrons are removed from an atom. While removing the electrons, we must remove the electrons from the highest principal quantum number.

12.8.2 Anions:

They are formed when electrons are added to the innermost empty shell.

12.8.3 Magnetic moment :

$$\sqrt{n(n+2)}\ \text{B.M.}$$

B.M. $\rightarrow$ Bohr Magneton

Where n is number of unpaired electrons.

Species with unpaired electrons are called **paramagnetic** and the species with no unpaired electrons are called **diamagnetic**

13. IMPORTANT RELATIONS

Rydberg equation : $\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

($R_H = 109678\ cm^{-1}$ and $n_2 > n_1$)

$c = \lambda \times \upsilon$ and $\bar{v} = \frac{1}{\lambda}$

$E = h\nu$ or $= \frac{hc}{\lambda}$

Bohr's Model

$$E_n = \frac{-1312Z^2}{n^2} kJ\ mol^{-1}$$

$$= \frac{-2.178 \times 10^{-18} Z^2}{n^2} J/atom = \frac{-13.6Z^2}{n^2} eV/atom$$

Velocity of electron, $\upsilon_n = \frac{2.165 \times 10^6 Z}{n} m/s$

Radius of orbit $= \frac{0.529\ n^2}{Z}$ Å

Photoelectric effect $= h\nu = h\nu_0 + \frac{1}{2}mv^2$

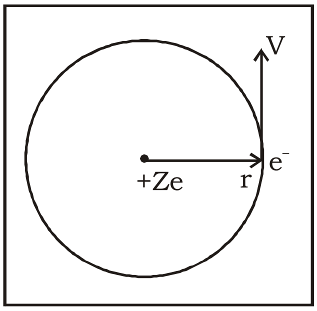
(ν_0 = Threshold frequency)

de-Broglie equation : $\lambda = \frac{h}{mv}$

Heisenberg's uncetainity principle : $\Delta x \times \Delta p \geq \frac{h}{4\pi}$

14. MATHEMATICAL MODELLING OF BOHR's POSTULATES

Consider an ion of atomic number (Z) containing single electron revolving its nucleus at a distance of 'r' as shown in the figure.



Note..

Atomic number = Number of protons on the nucleus = Z

$\Rightarrow$ Charge on the nucleus = + Ze

Electrostatic force of attraction (F) between the nucleus of charge + Ze and electron (–e) is given by :

$$F = \frac{K|q_1||q_2|}{r^2} \quad \text{where } K = \frac{1}{4\pi\varepsilon_0}$$

$$= 9 \times 10^9\ Nm^2C^{-2}$$

$$F = \frac{K|Ze||-e|}{r^2} = \frac{KZe^2}{r^2} \quad ....(i)$$

The centrifugal forces acting on the electron is $\frac{mv^2}{r}$ ....(ii)

This centrifugal force must be provided by the electrostatic force of attraction (F).

$\Rightarrow$ From (i) and (ii), we have :

$$\frac{KZe^2}{r^2} = \frac{mV^2}{r} \quad ....(iii)$$

Angular momentum of electron about the nucleus =

$$mVr = \frac{nh}{2\pi} \quad .... (iv)$$

where 'n' is a positive integer

($n = 1, 2, 3, ... \infty$)

Solve (iii) and (iv) to get :

$$V = \frac{2\pi KZe^2}{nh} \text{ and } r = \frac{n^2h^2}{4\pi^2\ Kme^2Z}$$

put $K = 9 \times 10^9\ Nm^2C^{-2}$, $e = 1.6 \times 10^{-19}C$ and $h = 6.63 \times 10^{-34}$ Js in the above expressions to get :

Velocity of an electron in n^{th} Bohr orbit ≡

$$\mathbf{V_n = 2.165 \times 10^6 \frac{Z}{n} ms^{-1}}$$

and Radius of the n^{th} Bohr orbit ≡ $r_n = 0.53 \frac{n^2}{Z}$ Å

Now, the Total Energy of the electron moving in n^{th} orbit ≡ $K.E._n + E.P.E._n$

$$T.E._n = \frac{1}{2}mV_n^2 + \frac{K(Ze)(-e)}{r} \quad \left[\because E.P.E. = \frac{K\ q_1\ q_2}{r} \right]$$

$$\Rightarrow \quad T.E_n = \frac{1}{2}\left(\frac{K\,Ze^2}{r_n}\right) + \frac{K(Ze)(-e)}{r_n} \quad \text{[Using (iii)]}$$

$$\Rightarrow \quad \mathbf{E_n \equiv T.\,E_n = \frac{-KZe^2}{2r_n}}$$

It can be shown from the above expressions that :

$$K.E._n = \frac{1}{2}\frac{K\,Ze^2}{r_n},\ P.E._n = \frac{-K\,Ze^2}{r_n} \text{ and } E_n = \frac{-K\,Ze^2}{2r_n}$$

or $K.E._n = -E_n$ and $E.P.E._n = 2E_n$

Using the value of r_n in the expression of E_n, we get :

$$E_n = \frac{-2\pi^2\,K^2me^4Z^2}{n^2h^2}$$

$$\mathbf{E_n \equiv -2.178 \times 10^{-18}\frac{Z^2}{n^2}\ J/atom}$$

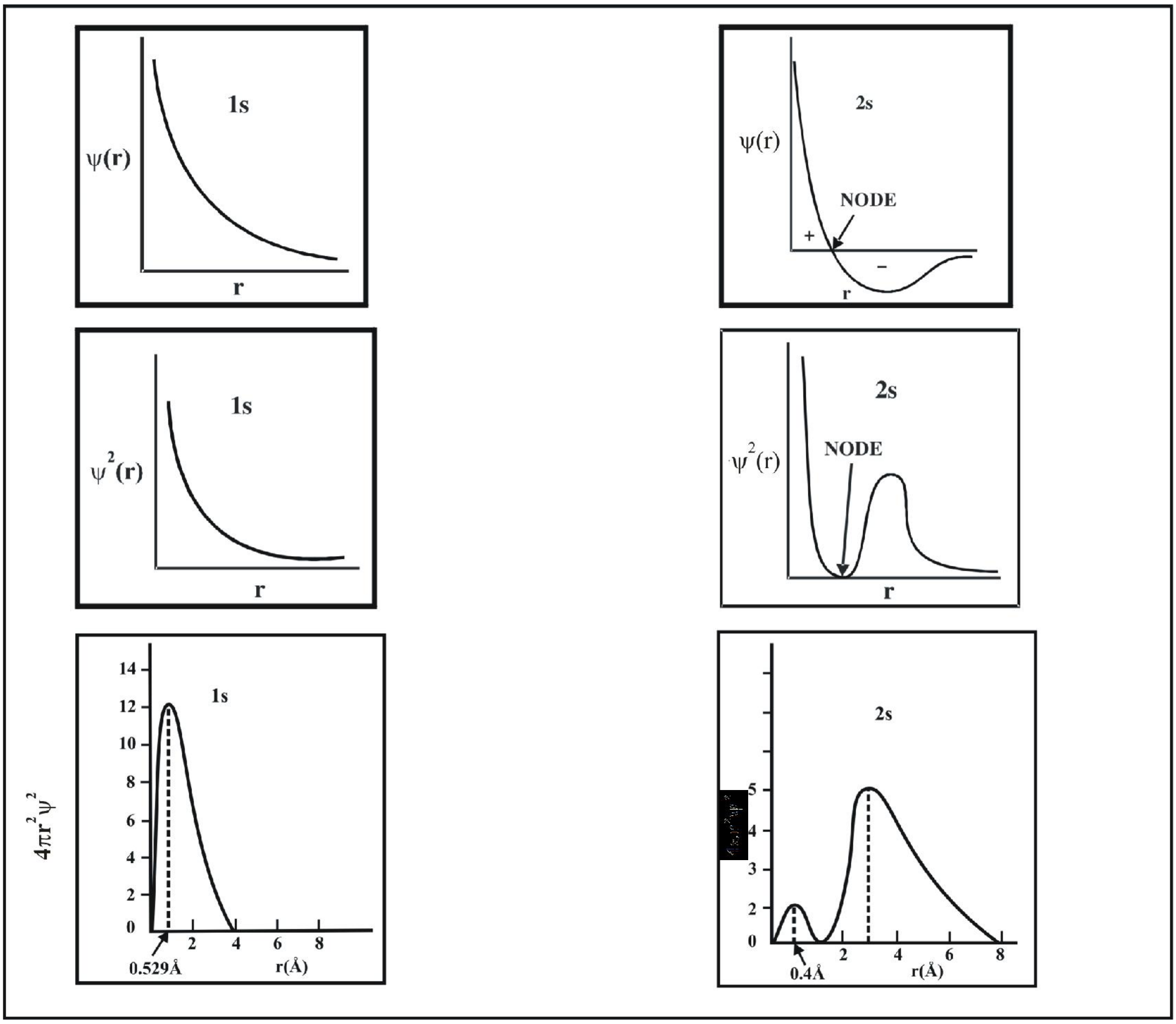
or $\mathbf{E_n \equiv -13.6\frac{Z^2}{n^2}\ eV/atom}$

$$\left[\because 1eV = 1.6 \times 10^{-19}\,J\right]$$

$$\equiv -2.178 \times 10^{-18}\frac{Z^2}{n^2} \times 6.02 \times 10^{23}\ J/mole$$

$$\mathbf{\equiv -1312\ \frac{Z^2}{n^2}\ kJ/mole}$$

15. PROBABILITY DISTRIBUTION DIAGRAMS FOR 1s AND 2s



SOLVED EXAMPLES

Example - 1

Distinguish between Proton, Neutron & Electron

Sol.

	Parameter	Proton	Neutron	Electron
1.	**Charge**	unit positive ($+1.6 \times 10^{-19}$ C)	no charge	unit negative (-1.6×10^{-19} C)
2.	**Mass**	1.672×10^{-27} Kg	1.674×10^{-27} Kg	9.1094×10^{-31} Kg
3.	**Denoted**	${}_1p^1$	${}_0n^1$	${}_{-1}e^0$
4.	**Location**	nucleus	nucleus	extra-nuclear region
5.	**Effected by**	Electric field & magnetic field (towards negative plate)	Remain undeflected in electric and magnetic fields.	Electric field & magnetic field (towards positive plate)
6.	**Discovered by**	Goldstein	Chadwick	Thomson

Example - 2

Give the difference between Isotopes, isobars & Isotones.

Sol.

	Isotopes	Isobars	Isotones
1.	They are atoms having the same number of protons but differ in the number of neutrons.	They are atoms having the same sum of protons and neutrons.	They are atoms having the same number of neutrons but differ in the number of protons.
2.	They have the same atomic number but differ in their mass number.	They have the same mass number but differ in their atomic number	They have different atomic number and mass number
3.	They are atoms of the same element hence they have identical chemical properties.	They are atoms of different elements hence there is no similarity in their chemical properties.	They are atoms of different elements hence there is no similarity in their chemical properties.

Example - 3

Why electrons are called Planetary according to Rutherford model?

Sol. This model was analgous to the solar system, where the nucleus may be compared to the sun and the electrons to the planets. The Coulombic force between the nucleus and the electron is $\frac{kq_1q_2}{r^2}$ where q_1 and q_2 are the charges, r is the distance of separation between the molecules and the electron and k is the proportionality constant. This is similar to gravitational force between two masses m_1 and m_2 as $G.\frac{m_1m_2}{r^2}$ where G is gravitational constant and r is the distance of separation between planet and the sun.

Example - 4

(a) Why e/m ratio of anode rays is different for different gases.

(b) Why was pressure of air inside the tube reduced to 10^{-2} atm in cathode ray experiment ?

Sol. (a) The charge to mass (e/m) ratio of anode rays is dependent upon the nature of the gas taken in the discharge tube. This is because positively charged particles are produced by the loss of one or more electrons from the neutral atoms of the gas contained in the discharge tube. Therefore, the mass of the positively charged particles will depend upon the nature of the gas. **In case of hydrogen, the charge to mass (e/m) ratio was maximum.**

(b) This is because at very low pressure gas gets ionized and become conducting.

Example - 5

Calculate the number of neutrons in

(a) dipositive zinc ion **(b) Mg^{+2}**

Sol. (a) Number of protons = 30

Mass Number = 65

$\Rightarrow p + n = 65$

$n = 65 - 30 = 35$

(b) Number of protons = 12

Mass Number = 24

$p + n = 24$

$n = 24 - 12 = 12.$

Example - 6

An atom of an element contains 29 electrons and 35 neutrons. Deduce

(a) the number of protons and

(b) the electronic configuration of the element

(c) Mass Number

Sol. Number of electrons = 29

Number of neutrons = 35

For neutral atom :

Number of protons = Number of electrons

(a) Therefore, number of protons = 29

(b) Atomic Number = 29

Element is Cu.

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$

(c) Mass Number = p + n

$= 29 + 35 = 64$

Example - 7

(i) Which model of an atom is called apple pie model?

(ii) Why are the atomic masses of most element is fractional?

Sol. (i) Thomson Model

(ii) It is due to the existence of various isotopes of an element in various percentage abundance.

Example - 8

Explain the Rutherford's scattering experiment and also give its drawbacks.

Sol. A very thin foil of gold (0.004nm) is bombarded by a fine stream of alpha particles. A fluorescent screen (ZnS) is placed behind the gold foil, where points were recorded which were emerging from α-particles. Polonium was used as the source of α-particles.

Observations

Rutherford carried out a number of experiments, involving the scattering of α-particles by a very thin foil of gold.

Observations were:

(i) Most of the α-particles (99%) passes through it, without any deviation or deflection.

(ii) Some of the α-particles were deflected through small angles.

(iii) Very few α-particles were deflected by large angles and occasionally an α-particle got deflected by 180°

Conclusions

(i) An atom must be extremely hollow and must consist of mostly empty space because most of the particles passed through it without any deflection.

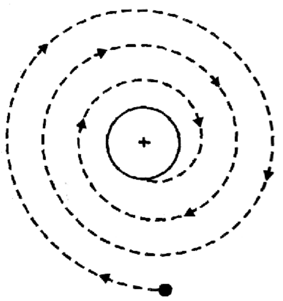
(ii) Very few particles were deflected to a large extent. This indicates that:

(a) Electrons because of their negative charge and very low mass cannot deflect heavy and positively charged α particles

(b) There must be a very heavy and positively charged body in the atom i.e. nucleus which does not permit the passage of positively charged α particles.

(c) Because, the number of α particles which undergo deflection of 180°, is very small, therefore the volume of positively charged body must be extremely small fraction of the total volume of the atom. This positively charged body must be at the centre of the atom which is called **nucleus.**

Drawbacks



ELECTRON FALLING INTO THE NUCLEUS

(i) According to classical mechanics, any charged body in motion under the influence of attractive forces should radiate energy continuously. If this is so, the electron will follow a spiral path and finally fall into the nucleus and the structure would collapse. This behaviour is never observed.

(ii) It says nothing about the electronic structure of atoms i.e. how the electrons are distributed around the nucleus and what are the energies of these electrons.

Example - 9

The nuclear radius is of the order of 10^{-13}cm while atomic radius is if the order 10^{-8}cm. Assuming nucleus to be spherical, what fraction of atomic volume is occupied by nucleus?

Sol. Volume of a sphere = $\frac{4}{3}\pi r^3$

Therefore, volume of nucleus = $\frac{4}{3}\pi (10^{-13})^3$

Volume of atom = $\frac{4}{3}\pi (10^{-8})^3$

Fraction of atomic volume occupied by

nucleus = $\dfrac{\frac{4}{3}\pi (10^{-13})^3}{\frac{4}{3}\pi (10^{-8})^3} = 10^{-15}$

Example - 10

(i) Calculate the total number of electrons in 1.6g of methane.

(ii) An ion with mass number 56 contains 3 units of positive charge and 30.4% more neutrons than electrons. Assign symbol to this ion.

Sol. (i) 1 molecule of CH_4 has = 10 electrons

16g CH_4 has = $10N_A$ electrons

1.6g CH_4 has = 1.0 N_A electrons

= 6.022×10^{23} electrons

(ii) Suppose number of electrons in the ion, M^{3+} = x

$\therefore$ No. of neutrons = $x + \frac{30.4}{100}x = 1.304x$

No. of electrons in the neutral atom = x + 3

$\therefore$ No. of protons = x + 3

Mass no. No. of protons + No. of neutrons

$56 = x + 3 + 1.304\,x$ or $2.304x = 53$ or $x = 23$

$\therefore$ No. of protons = Atomic no. = x + 3 = 23 + 3 = 26

Hence, the symbol of the ion will be ${}^{56}_{26}Fe^{3+}$.

Example - 11

(i) Arrange X-rays, cosmic rays and radiowaves according to frequency.

(ii) Calculate the wavenumber of yellow radiation having wavelength of 5800Å.

(iii) Define threshold frequency.

Sol. (i) Cosmic rays > x-rays > Radio waves

(ii) $\bar{\nu} = \frac{1}{\lambda} \Rightarrow \bar{\nu} = \frac{1}{5800 \times 10^{-10}} = 1.72 \times 10^{6}\ m^{-1}$

(iii) The minimum frequency required to eject an electron from the surface of metal.

Example - 12

Define

(a) Electromagnetic spectrum

(b) Black Body and Black body radiation.

(c) Photoelectric Effect

Sol.

(a) When all the electromagnetic radiations are arranged in increasing order of wavelength or decreasing frequency the band of radiations obtained is termed as electromagnetic spectrum.

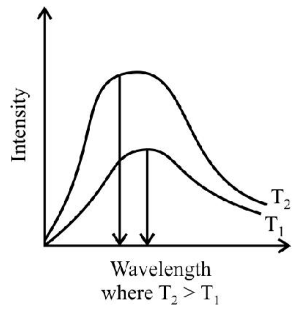
The visible spectrum is a subset of this spectrum (VIBGYOR) whose range of wavelength is 380-760nm.

The wavelengths increase in the order:

Gamma Rays < X-rays < Ultra-violet rays < Visible< Infrared < Micro-waves <Radio waves.

(b) (1) An ideal body, which emits and absorbs radiations of all frequencies is called **black body** and radiation emitted by a black body is called **black body radiation.**

(2) According to Maxwell's theory on heating a body the intensity should increase, that is, energy radiated per unit area should increase without having any effect on the wavelength or frequency.

(3) 

(4) So it is observed that with increasing temperature the dominant wavelength in the emitted radiations decreases and the frequency increases.

(c) When radiations with certain minimum frequency (ν_0) strike the surface of a metal, the electrons are ejected from the surface of the metal. This phenomena is called **photoelectric effect.** The electrons emitted are called **photoelectrons.**

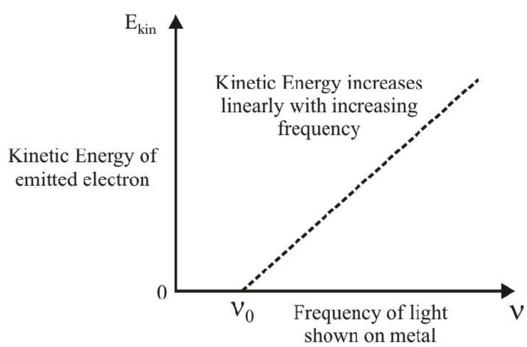
If the frequency of the incident light (ν) is more than the threshold frequency (ν_0), the excess energy is imparted to the electron as kinetic energy. Hence,

Energy of one quantum = Threshold Energy + Kinetic Energy

$$h\nu = h\nu_0 + (1/2)\,m_e v^2$$

When $\nu > \nu_0$, then on increasing the intensity the number of quanta incident increases thereby increasing the number of photoelectrons ejected.

When $\nu > \nu_0$, then on further increasing the frequency, the energy of each photon increases and thus kinetic energy of each ejected electron increases.



Example - 13

Differentiate between absorption spectrum and emission spectrum?

Sol. The main points of difference between absorption & amission spectra are summed up in the below :

Absorption Spectrum :

1. Absorption spectrum is obtained when the white light is first passed through the substance and the transmitted light is analysed in the spectroscope.
2. It consists of dark lines in the otherwise continous spectrum
3. Absorption spectrum is always discontinuous spectrum consisting of dark lines.

Emission Spectrum :

1. Emission spectrum is obtained when the radiation from the source are directly analysed in the spectroscope.
2. It consists of bright coloured lines separated by dark spaces.
3. Emission spectrum can be continuous spectrum (if source emits white light) or discontinuous, i.e., line spectrum if source emits some coloured radiation.

Example - 14

(a) Calculate the wavelength, frequency and wave number of light wave whose time period is 2×10^{-10} sec?

(b) Yellow light emitted from sodium lamp has wavelength of 580 nm. Calculate the frequency and wave number of yellow light.

(c) How long will it take for a radio wave of frequency 6×10^{13} Hz, sent by a path finder to travel from Mars to earth over a distance of 8×10^7 km

Sol. (a) Frequency $(\nu) = \dfrac{1}{\text{Period}} = \dfrac{1}{2.0 \times 10^{-10}\,\text{s}} = 5 \times 10^9\,\text{s}^{-1}$.

Wavelength, $\lambda = \dfrac{c}{\nu} = \dfrac{3.0 \times 10^8\,\text{m s}^{-1}}{5 \times 10^9\,\text{s}^{-1}} = 6.0 \times 10^{-2}\,\text{m}$

Wavenumber, $\bar{\nu} = \dfrac{1}{\lambda} = \dfrac{1}{6 \times 10^{-2}\,\text{m}} = 16.66\,\text{m}^{-1}$

(b) $C = \nu\lambda$

$$\Rightarrow \nu = \frac{3 \times 10^8}{580 \times 10^{-9}}$$

$$= 5.17 \times 10^{14}\,\text{Hz}$$

$$\bar{\nu} = \frac{1}{\lambda} = \frac{1}{580 \times 10^{-9}} = 1.7 \times 10^6\,\text{m}^{-1}$$

(c) All radiations in vacuum travel with the same speed,

$C = 3 \times 10^8$ m/s

Distance to be travelled $= 8 \times 10^7 \times 10^3$ m

$= 8 \times 10^{10}$ m

Time taken $= \dfrac{8 \times 10^{10}}{3 \times 10^8} = 2.66 \times 10^2$ sec

= 4 min. 26 sec.

Example - 15

A 100 watt bulb emits electromagnetic light of wavelength 400nm. Calculate the number of photons emitted per second by the bulb.

Sol. Power of the bulb = 100 watt

$= 100\,\text{J s}^{-1}$

Energy of one photon, $E = h\nu = \dfrac{hc}{\lambda}$

$$= \frac{(6.626 \times 10^{-34}\,\text{Js}) \times (3 \times 10^8\,\text{m s}^{-1})}{400 \times 10^{-9}\,\text{m}}$$

$= 4.969 \times 10^{-19}$ J

$\therefore$ Number of photons emitted

$$= \frac{100\,\text{J s}^{-1}}{4.969 \times 10^{-19}\,\text{J}}$$

$= 2.012 \times 10^{20}\,\text{s}^{-1}$.

Example - 16

The threshold frequency ν_0 for a metal is $7 \times 10^{14} s^{-1}$. Calculate the kinetic energy of an electron emitted when radiation of frequency $1 \times 10^{15} s^{-1}$ hits the metal.

Sol. $\nu_0 = 7 \times 10^{14}\ s^{-1}$; $\nu = 10^{15}\ s^{-1}$

According to photo electric effect

$h\nu = h\nu_0 + K.E.$

$K.E. = h(\nu - \nu_0)$

$= 6.626 \times 10^{-34} (10^{15} - 7 \times 10^{14})$

$= 1.9878 \times 10^{-19}$ J.

Example - 17

If photon of wavelength 150pm strikes an atom and one of its inner bound electron ejected out with a velocity of 1.5×10^7m/s. Calculate the energy with which it is bound to the nucleus.

Sol. Energy of incident photon

$$= \frac{hc}{\lambda}$$

$$= \frac{(6.626 \times 10^{-34} Js) \times (3.0 \times 10^8 ms^{-1})}{(150 \times 10^{-12} m)}$$

$= 1.325 \times 10^{-15}$ J

Energy of ejected electron

$$= \frac{1}{2} mv^2$$

$$= \frac{1}{2} \times (9.11 \times 10^{-31} kg) \times (1.5 \times 10^7\ ms^{-1})^2$$

$= 1.025 \times 10^{-16}$ J

Energy with which the electron was bound to the nucleus

$= 13.25 \times 10^{-16} - 1.025 \times 10^{-16}$

$= 12.225 \times 10^{-16}$ J

$$\text{or } = \frac{12.225 \times 10^{-16}}{1.602 \times 10^{-19}} = \mathbf{7.63 \times 10^3\ eV.}$$

Example - 18

The work function for caesium atom is 1.9eV. Calculate (a) threshold frequency (b) threshold wavelength of the radiation (c) If the cesium atom is irradiated with a wavelength of 500nm, calculate the kinetic energy and the velocity of ejected electron.

Sol. **(a)** Work function $(w_0) = h\nu_0$

$$\therefore \nu_0 = \frac{W_0}{h} = \frac{1.9 \times 1.602 \times 10^{-19} J}{6.626 \times 10^{-34}\ J\ s} (1\ eV = 1.602 \times 10^{-19})$$

$= \mathbf{4.59 \times 10^{14}\ s^{-1}}$

(b) $$\lambda_0 = \frac{c}{\nu_0} = \frac{3.0 \times 10^8\ m\ s^{-1}}{4.59 \times 10^{14} s^{-1}} = 6.54 \times 10^{-7} m$$

$= 654 \times 10^{-9} m = \mathbf{654\ nm}$

(c) Work function, $h\nu_0 = 1.9 \times 1.602 \times 10^{-19}$J

$= 3.04 \times 10^{-19}$ J

Energy of incident radiation, $h\nu = \dfrac{hc}{\lambda}$

$$= \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{500 \times 10^{-9}} = 3.97 \times 10^{-19} J$$

$h\nu = h\nu_0 + K.E.$

$\therefore K.E. = h\nu - h\nu_0$

$= 3.97 \times 10^{-19} - 3.04 \times 10^{-19}$

$\boxed{K.E. = 0.93 \times 10^{-19} J}$

$K.E. = \dfrac{1}{2} mv^2$

$$\therefore v^2 = \frac{2 \times K.E.}{m} = \frac{2 \times 0.93 \times 10^{-19}}{9.11 \times 10^{-31}} = 0.204 \times 10^{12}$$

$v = \sqrt{0.204 \times 10^{12}} = 0.452 \times 10^6$

$\boxed{\therefore v = 4.52 \times 10^5\ ms^{-1}}$

Example - 19

Electrons are emitted with zero velocity from a metal surface when it is exposed to radiations of wavelength 6800Å. Calculate the threshold frequency and work function of the metal.

Sol. K.E. = $h\nu - h\nu_0$

If the velocity is zero, K.E. = 0

$\therefore\ 0 = h\nu - h\nu_0$

or $h\nu = h\nu_0$ or $\nu = \nu_0$

$$\therefore\ \nu = \frac{c}{\lambda} = \frac{3.0\times10^{8}\,ms^{-1}}{6800\times10^{-10}\,m}$$

$= 4.41 \times 10^{14}\ s^{-1}$

Threshold frequency = $4.41 \times 10^{14}\ s^{-1}$

Work function = $h\nu_0 = 6.626 \times 10^{-34} \times 4.41 \times 10^{14}$

= **2.92×10^{-19} J.**

Example - 20

Why electronic energy is negative?

Sol. The negative sign of energy means that the energy of the electron in the atom is lower than the energy of a free electron at rest. A free electron at rest is an electron that is sufficiently far away from the nucleus and its energy is **assumed** to be **zero.** Mathematically, it corresponds to setting n equal to infinity in the equation so that $E_\infty = 0$. As the electron moves closer to the nucleus due to electrostatic attraction, work is done by the electron itself and hence energy is released. Consequently, its energy decreases and it takes energy values less than zero, which means negative values. The negative sign also indicates that the electron is bound to the nucleus and a hydrogen atom is in a stable state in comparison to a state where electron is sufficiently far away from the nucleus.

Example - 21

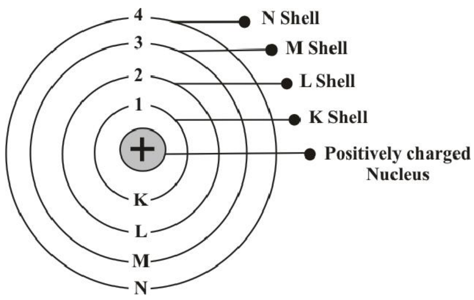
Describe postulates of Bohr's model.

Sol. **Postulates:**

1) An atom consists of a small, heavy positively charged nucleus in the centre and the electrons revolve around it in circular orbits.

2) Electrons revolve only in those orbits which have a fixed value of energy. Hence, these orbits are called energy levels or stationary states.

They are numbered as 1,2,3,...... These numbers are known as **Principal quantum Numbers**.



(a) **Energy** of an electron is given by:

$E_n = -R_H(Z^2/n^2)$ n = 1,2,3.......

where R_H is Rydberg's constant and its value is 2.18×10^{-18} J.

Z = atomic number

$$E_n = -2.18\times10^{-18}\frac{Z^2}{n^2}\ \text{J/atom}$$

$$E_n = -13.6\frac{Z^2}{n^2}\ \text{eV/atom}$$

$$E_n = -1312\frac{Z^2}{n^2}\ \text{kJ/mol}$$

Thus, energies of various levels are in the order:

K < L < M < N...... and so on.

Energy of the lowest state(n=1) is called **ground state.**

(b) **Radii** of the stationary states:

$$r_n = \frac{52.9n^2}{Z}\ \text{pm}$$

For H-atom (Z = 1), the radius of first stationary state is called **Bohr orbit** (52.9 pm)

(c) **Velocities** of the electron in different orbits:

$$V_n = \frac{2.188\times10^{6}Z}{n}\ \text{m/s}$$

3) Since the electrons revolve only in those orbits which have fixed values of energy, hence electrons in an atom can have only certain definite values of energy and not any of their own. Thus, **energy of an electron is quantised.**

4) Like energy, the angular momentum of an electron in an atom can have certain definite values and not any value of their own.

$$mvr = \frac{nh}{2\pi}$$

Where n=1,2,3...... and so on.

5) An electron does not lose or gain energy when it is present in the same shell.

6) When an electron gains energy, it gets excited to higher energy levels and when it de-excites, it loses energy in the form of electromagnetic radiations and comes to lower energy values.

Example - 22

What is meaning of "Quantisation of angular momentum"?

Sol. According to Bohr's Model, angular momentum of an electron, moving in an orbit is a fixed multiple of $\frac{h}{2\pi}$.

$$mvr = \frac{nh}{2\pi} \text{ where } n = 1, 2, 3, 4........$$

It means that when an electron gains or loses energy, it does so in such a way that n has a value which is a whole number. In other words, electron does not gain or lose energy in a continuous manner but in jumps (or bursts). This led to the concept of **quantisation of energy** which means that radiant energy is emitted or absorbed in bursts (or jumps) rather than as continuous flow.

Example - 23

What transition of Li^{+2} spectrum will have the same wavelength as that of second line of Balmer series of He^+ spectrum.

Sol. Using Rydberg's formula :

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)Z^2$$

$$\text{For } Li^{+2} : \frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)(3)^2 \qquad ....(1)$$

$$\text{For } He^{+} : \frac{1}{\lambda} = R_H\left(\frac{1}{2^2} - \frac{1}{4^2}\right)(2)^2 \qquad ....(2)$$

Compairing (1) and (2)

$$\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)(3)^2 = \left(\frac{1}{2^2} - \frac{1}{4^2}\right)(2)^2$$

$$\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = \left(\frac{1}{2^2} - \frac{1}{4^2}\right)\left(\frac{2}{3}\right)^2$$

$$\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = \left(\frac{1}{9} - \frac{1}{36}\right)$$

$$\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) = \left(\frac{1}{3^2} - \frac{1}{6^2}\right)$$

Hence, $n_1 = 3$ and $n_2 = 6$

Example - 24

Calculate the energy required for the process $He^+ \rightarrow He^{+2} + e^-$. the ionisation energy for the hydrogen atom in the ground state is 2.18×10^{-18} J/atom^{-1}.

Sol. For ionisation, $n_2 = \infty$

$$\text{Ionisation Energy, I.E.} = R_H.Z^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

$$= R_H.Z^2\left(\frac{1}{n_1^2} - \frac{1}{\infty^2}\right) = \frac{R_H.Z^2}{n_1^2}$$

$$\text{I.E. for H in ground state} = \frac{R_H.(1)^2}{1^2} = R_H$$

$$= 2.18 \times 10^{-18} \text{ J atom}^{-1}$$

$$\text{I.E. for } He^+ = \frac{R_H.2^2}{1^2}$$

$$= 2.18 \times 10^{-18}$$

$$= \boxed{8.72 \times 10^{-18} \text{ J}}$$

Example - 25

What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition,n=4 to n=2 of He^+ spectrum?

Sol. For He^+ ion,

$$\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

Now, $n_1 = 1, n_2 = 4$ and $Z = 2$

$$\frac{1}{\lambda} = R(2)^2\left[\frac{1}{2^2} - \frac{1}{4^2}\right] = \frac{3}{4}R \qquad ......(i)$$

For H atom $\frac{1}{\lambda} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$ .... (ii)

Equating equations (i) and (ii) (λ is the same)

$$\frac{1}{n_1^2} - \frac{1}{n_2^2} = \frac{3}{4}$$

Now, if $n_1 = 1$ and $n_2 = 2$

Therefore, the transition n = 2 to n = 1 in H atom will have the same wavelength as the transition from n = 4 to n = 2 in He^+.

Example - 26

The wavelength of the first line in the Balmer series is 6561Å. Calculate the wavelength of second line and limiting line in Balmer series.

Sol. According to Rhdberg equation,

$$\frac{1}{\lambda} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

For first line in Balmer series, $n_1 = 2, n_2 = 3$

$$\therefore \frac{1}{6561} = R\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = R\left(\frac{5}{36}\right) \quad \text{........ (i)}$$

For second line in Balmer series, $n_1 = 2,\ n_2 = 4$

$$\therefore \frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{4^2}\right) = R\left(\frac{3}{16}\right) \quad \text{....... (ii)}$$

Dividing eq. (i) by (ii),

$$\frac{\lambda}{6561} = \frac{5}{36} \times \frac{16}{3}$$

$$\therefore = \frac{6561 \times 5 \times 16}{36 \times 3} = \mathbf{4860\ Å}$$

For limiting line in Balmer series, $n_1 = 2$ & $n_2 = \infty$

$$\therefore \frac{1}{\lambda'} = R \times \left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) = \frac{R}{4} \quad \text{(iii)}$$

Dividing Eq. (i) by (iii)

$$\frac{\lambda'}{6561} = \frac{4 \times 5}{36}$$

$\boldsymbol{\lambda' = 3645}$ **Å**

Example - 27

Electromagnetic radiation of wavelength 242nm is just sufficient to ionize the sodium atom. Calculate the ionisation energy of sodium in kJ/mol.

Sol. $E = N\,h\nu = N\,h\frac{c}{\lambda}$

$$= \frac{(6.02 \times 10^{23}\,\text{mol}^{-1}) \times (6.626 \times 10^{-34}\,\text{J s}) \times (3 \times 10^8\,\text{m s}^{-1})}{242 \times 10^{-9}\ \text{m}}$$

$= 4.945 \times 10^5$ J mol^{-1} = **494.5 kJ mol^{-1}.**

Example - 28

The atomic spectrum of Li^{+2} arises due to transition of an electron from n_2 to n_1 level if $n_1+n_2=4$ and $n_2-n_1=2$. Calculate the wavelength (in nm)of transition.

Sol. Solving,

$n_1 + n_2 = 4$

$n_2 - n_1 = 2$

$n_2 = 3$ and $n_1 = 1$

Using Rydberg's formula :

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)^{Z^2}$$

$$\frac{1}{\lambda} = R_H\left(\frac{1}{1^2} - \frac{1}{3^2}\right)^{(3)^2}$$

$$\frac{1}{\lambda} = 8R_H$$

$$\lambda = \frac{1}{8R_H} = \frac{1}{8 \times 1.067 \times 10^7}$$

$$\lambda = 1.17 \times 10^{-8}\,\text{m}$$

$$\Rightarrow \quad \lambda = 11.7\,\text{nm.}$$

Example - 29

What is the wavelength of light emitted when electron in H-atom undergoes transition from energy level with n = 4 to an energy level n = 2?

Sol. $\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \times 1^2\left(\frac{1}{2^2} = \frac{1}{4^2}\right)$$

$$= 1.097 \times 10^7 \times \left(\frac{1}{4} - \frac{1}{16}\right)$$

$$= 1.097 \times 10^7 \times \frac{3}{16}$$

$$\lambda = \frac{16}{1.097 \times 10^7 \times 3}$$

$$= 4.8617 \times 10^{-17}$$

$$= 4861.7 \times 10^{-10}\,m$$

$$= \boxed{4861.7\ \text{Å}}$$

Example - 30

For Hydrogen atom calculate the energy required to remove the electron completely from n=2 orbit. What is the longest wavelength of light in cm that can be used to cause this transition.

Sol. $\Delta E = 2.18 \times 10^{-18} Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) J/atom$

$n_1 = 2 \qquad n_2 = \infty$

$$\Delta E = 2.18 \times 10^{-18} (1)^2 \left[\frac{1}{2^2} - \frac{1}{\infty}\right]$$

$$= 5.45 \times 10^{-19}\,J$$

$$\Delta E = \frac{hc}{\lambda}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{5.45 \times 10^{-19}}$$

$$= 3.647 \times 10^{-7}\,m$$

$$= 3.647 \times 10^{-5}\,cm.$$

Example - 31

The energy associated with first orbit in hydrogen atom is – 2.17 × 10^{-18}J/atom. What is the energy associated with fifth orbit?

Sol. Energy associated with nth orbit in hydrogen atom is

$$E_n = -\frac{2.17 \times 10^{-18} J}{n^2}\, atom^{-1}$$

$\therefore$ Energy associated with 5th orbit,

$$E_5 = -\frac{2.17 \times 10^{-18} J}{5^2}$$

$$= -8.68 \times 10^{-20}\,J$$

Example - 32

(a) Which tranistion between Bohr orbits corresponds to third line in the Balmer series of the hydrogen spectrum.

(b) Calculate the radius of Bohr's fifth orbit for hydrogen atom.

Sol. (a) $n_1 = 2 \qquad n_2 = 5$

Transition : $5 \to 2$

(b) Radius of Bohr's nth orbit is given as :

$r_n = 0.0529\, n^2\, nm$

$\therefore$ For n = 5

$r_5 = 0.0529 \times (5)^2\, nm = 1.3225\, nm$

Example - 33

Calculate the wavenumber for the longest wavelength transition in the Balmer series of atomic hydrogen.

Sol. For Balmer series,

$$\bar{\nu} = 109{,}677\left(\frac{1}{2^2} - \frac{1}{n^2}\right) cm$$

λ will be maximum of $\bar{\nu}$ is minimum $\left(\bar{\nu} = \frac{1}{\lambda}\right)$

for n = 3. The value will be

$$\bar{\nu} = 109{,}677\ cm^{-1}\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$= 109{,}677 \times \frac{5}{36}\ cm^{-1}$$

$= 15232.9\ cm^{-1} = 1.523 \times 10^6\ m^{-1}$.

Example - 34

Which state of triply ionised Be^{+3} has same orbital radius as that of ground state of H-atom.

Sol. For hydrogen atom, $r = 0.529 \frac{(1)^2}{(1)}$ Å

$= 0.529$ Å

For Be^{+3}, $r = \frac{0.529n^2}{(4)}$

Thus, $0.529 = \frac{0.529\, n^2}{4}$

$\Rightarrow n = 2.$

Example - 35

Give the difference between Electromagnetic waves and matter waves.

Sol :

Electromagnetic Waves :

1. The electromagnetic waves are associated with electric and magnetic fields, perpendicular to each other and to the direction of propagation.
2. They do not require any medium for propagation, i.e., they can pass through vacuum.
3. They travel with the same speed as that of light.
4. They leave the source, i.e., they are emitted by the source.
5. Their wavelength is given by $\lambda = \frac{c}{\nu}$

Matter waves :

1. Matter waves are not associated with electric and magnetic fields.
2. They require medium for their propagation, i.e., they cannot pass through vaccum.
3. They travel with lower speeds. Moreover, it is not constant for all matter waves.
4. They do not leave the moving particle, i.e., they are not emitted by the particle.
5. Their wevelength is given by $\lambda = \frac{h}{mv}$

Example - 36

Give the difference between particle and a wave.

Sol.

Particle :

1. A particle occupies a well-defined position in space, i.e., a particle is localized in space, e.g., a grain of sand, cricket ball, etc.
2. When a particular space is occupied by one particle, the same space cannot be occupied simultaneously by any other particle. In other words, particles **do not interfere**.
3. When a number of particles are present in a given region of space, their total value is equal to their sum, i.e., it is neither less nor more.

Wave :

1. A wave is spread out in space, e.g., on throwing a stone in a pond of water, the waves start moving out in the form of concentric circles. Similarly, the sound of the speaker reaches everybody in the audience. Thus, a wave is delocalized in space.
2. Two or more waves can coexist in the same region of space and hence interfere.
3. When a number of waves are present in a given region of space, due to interference, **the resultant wave can be larger or smaller** than the individual waves, i.e., inter-ference may be constructive or destructive.

Example - 37

(i) State and illustrate Heisenberg's uncertainty principle

(ii) Why electron cannot exist in the nucleus?

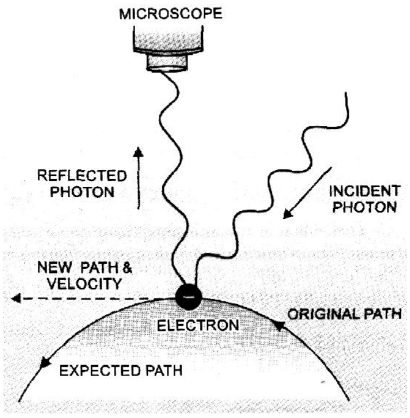
Sol. (i) It is impossible to measure simultaneously the position and momentum of a small particle with absolute accuracy. If an attempt is made to measure any of these two quantities with higher accuracy, the other becomes less accurate. The product of the uncertainty in the position (Δx) and the uncertainty in momentum (Δp) is always a constant and is equal to or greater than $h/4\pi$.

$(\Delta x).(\Delta p) \geq h/4\pi$

Or $(\Delta x).(m\Delta v) \geq h/4\pi$

Or $(\Delta x).(\Delta x) \geq h/4\pi m$

Explaination



Change of momentum and position of electron on impact with a photon

Suppose we attempt to measure both the position and momentum of an electron. To pin point the position of the electron we have to use light so that the photon of light strikes the electron and the reflected photon is seen in the microscope. As a result of the hitting, the position as well as the velocity of the electron are disturbed.

It rules out the existence of definite paths or trajectories of electrons as stated in Bohr's Model.

(ii) On the basis of Heisenberg's uncertainty principle, it can be shown why electrons cannot exist within the atomic nucleus. This is because the diameter of the atomic nucleus is of the order of 10^{-15}m. Hence, if the electron were to exist within the nucleus, the maximum uncertainty in its position would have been 10^{-15}m (i.e., $\Delta x = 10^{-15}$ m) Taking the mass of electron as 9.1×10^{-31} kg, the minimum uncertainty in velocity can be calculated by applying uncertainty principle as follows :

$$\Delta x.\,\Delta p = \frac{h}{4\pi} \quad \text{or} \quad \Delta x.\,(m \times \Delta v) = \frac{h}{4\pi}$$

or $$\Delta v = \frac{h}{4\pi \times \Delta x \times m} = \frac{6.6 \times 10^{-34}\,\text{kg m}^2\text{s}^{-1}}{4 \times 3.14 \times (10^{-15}\,\text{m}) \times (9.1 \times 10^{-31}\,\text{kg})}$$

$= 5.77 \times 10^{10}$ m s^{-1}

This value is much higher than the velocity of light (viz, 3×10^8 m s^{-1}) and hence is not possible.

Example - 38

Show that the circumference of the Bohr orbit for hydrogen atoms is an integral multiple of the de-Beoglie wavelength associated with the electron moving around the orbit.

Sol. According to Borh postulate of angular momentum,

$$mvr = n\frac{h}{2\pi} \qquad \text{or} \quad 2\,\pi\,r = n\frac{h}{mv} \quad ......\text{(i)}$$

According to de Broglie equation, $\lambda = \dfrac{h}{mv}$ .... (ii)

Substituting this value in eqn. (i), we get $2\,\pi\,r = n\lambda$

Thus, the circumference ($2\,\pi\,r$) of the Bohr orbit for hydrogen atom is an integral multiple of de Broglie wavelength.

Example - 39

Find velocity of electron for H-atom in its first Bohr orbit of radius a_0. Also, find the de-broglie wavelength.

Sol. According to Bohr's model, angular momentum is quantised.

$$mvr = \frac{nh}{2\pi}$$

$$v = \frac{nh}{2\pi\, ma_0}$$

de-Broglie wavelength,

$$\lambda = \frac{h}{mv} = \frac{h\,(2\pi\, ma_0)}{m\,(nh)}$$

$$\lambda = \frac{2\,\pi a_0}{n}$$

Example - 40

Does Bohr model satisfy Heisenberg principle ?

Sol. No, it rules out the well defined circular paths (orbits) or trajectories proposed by Bohr. Since for a subatomic particle like an electron, it is not possible to simultaneously determine the position and velocity at any moment with good degree of precision, therefore, it is not possible to talk about the trajectroy of an electron or well defined circular orbits.

Example - 41

Calculate the mass of a photon with wavelength 3.6Å

Sol. Here, $\lambda = 3.6$ Å $= 3.6 \times 10^{-10}$ m. As photon travels with the velocity of light,

$v = 3.0 \times 10^8 \text{ m s}^{-1}$

By de Broglie equation, $\lambda = \dfrac{h}{mv}$

or $m = \dfrac{h}{\lambda v}$

$$= \frac{6.626 \times 10^{-34} \text{ J s}}{(3.6 \times 10^{-10} \text{ m})\,(3.0 \times 10^8 \text{ m s}^{-1})}$$

= 6.135×10^{-29} kg.

Example - 42

The kinetic energy of an electron is 4.55×10^{-25} J. Calculate the wavelengthof the electron.

Sol. Here, we are given

kinetic energy

i.e., $\frac{1}{2}mv^2 = 4.55 \times 10^{-25}$ J

$m = 9.1 \times 10^{-31}$ kg

$h = 6.6 \times 10^{-34}$ kg m^2 s^{-1}

$$\therefore \frac{1}{2} \times (9.1 \times 10^{-31})\, v^2 = 4.55 \times 10^{-25}$$

or $v^2 = \dfrac{4.55 \times 10^{-25} \times 2}{9.1 \times 10^{-31}} = 10^6$

or $v = 10^3 \text{ m sec}^{-1}$ $\therefore$ $\lambda = \dfrac{h}{mv}$

$$= \frac{6.6 \times 10^{-34} \text{ kg m}^2\text{s}^{-1}}{(9.1 \times 10^{-31} \text{ kg}) \times 10^3 \text{ m s}^{-1}}$$

= 7.25×10^{-7} m.

Example - 43

(a) Calculate the de-Broglie wavelength of an electron moving with 1% speed of light.

(b) A molecule of O_2 and that of SO_2 travel with the same velocity. What is the ratio of their wavelengths ?

Sol. We know that $\lambda = \dfrac{h}{mv}$

$m = 9.1 \times 10^{-31}$ kg, $h = 6.63 \times 10^{-34}$ kg m^2 s^{-1}

v = 1% of speed of light $= \dfrac{1 \times 3.0 \times 10^8}{100}$ m s^{-1}

$= 3.0 \times 10^6$ m s^{-1} ($\because$ speed of light $= 3.0 \times 10^8$ m s^{-1})

$$\lambda = \frac{6.63 \times 10^{-34} \text{ kg m}^2\text{s}^{-1}}{(9.1 \times 10^{-31} \text{ kg}) \times (3.0 \times 10^6 \text{ m s}^{-1})}$$

= 2.43×10^{-10} m.

(b) $\lambda = \dfrac{h}{mv}$

$$\frac{\lambda_{O_2}}{\lambda_{SO_2}} = \frac{m_{SO_2}}{m_{O_2}} = \frac{64}{32}$$

$$\frac{\lambda_{O_2}}{\lambda_{SO_2}} = \frac{2}{1}$$

Example - 44

An electron is moving with KE of 3×10^{-25} J. Calculate its wavelength and frequency.

Sol. K.E. $= \frac{1}{2}mv^2$

$$\therefore v = \sqrt{\frac{2\text{ K.E.}}{m}} = \sqrt{\frac{2 \times 3.0 \times 10^{-25}\text{ J}}{9.1 \times 10^{-31}\text{ kg}}}$$

$= 812 \text{ m}^{\text{s}-1}$ (1J = 1 kg m^2 s^{-2})

By de Broglie equation, $\lambda = \dfrac{h}{mv}$

$$= \frac{6.626 \times 10^{-34} \text{ J s}}{(9.1 \times 10^{-31} \text{ kg})\,(812 \text{ m s}^{-1})} = 8.967 \times 10^{-7} \text{ m}$$

= 8967 Å.

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8}{8.967 \times 10^{-7}} = 3.34 \times 10^{14} \text{ Hz}$$

Example - 45

Calculate the accelerating potential that must be applied on a proton beam to give it an effective wavelength of 0.005nm.

Sol. Kinetic Energy $= \frac{1}{2}mv^2 = qV$

$$V = \frac{mV^2}{2q} \qquad .... (1)$$

According to de-Broglie, $\lambda = \frac{h}{mv}$

$$\Rightarrow v = \frac{h}{m\lambda}$$

Putting in (1)

$$V = \frac{m}{2q}\left(\frac{h}{m\lambda}\right)^2$$

$$V = \frac{h^2}{2mq\lambda^2}$$

$$V = \frac{(6.626\times10^{-34})^2}{(2\times1.67\times10^{-27})\times(1.6\times10^{-19})\,(0.005\times10^{-9})^2}$$

V = 32.8 Volts

Example - 46

What will happen to the wavelength associated with a moving particle if its velocity is doubled?

Sol. Wavelength becomes half of the original value.

(because $\lambda = \frac{h}{mv}$, i.e., $\lambda \propto \frac{1}{v}$)

Example - 47

The mass of an electron is 9.1 × 10^{-31}kg. If its kinetic energy is 10MeV, calculate its wavelength.

Sol. $1eV = 1.6 \times 10^{-19}$ J

$KE = 10 \times 10^6$ eV

$K.E. = 10 \times 10^6 \times 1.6 \times 10^{-19}]$

$K.E = 1.6 \times 10^{-12}$ J

$$K.E = \frac{1}{2}mv^2$$

$$v = \sqrt{\frac{2\,(K.E.)}{m}}$$

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2m\,(KE)}}$$

$$\lambda = \frac{6.626\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\,(1.6\times10^{-12})}}$$

$$\lambda = 3.88\times10^{-13}\,m$$

Example - 48

A golf ball has a mass of 40g and a speed of 45m/s. If the speed can be measured within accuracy of 2%. Calculate uncertainty in the position.

Sol. Uncertainty in speed = 2% of 40 m s^{-1}

i.e., $\Delta v = \frac{2}{100}\times 45 = 0.9\ m\ s^{-1}$

Applying uncertainty principle

$$\Delta x\ (m\times\Delta v) = \frac{h}{4\pi}$$

or $\Delta x = \frac{h}{4\,\pi\,m\,\Delta\,v}$

$$= \frac{6.626\times10^{-34}\,kg\ m^2\ s^{-1}}{4\times3.14\times(40\times10^{-3}\,kg)\,(0.9\ m\ s^{-1})}$$

= 1.46 × 10^{-33} m

Example - 49

Calculate the uncertainty in position of dust particle with mass equal to 1mg if the uncertainity in velocity is 5.5×10^{-20}m/s.

Sol. $m = 10^{-3}\,g = 10^{-6}$ kg

$\Delta v = 5.5 \times 10^{-20}$ m/s.

$$\Delta x \times \Delta v = \frac{h}{4\pi m}$$

$$\Delta x = \frac{h}{4\pi\, m\Delta v}$$

$$\Delta x = \frac{6.626\times10^{-34}}{4\times3.14\times10^{-6}\times5.5\times10^{-20}}$$

$\Delta x = 9.59 \times 10^{-10}$m

Example - 50

The approximate mass of an electron is 10^{-27}g. Calculate the uncertainty in its velocity if the uncertainty in its position were of the order 10^{-11}m.

Sol. $m = 10^{-27}g = 10^{-30}$ kg

$$\Delta x \,.\, \Delta v = \frac{h}{4\pi m}$$

$$\Delta v = \frac{h}{4\pi\, m\Delta x}$$

$$\Delta v = \frac{6.626\times10^{-34}}{4\times3.14\times10^{-30}\times10^{-11}}$$

$\Delta v = 5.25 \times 10^{6}$ m/s.

Example - 51

An electron has a speed of 500m/s with an uncertainty of 0.02% what is the uncertainty in locating its position.

Sol. Uncertainity in speed $= \dfrac{0.02}{100}\times 500 = 0.1\, m/s$

$$\Delta x \,.\, \Delta v = \frac{h}{4\pi m}$$

$$\Delta x = \frac{h}{4\pi\, m\Delta v} = \frac{6.626\times10^{-34}}{4\times3.14\times9.1\times10^{-31}\times0.1}$$

$\Delta x = 5.77\times10^{-4}$ m.

Example - 52

(a) What is the physical significance of ψ and ψ^2

(b) What is quantum mechanics ?

Sol. (a) In the phsyical sense, ψ gives the amplitude of the wave associated with the electron. We know that in the case of light waves, the square of the amplitude of the wave at a point is proportional to the intensity of light. Extending the same cocept to electron wave motion, the square of the wave function, ψ^2 may be taken as intensity of electron at any point. In other words, ψ^2 determines the probability of finding the moving electron in a given region i.e. it gives the probability density. Thus, ψ^2 has been called the **probability density** and ψ **the probability amplitude.** Large value of ψ^2 means a high probability of finding the electron at that place and a small value of ψ^2 means low probability. If ψ^2 is almost zero at a particular point, it means that the probability of finding the electron at that point is negligible.

(b) A branch of science that takes dual nature of matter into consideration is known as quantum mechanics.

Example - 53

Distinguish between Orbit and Orbital

Sol : Orbit

(i) An orbit is a well defined circular path around the nucleus in which the electron revolves.

(ii) An orbit represents the planar motion of an electron around the nucleus.

(iii) All the orbits are circular.

(iv) The concept of an orbit is not in accordance with the wave character of electrons (de Broglie's hypothesis and Heisenberg's uncertainity principle)

(v) The orbits do not have any directional characteristics.

(vi) The maximum number of electrons in any orbit is given by $2n^2$, where n is the number of orbit.

Orbital

(i) An orbital is the three dimensional space around the nucleus within which the probability of finding an electron is maximum.

(ii) An orbital represents a three dimensional region of space around the nucleus.

(iii) Different orbitals have different shapes. e.g. s-orbitals are spherically symmetric, p-orbitals are dumb-bell shaped and so on.

(iv) The concept of an orbital is in accordance with the wave character of electrons and uncertainty principle.

(v) All the orbitals, except s-orbitals, have directional characteristics.

(vi) The maximum number of electrons present in any orbital is two.

Example - 54

(i) What is an orbital?

(ii) Which d-orbital does not have four lobes? Draw its shape?

(iii) Compare the shapes of 1s and 2s orbital.

(iv) Why the shape of s-orbital is spherically symmetric?

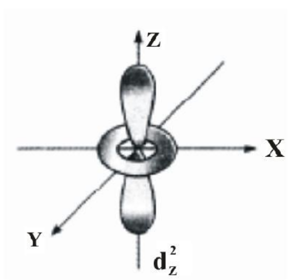
(v) Explain the shapes of s,p,d orbitals

(vi) Outline the shapes of: (a) 3s (b) $3p_z$ (c) $3d_{xz}$ (d) $3d_{x^2-y^2}$

Sol. :

(i) An orbital may be defined as a region in space around the nucleus where the probability of finding the electron is maximum.

(ii)



(iii) Similarities : (i) Both have spherical shape (ii) Both have same angular momentum as it is = $\sqrt{l(l+1)}\frac{h}{2\pi}$.

Differences :

(i) 1s has no node while 2s has one node.

(ii) Energy of 2s is greater than that of 1s.

(iii) Size of 2s is larger than that of 1s.

(iv) Shape of s-orbital is spherically symmetric because the probability of finding the electron is sam in all the directions at a particular distance from the nucleus.

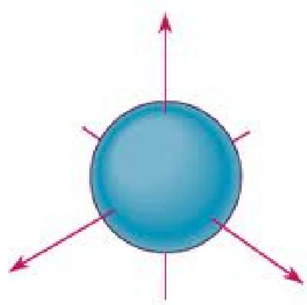
(v) (1) Shapes of s-orbitals:

(a) They are non-directional and spherically symmetric i.e. probability of finding the electron at a given distance is equal in all directions.

(b) 1s orbital and 2s orbital have same shape but size of 2s is larger.

(c) There is a spherical shell within 2s orbital where electron density is zero and is called a node.

(d) The value of azimuthal quantum number (l) is zero (l =0) and magnetic quantum number can have only one value i.e. m=0



(2) Shapes of p-orbitals:

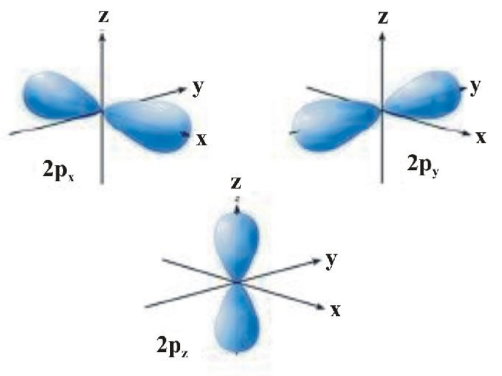
(a) It consists of two lobes present on either side of the plane that passes through the nucleus. The p-orbital is dumb-bell shaped.

(b) There are three possible orientations of electron cloud in p-orbitals. Therefore, the lobes of p-orbital may be considered to be along x,y and z axis. Hence they are designated as p_x, p_y, p_z. The three p-orbitals are oriented at right angles to one another.

(c) First main energy level(Principal quantum number=1) does not contain any p-orbital.

(d) The three p-orbitals of a particular energy level have same energy in absence of an external electric and magnetic field and are called degenerate orbitals.

(e) Like s orbitals, p-orbitals increase in size with increase in the energy of main shell of an atom. Thus, value of azimuthal quantum number is one (l =1) and magnetic quantum number has three values (m= –1, 0, +1)



(3) Shapes of d-orbitals:

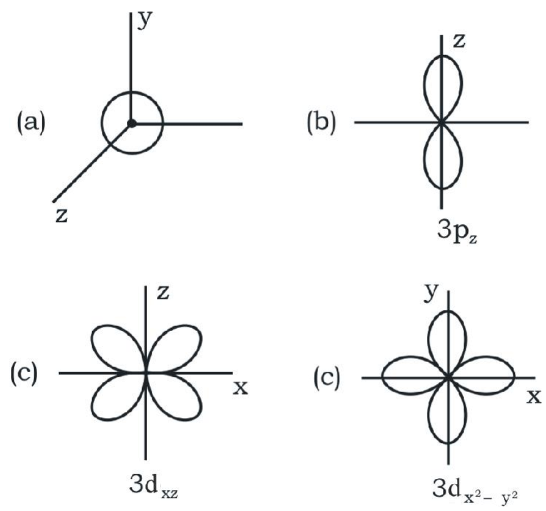
(a) They are designated as d_{xy}, d_{yz}, d_{zx} and d_{x2-y2}. They have a shape like a four leaf clover. The fifth d orbital designated as d_z^2 looks like a doughnut.

(b) All five d orbitals have same energy in the absence of magnetic field.

(c) The d orbitals have azimuthal quantum number l = 2 and magnetic quantum number values –2, –1,0,+ 1,+ 2.

(d) For principal shell number 1 and 2, there are no d orbitals.

(vi)



Example - 55

(a) What is the lowest value of n which allows a g-orbital to exist?

(b) What are degenerate orbitals ?

Sol. (a) For g-subshell, l = 4. As l = 0 to n–1, hence to have l = 4, minimum value of n = 5, i.e., 5th shell.

For l = 4, m = –4, –3, – 2, –1, 0, +1, +2, +3, + 4, i.e., 9 values which means 9 orbitals.

(b) The orbitals of same shell and sub-shell having equal energy are called degenerate orbitals.

eg. 3px, 3py, 3pz

Example - 56

(i) Designate the orbital with n = 4, l = 2 and m = 0

(ii) List the quantum numbers of electrons for 3d orbital

(iii) An atomic orbital has n=3. What are the possible values of l and m.

(iv) Which of the following orbitals are possible? 1p, 2s,2p,3f

(v) Using s,p,d notationsdescribe the following quantum numbers:

(a) n=1, l = 0 **(b)** n=3, l = 2

(c) n=3, l = 1 **(d)** n=4, l = 3

(e) n=2, l = 1

(vi) Write the values of n,l,m,s for 4p

(vii) What is the total number of orbitals in the 4f sub-shell?

(viii) What is the maximum number of electrons that can occupy the 4d sub-shell

(ix) How many electrons will be present in possible orbital having n = 3, l = 1, m = –1

(x) Calculate the number of electrons in

(a) 3pz orbital (b) 3d subshell (c) 7s subshell.

(xi) How many electrons in an atom may have the following quantum numbers

(a) n = 4, $m_s = -\frac{1}{2}$ **(b) n = 3, l = 0**

Sol. (i) $4d_z^2$

(ii) n = 3 l = 2 m = –2, –1, 0, +1, +2 $s = +\frac{1}{2}, \frac{-1}{2}$

(iii) $n = 3$ $\quad l = 0, 1, 2$

when $\quad l = 0 \quad m = 0$

$l = 1 \quad m = -1, 0, +1$

$l = 2 \quad -2, -1, 0, +1, +2$

(iv) 1p is not possible because when n = 1, l = 0 only (for p, l = 1)

2s is possible becuase when n = 2, l = 0 (for s, l = 0)

2p is possible because when n = 2, l = 0, 1 (for p, l = 1)

3f is not possible because when n = 3, l = 0, 1, 2 (for f, l = 3)

(v) (a) 1s, (b) 3d, (c) 3p, (d) 4f, (e) 2p

(vi) $n = 4\ ;\ l = 1\ ;\ m = -1, 0, +1 \quad s = +\frac{1}{2}, \frac{-1}{2}$

(vii) Number of orbitals in f-subshell = 7

(viii) Maximum number of electrons in d-subshell - 10

(ix) m = –1 represents an orbital and orbital can have maximum of two electrons.

(x) (a) An orbital can have maximum of 2 electrons

(b) d-subshell has maximum of 10 electrons

(c) s-subshell has maximum of two electrons

(xi) **(a)** Total electrons in n = 4 are $2n^2$, i.e., $2 \times 4^2 = 32$.

Half of them, i.e., 16 electrons have $m_s = -\frac{1}{2}$.

(b) n = 3, l = 0 means 3s orbital which can have 2 electrons.

Example - 57

What is the angular momentum of an electron in (i) 2s orbital (ii) 4f orbital (iii) 2p angular momentum $= \frac{h}{2\pi}\sqrt{l\,(l+1)}$

Sol. (i) for 2s, l = 0, thus angular momentum = 0

(ii) For 4f orbital, l = 3, angular momentum $= \frac{h}{2\pi}\sqrt{12}$

(iii) For 2p orbital, l = 1, angular momentum $= \frac{\sqrt{3}h}{\pi} = \frac{\sqrt{2}h}{2\pi}$

Example - 58

(i) How many electrons will be present in all the possible orbital having (n+l)=4 (b) in sub-shell having n + l =5

(ii) How many electrons in sulphur (Z=16) can have n + l = 3

(iii) How many electrons in Cl have n + l = 3

Sol

(i) (a) Subshells with n + l = 4 are 4s, 3p

Hence, electrons present = 2 + 6 = 8

(b) Subshells with n + l = 5 are 5s, 4p, 3d. Hence, electrons present = 2 + 6 + 10 = 18.

(ii) For 1 s^2, n + l = 1 + 0 = 1

for 2 s_2, n + l = 2 + 0 = 2

For 2 p^6, n + l = 2 + 1 = 3

For 3 s^2, n + l = 3 + 0 = 3

For 3 p^4, n + l = 3 + 1 = 4

Thus, n + l = 3 for 2 p^6 and 3 s^2 electrons, i.e., for 8 electrons.

(iii) Cl $\rightarrow$ $1s^2 2s^2 2p^6 3s^2 3p^5$

For $1s^2$, n + l = 1 + 0 = 1

For $2s^2$, n + l = 2 + 0 = 2

For $2p^6$, n + l = 2 + 1 = 3

For $3s^2$, n + l = 3 + 0 = 3

For $3p^5$, n + l = 3 + 1 = 4

Thus, (n + l) = 3 for $2p^6$ & $3s^2$ electrons, i.e. for 8 electrons.

Example - 59

What are quantum number of the valence electrons in potassium atom[z=19] in ground state?

Sol. K[19] : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$

Therefore, n = 4, l = 0, m = 0, $s = +\frac{1}{2}$ or $-\frac{1}{2}$

Example - 60

What information do you get from the principal quantum number about an atom ?

Sol. (i) It gives us the average distance of the electron from the nucleus.

(ii) It determines the energy of the electron in H-atom and hydrogen like particles.

(iii) The maximum number of electrons present in any shell is given by $2n^2$ where 'n' is the number of principal shell.

Example - 61

Explain pauli exclusion principle & Why Pauli exclusion principle is called exclusion principle?

Sol. No two electrons in an atom can have the same set of four quantum numbers.

If one electron in an atom has some particular values for the four quantum numbers, then all the other electrons in that atom are excluded from having the same set of values. It is because of this reason that this principle is called exclusion principle.

Example - 62

Write short note on Hund's rule of maximum multiplicity. Why it is called multiplicity rule?

Sol. Electron pairing in p, d and f orbitals cannot occur until each orbital of a given subshell contains one electron each or is singly occupied.

This is due to the fact that electrons being identical in charge, repel each other when present in the same orbital. This repulsion can, however, be minimised if two electrons move as far apart as possible by occupying different degenerate orbitals.

Further, all the singly occupied will have parallel spins, i.e., in the same direction, viz., either clockwise or anticlockwise. This is due to the fact that two electrons with parallel spins (of course in different orbitals) will encounter less inter-electronic repulsions in space than when they have opposite spins and total spin of unpaired electrons is maximum.

Example - 63

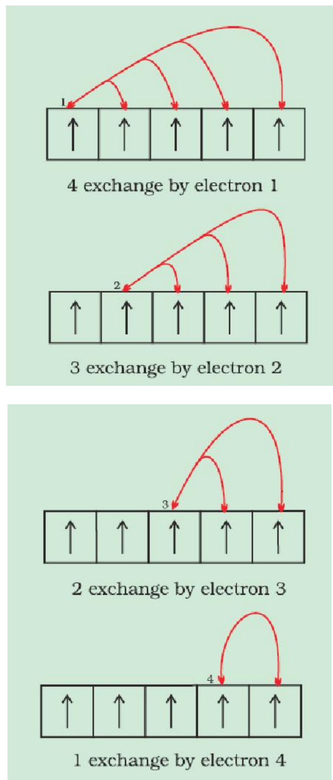
Why half filled and fully filled orbitals are stable?

Sol. The completely filled and completely half filled sub-shells are stable due to the following reasons:

1. Symmetrical distribution of electrons: It is well known that symmetry leads to stability. The completely filled or half filled subshells have symmetrical distribution of electrons in them and are therefore more stable. Electrons in the same subshell (here *3d*) have equal energy but different spatial distribution. Consequently, their shielding of one another is relatively small and the electrons are more strongly attracted by the nucleus.

2. **Exchange Energy :** The stabilizing effect arises whenever two or more electrons with the same spin are present in the degenerate orbitals of a subshell. These electrons tend to exchange their positions and the energy released due to this exchange is called exchange energy. The number of exchanges that can take place is maximum when the subshell is either half filled or completely filled. As a result the exchange energy is maximum and so is the stability.

eg. Cr (24) : [Ar] $4s^1\ 3d^5$



Thus, total number of exchanges=10

Example - 64

Why the three electrons present in 2p subshell of nitrogen remain unpaired?

Sol. According to Hund's rule, electron pairing in p, d and f orbitals cannot occur until each orbital of a given subshell contains one electron each. so that total spin of unpaired electrons is maximum.

For the element nitrogen, which contains 7 electrons, the following configurations can be written :

1s	2s	$2p_x$	$2p_y$	$2p_z$
↑↓	↑↓	↑	↑	↑

Total spin of unpaired electrons

$$=\frac{1}{2}+\frac{1}{2}+\frac{1}{2}=1\frac{1}{2}$$

Example - 65

Write the electronic configuration of

(i) Cu (ii) Cu^+ (iii) Cu^{+2} (iv) Cr (v) Cr^{+3} (vi) Co^{+3}

(vii) O^{-2}(viii) Fe^{+3} (ix) Fe^{+2} (x) Zn^{+2} (xi) H^- (xii) Na^+

(xiii) O^{-2} (xiv) F^- (xv) Al^{+3} (xvi) Sc (xvii)Cl^-

Sol. (i) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$

(ii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$

(iii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^9$.

(iv) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$.

(v) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3$

(vi) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$

(vii) $1s^2\ 2s^2\ 2p^6$

(viii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5$

(ix) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$

(x) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$

(xi) $1s^2$

(xii) $1s^2\ 2s^2\ 2p^6$

(xiii) $1s^2\ 2s^2\ 2p^6$

(xiv) $_9F = 1s^2\ 2s^2\ 2p^5 \therefore F^- = 1s^2\ 2s^2\ 2p^6$

(xv) $1s^2\ 2s^2\ 2p^6$

(xvi) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^1$

(xvii) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

Example - 66

(a) Indicate the number of unpaired electron in

(i) P (ii) Cr(iii) Si (iv) Kr (v)Fe^{+2}

(b) Which is more paramagnetic:

Fe^{+2} or Fe^{+3}

(c) Which is more stable Fe^{+2} or Fe^{+3}

Sol.

(a) (i) $_{15}P = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p_x^1\ 3p_y^1\ 3p_z^1$.

No. of unpaired electron = 3

(ii) $_{24}Cr = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$.

No. of unpaired electrons = 6

(iii) $_{14}Si = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p_x^1\ 3p_y^1$.

No. of unpaired electrons =2

(iv) $_{36}Kr$ = Noble gas. All orbitals are filled. Unpaired electrons = 0.

(v) $_{26}Fe = 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2$. No. of unpaired electrons = 4 (in 3 d)

(b) As Fe^{3+} contains 5 unpaired electrons while Fe^{2+} contains only 4 unpaired electrons, Fe^{3+} is more paramagnetic.

(c) Fe^{+2} : [Ar] $4s^0\ 3d^6$

Fe^{+3} : [Ar] $4s^0\ 3d^5$

Since d-subshell is half filled in Fe^{+3}, hence it is more stable

Example - 67

(i) What are the atomic numbers of the elements whose outermost electrons are represented by (a)$3s^1$ (b) $2p^3$ (c) $3d^5$

(ii) Which atoms are indicated by the following configurations? (a) [He]$2s^1$ (b) [Ne]$3s^2 3p^3$ (c) [Ar]$4s^2 3d^1$

Sol.

(i) (a) Total electrons : 2 + 2 + 6 + 1 = 11

Atomic Number = 11

(b) Total electrons : 2 + 2 + 3 = 7

Atomic Number = 11

(c) There can be two cases :

For Cr : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$

Thus atomic number = 24

For Mn : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3s^5$

Thus atomic Number = 25

(ii) (a) $1s^2\ 2s^1$, Thus $_3Li$

(b) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$, Thus $_{15}P$

(c) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^1$, Thus $_{21}Sc$

Example - 68

(i) Which orbital in the following pair is lower in energy in a many electron atom : 3p and 3d

(ii) Why 4s orbital is filled before 3d?

Sol. (i) Orbital having lower value of $(n + l)$ will have lower energy.

For 3p : $n + l = 4$ $(n = 3, l = 1)$

3d : $n + l = 5$ $(n = 3, l = 2)$

Thus 3p has lower energy.

(ii) Orbitals are filled with electrons starting with the orbital of lowest energy.

Orbital having lower value of $(n + l)$ will have lower energy.

For 4s : $n + l = 4$ $(n = 4, l = 0)$

For 3d : $n + l = 5$ $(n = 3, l = 2)$

Thus 4s is filled before 3d.

Example - 69

(i) What is the total number of electrons in 2p subshell of a chlorine atom in the ground state.

(ii) Which sub-shells are occupied in the outermost principal energy level of Argon atom in the ground state.

(iii) How many electrons are there in the valence quantum level of copper (Atomic number=29)?

Sol. (i) Cl (17) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$

Total electrons in 2p = 6

(ii) Ar (18) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

p-subshell is occupied

(iii) Cu (29) : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$

Valence quantum level of Cu = 4

It has 1 electron

Example - 70

How many radial nodes & planar nodes are present in (a) 5d (b) 4p (c) 6d

Sol. **(a)** Radial nodes : $n - l - 1$

$= 5 - 2 - 1 = 2$

Planar nodes = $l = 2$

(b) Radial nodes = $n - l - 1$

$= 4 - 1 - 1 = 2$

Planar nodes = $l = 1$

(c) Radial nodes = $n - l - 1$

$= 6 - 2 - 1 = 3$

Planar nodes = $l = 2$

Example - 71

Estimate the highest velocity of the electron being ejected by light of $\nu = 2.4\times10^{15}$Hz for a metal with a work function of 10 eV.

Sol. Energy of incident photon = $h\nu$

$= 6.63 \times 10^{-34} \times 2.4 \times 10^{15}$

$= 1.63 \times 10^{-18}$ J

Threshold Energy = 10eV = 1.6×10^{-18}J

Therefore, KE = $1.63 \times10^{-18} - 1.6\times10^{-19}$ J

$= 1.47 \times 10^{-18}$J

$\frac{1}{2} m_e v^2 = 1.47 \times 10^{-18}$

$v = \sqrt{(3.23\times10^{12})}\ m/s$ = $\mathbf{1.79\times10^6}$ **m/s Ans.**

Example - 72

A H-like species is in their excited state (A) and absorbs a photon of energy 3.868 eV and get excited to a new state (B). The electron from B on returning to a lower orbit, can give a maximum of ten different emissions. Some of the radiations have energies greater than it and some equal to 3.868 eV. Exactly 2 radiations have energies less than 3.868 eV. Determine the orbit numbers of states A and B and also identify the species.

Sol. Total number of emissions from state B = 10.

Thus, $n_B (n_B - 1)/2 = 10 \Rightarrow \mathbf{n_B = 5}$ **Ans.**

Also, number of spectral lines with energies less than 3.868 eV = 2. So, there are two transitions possible between n_B to n_A excluding the direct transition from B to A. Therefore, $\mathbf{n_A = 3}$ **Ans.**

Also I.P = $13.6Z^2$ eV and from the given information 3.868 $= 13.6Z^2([1/3^2) - [1/5^2])$ Thus, $\mathbf{Z = 2}$ **Ans.**

Example - 73

Estimate the Debroglie wavelength of

(a) An electron moving with a velocity of 7.28×10^7m/s.

(b) A 100 kg motorbike moving at 6.63 m/s

Sol. **(a)** mass of electron = 9.1×10^{-31}kg and given velocity $= 7.28\times10^7$m/s

Thus, momentum of electron

$= m \times v = 9.1\times7.28\times10^{-24}$ kgm/s $= 6.625\times10^{-23}$

Therefore, wavelength = h/mv = 10^{-8}m

= 10^{-11}m Ans.

(b) Momentum = 100×6.63 kgm/s = 663 kgm/s

Therefore, wavelength = h/mv = $\mathbf{10^{-36}}$**m Ans.**

Example - 74

A electron is moving with a velocity of $10^8 \pm 10^5$ m/s. Calculate the uncertainty in its position.

Sol. $\Delta v = 2\times10^5$. Therefore $p = m\Delta v = 1.82\times10^{-25}$

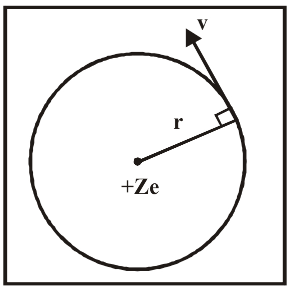
Thus, $\Delta x = h/4\pi\Delta p$ = $\mathbf{2.9 \times 10^{-10}}$ **m Ans.**

Example - 75

Determine the frequency of revolution of the electron in 2nd Bohr's orbit in hydrogen atom.

Sol. The frequency of revolution of electron is given by :

$$\text{Frequency} = \frac{1}{\text{time period}}$$



Time period

$$= \frac{\text{Total distance covered in 1 revolution}}{\text{velocity}} = \frac{2\pi r}{v}$$

Hence frequency = $\frac{v}{2\pi r}$

Calculate velocity (v_2) and radius (r_2) for electron in 2nd Bohr orbit in H-atom (Z = 1)

Z = 1 for H-atom

Using $r_n = 0.529\frac{n^2}{Z}$ Å

$$r_2 = 0.529 \times 10^{-10}\frac{(2)^2}{1}\text{m} = 1.12 \times 10^{-10}\text{m}$$

$v_n = 2.165 \times 10^6$ (1/n) m/s

$v_2 = 2.165 \times 10^6$ (1/2) = 1.09×10^6 m/s

Hence frequency $= \frac{v_2}{2\pi r_2} = \frac{1.09 \times 10^6}{2(\pi)(2.12 \times 10^{-10})}$

$v = 8.18 \times 10^{14}$ Hz Ans.

Example - 76

Determine the maximum number of lines that can be emitted when an electron in H atom in n = 6 state drops to the ground state. Also find the transitions corresponding to the lines emitted.

Sol. The maximum number of lines can be calculated by using the formula $\frac{(n_2 - n_1 + 1)(n_2 - n_1)}{2}$ where $n_2 = 6$ and $n_1 = 1$ are 15

The distinct transition corresponding to these line are

$6 \to 1$

$6 \to 2$, $2 \to 1$

$6 \to 3$, $3 \to 2$, $3 \to 1$

$6 \to 4$, $4 \to 3$, $4 \to 2$, $4 \to 1$

$6 \to 5$, $5 \to 4$, $5 \to 3$, $5 \to 2$, $5 \to 1$

Example - 77

Find the wavelength of radiation required to excite the electron in ground level of Li^{++} (Z = 3) to third energy level. Also find the ionisation energy of Li^{2+}. (R = 109, 677 cm^{-1})

Sol. Ground level : n = 1

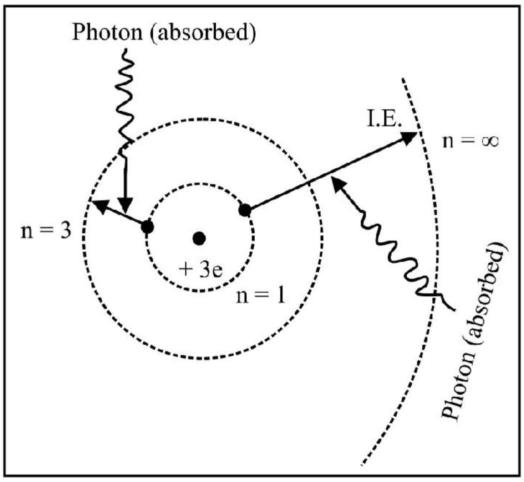
Use : $\frac{1}{\lambda} = RZ^2\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

Putting the values : $n_1 = 1$, $n_2 = 3$, Z = 3

We get : $\frac{1}{\lambda} = 1.09 \times 10^7 \times 3^2 \times \left(\frac{1}{1^2} - \frac{1}{3^2}\right)$

$\frac{1}{\lambda} = 8.776 \times 10^7\ m^{-1} \Rightarrow \lambda = \frac{1}{\bar{v}}$ **= 113.97Å Ans.**

Ionisation energy is the energy required to remove the electron from ground state to infinity i.e. corresponding transition responsible is $1 \to \infty$.



i.e. $\Delta E_{(1\to\infty)} = 13.6 \times 3^2\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right)$ eV

Ionisation energy = $\Delta E_{(1\to\infty)} = 122.4$ eV

= 1.95×10^{-17} J Ans. $[\because 1\text{eV} = 1.6 \times 10^{-19}\text{J}]$

Example - 78

In all, how many nodal planes are there in the atomic orbitals for the principal quantum number n = 3.

Sol. Shell with n = 3 has 1 's' (3s), 3 'p' (p_x, p_y, p_z) and 3 'd' (d_{xy}, d_{xz}, d_{yz}, $d_{(x^2-y^2)}$ and d_{z^2}) orbitals.

- 's' has no nodal plane.
- Each of p_x, p_y, p_z has one nodal plane, which means a total of 3 nodal planes.

  Each of d_{xy}, d_{xz}, d_{yz}, $d_{(x^2-y^2)}$ d_{z^2} has 2 nodal planes, which means a total of 10 nodal planes.

  Hence for n = 3, a total of **13 nodal planes** are there.

Example - 79

Write down the electronic configuration of following species. Also find the number of unpaired electrons in each.

(a) Fe, Fe^{2+}, Fe^{3+} (Z of Fe = 26)

(b) Br, Br^- (Z of Br = 35)

Sol. Follow the order of increasing energy (Aufbau Rule) : 1s, 2s, 2p, 3s, 3p, 4s, 3p, 4d, 5s, 4d ......

(a) Fe (Z = 26) : $\mathbf{1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6}$

Note that 3d orbital are not fully filled.

$3d^6 \equiv$ $\uparrow\downarrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$

Orbitals filled as per Hund's Rule.

Clearly the number of unpaired electrons is **4.**

- Fe^{2+} : (Z = 24)

  While Writing electronic configuration (e.c.) of cations, first write e.c. of neutral atom and then "remove desired number of electrons from outermost orbital".

  In Fe^{2+}, remove $2e^-$ from $4s^2$ since 4s orbital (through lower in energy then 3d) is the outermost. Hence e.c. of Fe^{2+} is : $\mathbf{1s^2\ 2s^2\ 2p\ 3p^6\ 3d^6\ 4s^0}$

  Note that number of unpaired electrons remains same as that in Fe, i.e. **4.**

- Fe^{3+} (Z = 23)

  Now remove $2e^-$ from $4s^2$ and $1e^-$ from $3d^6$ to get e.c. as : $\mathbf{1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^0}$

  Note that, now all 'd' orbits have an odd electron (i.e. are half filled).

  $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$

  Hence number of unpaired electrons in Fe^{3+} is **5.**

(b) Br (Z = 35)

Following Aufbau rule, e.c. is :

$\mathbf{1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^5}$

Clearly one of $4p^5$ orbitals contains unpaired electrons :

$4p^5 \equiv$ $\uparrow\downarrow$ $\uparrow\downarrow$ $\uparrow$

Orbitals filled as per Hund's Rule.

Hence Br has only **one** unpaired electron.

- Br^- (Z = 36)

  Since anion (s) is (are) formed by adding electron (s), so simply write e.c. as per total number of electrons finally.

  For Z = 36, e.c. is : $\mathbf{1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6}$

  Clearly there are **no** unpaired electrons.

Example - 80

A compound of Vanadium has a magnetic moment of 1.73 B.M. Work out the electronic configuration of vanadium in the compound.

Sol. The magnitude of magnetic moment (μ) of a compound/ species/ion is given by :

$$\mu = \sqrt{n(n+2)} \text{ B.M}$$

(n = number of unpaired electrons ; BM : unit of magnetic moment in Bohr's Magneton)

$$\Rightarrow \quad 1.73 = \sqrt{n(n+2)}$$

On solving for n, we get n = 1. This means that vanadium ion (Z = 23) in the compound has one unpaired electron.

3d : $\uparrow$ ○ ○ ○ ○

So its electronic configuration (e.c.) must be :

$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^1$

i.e vanadium exists as V^{4+} ion in the compound since the ground state e.c. of ${}_{23}V$ is :

$\mathbf{1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2}$

3d : $\uparrow$ $\uparrow$ $\uparrow$ ○ ○ 4s : $\uparrow\downarrow$

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

Dalton Theory

1. Which is not basic postulate of Dalton's atomic theory ?

(a) Atoms are neither created nor destroyed in a chemical reaction

(b) In a given compound, the relative number and kinds of atoms are constant.

(c) Atoms of all elements are alike, including their masses.

(d) Each element is composed of extremely small particles called atoms.

Fundamental Particles

2. The number of electrons in a neutral atom of an element is equal to its :

(a) atomic weight (b) atomic number

(c) equivalent weight (d) electron affinity

3. The e/m for positive rays in comparison to cathode rays is :

(a) very low (b) high

(c) same (d) none

4. Which has highest e/m ratio ?

(a) He^{2+} (b) H^+

(c) He^+ (d) H

5. Cathode rays have :

(a) mass only (b) charge only

(c) neither mass nor charge

(d) mass and charge both

6. Mass of neutron is .... times the mass of electron.

(a) 1840 (b) 1480

(c) 2000 (d) None of these

7. Positive rays or canal rays are :

(a) electromagnetic waves

(b) a steam of positively charged gaseous ions

(c) a stream of electrons

(d) neutrons

Atomic Number & Mass Number

8. The triad of nuclei that is isotonic is :

(a) $^{14}_{6}C, ^{15}_{7}N, ^{17}_{9}F$ (b) $^{12}_{6}C, ^{14}_{7}N, ^{19}_{9}F$

(c) $^{14}_{6}C, ^{14}_{7}N, ^{17}_{9}F$ (d) $^{14}_{6}C, ^{14}_{7}N, ^{19}_{9}F$

9. The ion that is isoelectronic with CO is :

(a) CN^- (b) O_2^+

(c) O_2 (d) N_2^+

10. An isotone of $_{32}Ge^{76}$ is

(i) $_{32}Ge^{77}$ (ii) $_{33}As^{77}$

(iii) $_{34}Se^{77}$ (iv) $_{34}Se^{78}$

(a) Only (i) and (ii) (b) Only (ii) and (iii)

(c) Only (ii) and (iv) (d) (ii), (iii), and (iv)

11. Which of the following atoms and ions are isoelectronic i.e. have the same number of electrons with the neon atom

(a) F^- (b) Oxygen atom

(c) Mg (d) N^-

Rutherford's Model

12. When a gold sheet is bombarded by a beam of α–particles, only a few of them get deflected whereas most go straight, undeflected. This is because

(a) The force of attraction exerted on the α–particles by the oppositely charged electrons is not sufficient.

(b) A nucleus has a much smaller volume than that of an atom.

(c) The force of repulsion acting on the fast moving α–particles is very small.

(d) The neutrons in the nucleus do not have any effect on the α–particles.

13. Discovery of the nucleus of an atom was due to theexperiment carried out by

(a) Bohr (b) Mosley

(c) Rutherford (d) Thomson

14. Rutherford's scattering experiment is related to the size of the

(a) Nucleus (b) Atom

(c) Electron (d) Neutron

Maxwell's EM Wave Theory

15. The velocity of light is 3.0×10^8 ms^{-1}. Which value is closest to the wavelength in nanometres of a quantum of light with frequency of 8×10^{15} s^{-1}

(a) 3×10^7 (b) 2×10^{-25}

(c) 5×10^{-18} (d) 3.7×10^1

16. The frequency of a wave of light is 12×10^{14} s^{-1}. The wave number associated with this light is

(a) 5×10^{-7} m (b) 4×10^{-8} cm^{-1}

(c) 2×10^{-7} m^{-1} (d) 4×10^4 cm^{-1}

17. Rank the following types of radiations from the highest energy to the lowest.

ultraviolet/visible/X-ray/microwave/infrared

(a) X-ray, ultraviolet, microwave, infrared, visible

(b) ultraviolet, X-ray, visible, infrared, microwave

(c) infrared, microwave, ultraviolet, visible, X-ray

(d) X-ray, ultraviolet, visible, infrared, microwave

18. The frequency of a green light is 6×10^{14} Hz. Its wavelength is :

(a) 500 nm (b) 5 nm

(c) 50,000 nm (d) None of these

Planck's Quantum Theory

19. Which wave property is directly proportional to energy of electromagnetic radiation :

(a) velocity (b) frequency

(c) wave number (d) all of these

20. The number of photons of light of $\bar{\nu} = 2.5 \times 10^6$ m^{-1} necessary to provide 1 J of energy are

(a) 2×10^{18} (b) 2×10^{17}

(c) 2×10^{20} (d) 2×10^{19}

21. The number of photons emitted in 10 hours by a 60 W sodium lamp (λ of photon = 6000 Å)

(a) 6.50×10^{24} (b) 6.40×10^{23}

(c) 8.40×10^{23} (d) 3.40×10^{23}

22. Minimum number of photons of light of wavelength 4000Å, which provide 1J energy :

(a) 2×10^{18} (b) 2×10^9

(c) 2×10^{20} (d) 2×10^{10}

23. The energy ΔE corresponding to intense yellow line of sodium of λ 589 nm is :

(a) 2.10 eV (b) 43.37 eV

(c) 47.12 eV (d) 2.11 kcal

24. The relation between energy of a radiation and its frequency was given by :

(a) de Broglie (b) Einstein

(c) Planck (d) Bohr

25. Which is not characteristics of Planck's quantum theory of radiation ?

(a) Radiation is associated with energy.

(b) Energy is not absorbed or emitted in whole number or multiples of quantum.

(c) The magnitude of energy associated with a quantum is proportional to the frequency.

(d) Radiation energy is neither emitted nor absorbed continuously but in small packets called quanta.

26. Which of the following is not a characteristic of Planck's quantum theory of radiation ?

(a) Energy is not absorbed or emitted in whole number multiples of quantum.

(b) Radiation is associated with energy.

(c) Radiation is associated with energy emitted or absorbed continuously but in the form of small packets called quanta.

(d) The magnitude of energy associated with quantum is proportional to frequency.

Blackbody Radiation and Photoelectric Effect

27. Einstein's theory of photoelectric effect is based on :

(a) Newtons corpuscular theory of light

(b) Huygen's wave theory of light

(c) Maxwell's electromagnetic theory of light

(d) Planck's quantum theory of light

28. In photoelectric effect the number of photo-electrons emitted is proportional to :

(a) intensity of incident beam

(b) frequency of incident beam

(c) velocity of incident beam

(d) work function of photo cathode

29. Increase in the frequency of the incident radiations increase the :

(a) rate of emission of photo-electrons

(b) work function

(c) kinetic energy of photo-electrons

(d) threshold frequency

30. Threshold wavelength depends upon :

(a) frequency of incident radiation

(b) velocity of electrons

(c) work function

(d) None of the above

31. Photoelectric effect shows

(a) particle-like behaviour of light

(b) wave-like behaviour of light

(c) both wave-like and particle-like behaviour of light

(d) neither wave-like nor particle-like behaviour of light

32. Ultraviolet light of 6.2 eV falls on aluminium surface (work function = 4.2 eV). The kinetic energy (in joule) of the fastest electron emitted is approximately :

(a) 3×10^{-21} (b) 3×10^{-19}

(c) 3×10^{-17} (d) 3×10^{-15}

33. The threshold wavelength for photoelectric effect on sodium is 5000 Å. Its work function is :

(a) 4×10^{-19} J (b) 1J

(c) 2×10^{-19}J (d) 3×10^{-10} J

34. The kinetic energy of the photoelectrons does not depend upon

(a) Intensity of incident radiation

(b) Frequency of incident radiation

(c) Wavelength of incident radiation

(d) Wave number of incident radiation.

35. If the threshold frequency of a metal for photoelectric effect is ν_0, then which of the following will not happen ?

(a) If the frequency of the incident radiation is ν_0, the kinetic energy of the electrons ejected is zero.

(b) If the frequency of the incident radiation is ν, the kinetic energy of the electrons ejected will be $h\nu - h\nu_0$.

(c) If the frequency is kept same at ν but intensity is increased, the number of electrons ejected will increase.

(d) If the frequency of incident radiation is further increased, the number of photoelectrons ejected will increase.

Spectra

36. The line spectrum of two elements is not identical because

(a) They do not have same number of neutrons

(b) They have dissimilar mass number

(c) They have different energy level schemes

(d) They have different number of valence electrons

Bohr's Atomic Model

37. The energy of electron in 3rd orbit of hydrogen atom is

(a) –1311.8 kJ mol^{-1} (b) –82.0 kJ mol^{-1}

(c) –145.7 kJ mol^{-1} (d) –327.9 kJ mol^{-1}

38. The ionization energy of H atom is 13.6 eV. The ionization energy of Li^{2+} ion will be

(a) 54.4 eV (b) 40.8 eV

(c) 27.2 eV (d) 122.4 eV

39. The ratio of the difference in energy between the first and the second Bohr orbit to that between second and third Bohr orbit is

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$

(c) $\frac{27}{5}$ (d) $\frac{4}{9}$

40. Energy of electron of hydrogen atom in second Bohr orbit is

(a) -5.44×10^{-19} J (b) -5.44×10^{-19} kJ

(c) -5.44×10^{-19} cal (d) -5.44×10^{-19} eV

41. The energy of second Bohr orbit in the hydrogen atom is –3.4 eV. The energy of fourth orbit of He^+ ion would be

(a) –3.4 eV (b) –0.85 eV

(c) –13.64 eV (d) +3.4 eV

42. The energy of an electron in the first Bohr orbit of H atom is –13.6 eV. The possible energy value(s) of the excited state(s) for electrons in Bohr orbits to hydrogen is (are)

(a) –3.4 eV (b) –4.2 eV

(c) –6.8 eV (d) +6.8 eV

43. The ionization energy of hydrogen atom is 13.6 eV. The energy required to excite the electron in a hydrogen atom from the ground state to the first excited state is

(a) 1.69×10^{-18} J (b) 1.69×10^{-23} J

(c) 1.69×10^{23} J (d) 1.69×10^{25} J

44. In a Bohr's model of atom when an electron jumps from n = 1 to n = 3, how much energy will be emitted or absorbed (1erg = 10^{+} J)

(a) 2.15×10^{-11} erg (b) 0.1911×10^{-10} erg

(c) 2.389×10^{-12} erg (d) 0.239×10^{-10} erg

45. An electron in H-atom is moving with a kinetic energy of 5.45×10^{-19}J. What will be energy level for this electron ?

(a) 1 (b) 2

(c) 3 (d) None of these

46. The energy required to dislodge electron from excited isolated H-atom, IE_1 = 13.6 eV is

(a) = 13.6 eV (b) > 13.6 eV

(c) < 13.6 and > 3.4 eV (d) $\leq$ 3.4 eV

47. The radius of first Bohr's orbit for hydrogen is 0.53 Å. The radius of third Bohr's orbit would be

(a) 0.79 Å (b) 1.59 Å

(c) 3.18 Å (d) 4.77 Å

48. The Bohr orbit radius for the hydrogen atom (n = 1) is approximately 0.530 Å. The radius for the first excited state (n = 2) orbit is

(a) 0.13 Å (b) 1.06 Å

(c) 4.77 Å (d) 2.12 Å

49. According to Bohr model, angular momentum of an electron in the 3rd orbit is :

(a) $\frac{3h}{\pi}$ (b) $\frac{1.5h}{\pi}$

(c) $\frac{3\pi}{h}$ (d) $\frac{9h}{\pi}$

50. Electronic energy is a negative energy because

(a) Electron carries negative charge.

(b) Energy is zero near the nucleus and decreases as the distance from the nucleus increases.

(c) Energy is zero at an infinite distance from the nucleus and decreases as the electron comes closer to the nucleus.

(d) There are interelectronic repulsions.

51. Ratio of frequency of revolution of electron in the second excited state of He^+ and second state of hydrogen is

(a) $\frac{32}{27}$ (b) $\frac{27}{32}$

(c) $\frac{1}{54}$ (d) $\frac{27}{2}$

Energy Levels of Hydrogen Atom

52. The line spectrum observed when electron jumps from higher level to M level is known as

(a) Balmer series (b) Lyman series

(c) Paschen series (d) Brackett series

53. How many spectral lines are produced in the spectrum of hydrogen atom from 5th energy level ?

(a) 5 (b) 10

(c) 15 (d) 4

54. What transition in He^+ ion shall have the same wave number as the first line in Balmar series of H atom ?

(a) $7 \rightarrow 5$ (b) $5 \rightarrow 3$

(c) $6 \rightarrow 4$ (d) $4 \rightarrow 2$

55. An electron jumps from 6th energy level to 3rd energy level in H-atom, how many lines belong to visible region ?
(a) 1 (b) 2
(c) 3 (d) Zero

56. The wavenumber for the shortest wavelength transition in the Balmer series of atomic hydrogen is
(a) 27420 cm^{-1} (b) 28420 cm^{-1}
(c) 29420 cm^{-1} (d) 12186 cm^{-1}

57. The difference in wavelength of second and third lines of Balmer series in the atomic spectrum is
(a) 131 Å (b) 524 Å
(c) 324 Å (d) 262 Å

58. The third line in Balmer series corresponds to an electronic transition between which Bohr's orbits in hydrogen atom
(a) $5 \rightarrow 3$ (b) $5 \rightarrow 2$
(c) $4 \rightarrow 3$ (d) $4 \rightarrow 2$

59. When electrons in N shell of excited hydrogen atom return to ground state, the number of possible lines spectrum is :
(a) 6 (b) 4
(c) 2 (d) 3

60. The wave number of the first line of Balmer series of H atom is 15200 cm^{-1}. What is the wave number of the first line of Balmer series of Li^{2+} ion ?
(a) 15200 cm^{-1} (b) 6080 cm^{-1}
(c) 76000 cm^{-1} (d) 136800 cm^{-1}

61. In hydrogen spectrum, the series of lines appearing in ultra violet region of electromagnetic spectrum are called
(a) Balmer lines (b) Lyman lines
(c) Pfund lines (d) Brackett lines

62. The wave number of the first line of Balmer series of hydrogen is 15200 cm^{-1}. The wave number of the first Balmer line of Li^{2+} ion is
(a) 15200 cm^{-1} (b) 60800 cm^{-1}
(c) 76000 cm^{-1} (d) 136800 cm^{-1}

63. A certain transition in H spectrum from an excited state to the ground state in one or more steps gives rise to a total of 10 lines. How many of these belong to the UV spectrum ?
(a) 3 (b) 4
(c) 6 (d) 5

de-Broglie Concept

64. The de–Broglie wavelength associated with a material particle is
(a) Directly proportional to its energy
(b) Directly proportional to momentum
(c) Inversely proportional to its energy
(d) Inversely proportional to momentum

65. The de Broglie wavelength of a tennis ball of mass 66 g moving with the velocity of 10 metres per second is approximately
(a) 10^{-35} metres (b) 10^{-33} metres
(c) 10^{-31} metres (d) 10^{-36} metres

66. The wavelength of a cricket ball weighing 100 g and travelling with a velocity of 50 m/s is
(a) 1.3×10^{-28} m (b) 1.3×10^{-37} m
(c) 1.3×10^{-34} m (d) 1.3×10^{-30} m

67. An electron has kinetic energy 2.8×10^{-23} J de–Broglie wavelength will be nearly ($m_e = 9.1 \times 10^{-31}$ kg)
(a) 9.24×10^{-4} m (b) 9.24×10^{-7} m
(c) 9.24×10^{-8} m (d) 9.24×10^{-10} m

68. A cricket ball of 0.5 kg is moving with a velocity of 100 ms^{-1}. The wavelength associated with its motion is
(a) 1/100 cm (b) 66×10^{-34} m
(c) 1.32×10^{-35} m (d) 6.6×10^{-28} m

69. An electron with velocity v is found to have a certain value of de Broglie wavelength. The velocity that the neutron should possess to have the same de Broglie wavelength is
(a) v (b) v/1840
(c) 1840v (d) 1840/v

Heisenberg's Principle

70. If uncertainty in the position of an electron is zero, the uncertainty in its momentum would be
(a) zero (b) $\geq \frac{h}{4\pi}$
(c) $< \frac{h}{4\pi}$ (d) infinite

71. For an electron, if the uncertainty in velocity is Δv, the uncertainty in its position (Δx) is given by :

(a) $\frac{h}{2}\pi m\Delta v$ (b) $\frac{2\pi}{hm\Delta v}$

(c) $\frac{h}{4\pi m\Delta v}$ (d) $\frac{2\pi m}{h\Delta v}$

72. A ball of mass 200g is moving with a velocity of 10m sec^{-1}. If the error in measurement of velocity is 0.1%, the uncertainty in its position is :

(a) 3.3×10^{-31} m (b) 3.3×10^{-27} m

(c) 5.3×10^{-25} m (d) 2.64×10^{-32} m

73. The Heisenbergs Uncertainty Principle states that ....

(a) no two electrons in the same atom can have the same set of four quantum numbers

(b) two atoms of the same element must have the same number of protons

(c) it is impossible to determine accurately both the position and momentum of an electron simultaneously

(d) electrons of atoms in their ground states enter energetically equivalent sets of orbitals singly before they pair up in any orbital of the set

74. If the uncertainty in the position of an electron is zero, the uncertainty in its momentum be

(a) Zero (b) $\frac{h}{2\pi}$

(c) $\frac{h}{4\pi}$ (d) Infinity

75. If uncertainty in the measurement of position and momentum of an electron are equal then uncertainly in the measurement of its velocity is approximately :

(a) 8×10^{12} m s^{-1} (b) 6×10^{12} m s^{-1}

(c) 4×10^{12} m s^{-1} (d) 2×10^{12} m s^{-1}

Quantum Mechanical Model of an Atom

76. In the Schrodinger's wave equation ψ represents

(a) Orbit (b) Wave function

(c) Wave (d) Radial probability

Quantum Numbers

77. For each value of ℓ, the number fo m_s values are

(a) 2ℓ (b) $n\ell$

(c) $2\ell + 1$ (d) $n - \ell$

78. A subshell with $n = 6$, $\ell = 2$ can accommodate a maximum of

(a) 10 electrons (b) 12 electrons

(c) 36 electrons (d) 72 electrons

79. Which of the following sets of quantum number is correct for an electron in 4f orbital ?

(a) $n = 3, \ell = 2, m = -2, s = +1/2$

(b) $n = 4, \ell = 4, m = -4, s = -1/2$

(c) $n = 4, \ell = 3, m = +1, s = +1/2$

(d) $n = 4, \ell = 3, m = +4, s = +1/2$

80. For a d-electron, the orbital angular momentum is

(a) $\sqrt{6}\,(h/2\pi)$ (b) $\sqrt{2}\,(h/2\pi)$

(c) $h/2\pi$ (d) zero

81. The correct designation of an electron with $n = 4$, $l = 3$, $m = 2$, and $s = 1/2$ is :

(a) 3d (b) 4f

(c) 5p (d) 6s

82. A 3d-electron having $s = +1/2$ can have a magnetic quantum no :

(a) +2 (b) +3

(c) –3 (d) +4

Electronic Configuration

83. The electrons identified by n and l

(i) $n = 4, l = 1$ (ii) $n = 4, l = 0$

(iii) $n = 3, l = 2$ (iv) $n = 3, l = 1$

can be placed in order of increasing energy, from lowest to highest

(a) (iv) < (ii) < (iii) < (i) (b) (ii) < (iv) < (i) < (iii)

(c) (i) < (iii) < (ii) < (iv) (d) (iii) < (i) < (iv) < (ii)

84. According to $(n + l)$ rule after completing 'np' level the electron enters to :

(a) $(n - 1)$ d (b) $(n + 1)$ s

(c) nd (d) $(n + 1)$ p

85. The correct ground state electronic configuration of chromium atom (Z = 24) is

(a) $[Ar]\ 3d^5\ 4s^1$ (b) $[Ar]\ 3d^4\ 4s^2$

(c) $[Ar]\ 3d^6\ 4s^0$ (d) $[Ar]\ 4s^1\ 4p^5$

86. In manganese atom, Mn (Z = 25), the total number of orbitals populated by one or more electrons (in ground state) is

(a) 15 (b) 14

(c) 12 (d) 10

87. The correct set of quantum numbers for the unpaired electron of chlorine atom is

	n	ℓ	m
(a)	2	1	0
(b)	2	1	1
(c)	3	1	1
(d)	3	0	0

88. The maximum number of 4d-electrons having spin quantum number $s = +\frac{1}{2}$ are

(a) 10 (b) 7

(c) 1 (d) 5

89. Which of the following has maximum number of unpaired electrons ?

(a) Mg^{2+} (b) Ti^{3+}

(c) V^{3+} (d) Fe^{3+}

90. Azimuthal quantum number for the last electron in Na atom is

(a) 1 (b) 0

(c) 2 (d) 3

91. Presence of three unpaired electrons in phosphorus atom can be explained by

(a) Pauli's rule

(b) Uncertainty principle

(c) Aufbau's rule

(d) Hund's rule

92. Consider the following ions

1. Ni^{2+} 2. Co^{2+}

3. Cr^{2+} 4. Fe^{3+}

(Atomic number : Cr = 24, Fe = 26, Co = 27 and Ni = 28)

The correct sequence of increasing number of unpaired electrons in these ions is

(a) 1, 2, 3, 4 (b) 4, 2, 3, 1

(c) 1, 3, 2, 4 (d) 3, 4, 2, 1

93. The quantum numbers for the outermost electron of an element are given below

$$n = 2, \ell = 0, m = 0, m_s = +\frac{1}{2}$$

The atom is

(a) hydrogen (b) lithium

(c) beryllium (d) boron

94. For which one of the following sets of four quantum numbers an electron will have the highest energy

	n	ℓ	m	s
(a)	3	2	1	1/2
(b)	4	1	0	–1/2
(c)	4	2	–1	1/2
(d)	5	0	0	–1/2

95. Which electronic configuration does not follow the Pauli's exclusion principle ?

(a) $1s^2, 2s^2 2p^4$ (b) $1s^2, 2s^2 2p^4, 3s^2$

(c) $1s^2, 2p^4$ (d) $1s^2, 2s^2 2p^6, 3s^3$

Magnetic Moment

96. Magnetic quantum number for the last electron in sodium is :

(a) 3 (b) 1

(c) 2 (d) zero

97. Nitrogen has the electronic configuration $1s^2, 2s^2 2p_x^1 2p_y^1 2p_z^1$ and not $1s^2, 2s^2 2p_x^2 2p_y^1 2p_z^0$. It was proposed by :

(a) Aufbau principle

(b) Pauli's exclusion principle

(c) Hund's rule

(d) Uncertainty principle

98. Which of the following has maximum number of unpaired electron (atomic number of Fe 26)

(a) Fe (b) Fe (II)

(c) Fe (III) (d) Fe (IV)

99. A compound of vanadium has magnetic moment of 1.73 B.M. The electronic configuration of the vanadium ion in the compound is

(a) $[Ar]\,4s^0 3d^1$ (b) $[Ar]\,4s^1 3d^0$

(c) $[Ar]\,4s^2 3d^0$ (d) $[Ar]\,4s^0 3d^3$

100. The total spin and magnetic moment for the atom with atomic number 7 are

(a) $\pm 3, \sqrt{3}$ BM (b) $\pm 1, \sqrt{8}$ BM

(c) $\pm \frac{3}{2}, \sqrt{15}$ BM (d) $0, \sqrt{8}$ BM

Nodes

101. How many spherical nodes are present in 4s orbital in a hydrogen atom ?

(a) 0 (b) 2

(c) 3 (d) 4

102. The number of nodes possible in radial wave function of 3d orbital is

(a) 1 (b) 2

(c) 0 (d) 3

103. The d-orbital with the orientation along X and Y axes is called :

(a) d_{z^2} (b) d_{zy}

(c) d_{yz} (d) $d_{x^2-y^2}$

104. Which d-orbital does not have four lobes ?

(a) $d_{x^2-y^2}$ (b) d_{xy}

(c) d_{z^2} (d) d_{xz}

105. $3p_y$ orbital has .... nodal plane

(a) XY (b) YZ

(c) ZX (d) Any

106. The number of angular nodes in a 3s atomic orbital is

(a) 0 (b) 1

(c) 2 (d) 3

107. The number of radial nodes in a 3s atomic orbital is

(a) 0 (b) 1

(c) 2 (d) 3

Schrodinger Equation

108. The quantum number not obtained from Schrodinger equation is

(a) n (b) *l*

(c) m (d) s

EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTION

1. Energy of H-atom in the ground state is –13.6 eV, hence energy in the second excited state is **(2002)**
(a) –6.8 eV (b) –3.4 eV
(c) –1.51 eV (d) –4.53 eV

2. Which of the following ions has the maximum magnetic moment? **(2002)**
(a) Mn^{2+} (b) Fe^{2+}
(c) Ti^{2+} (d) Cr^{2+}

3. In a hydrogen atom, if the energy of an electron in the ground state is – 13.6 eV, then that in the 2nd excited state is **(2002)**
(a) – 1.51 e V (b) – 3.4 eV
(c) – 6.04 e V (d) – 13.6 eV

4. Uncertainty in position of a particle of 25 g in space is 10^{-5} m. Hence, uncertainty in velocity ($m\,s^{-1}$) is (Planck's constant, $h = 6.6 \times 10^{-34}$ Js) **(2002)**
(a) 2.1×10^{-28} (b) 2.1×10^{-34}
(c) 0.5×10^{-34} (d) 5.0×10^{-24}

5. The de-Broglie wavelength of a tennis ball of mass 60 g moving with a velocity of 10 m/s is approximately (Planck's constant, $h = 6.63 \times 10^{-34}$ Js) **(2003)**
(a) 10^{-33} m (b) 10^{-31} m
(c) 10^{-16} m (d) 10^{-25} m

6. The number of unpaired d-electrons retained in Fe^{2+} (At. no. of Fe = 26) ion is **(2003)**
(a) 6 (b) 3
(c) 4 (d) 5

7. The orbital angular momentum for an electron revolving in an orbit is given by $\sqrt{\ell(\ell+1)}\,h/2\pi$. This momentum for an s-electron will be given by **(2003)**
(a) $+\frac{1}{2}\cdot\frac{h}{2\pi}$ (b) zero
(c) $\frac{h}{2\pi}$ (d) $\sqrt{2}.\frac{h}{2\pi}$

8. In Bohr series of lines of hydrogen spectrum, the third line from the red end corresponds to which one of the following inner-orbit jumps of the electron for Bohr orbits in an atom of hydrogen? **(2003)**
(a) $3 \to 2$ (b) $5 \to 2$
(c) $4 \to 1$ (d) $2 \to 5$

9. The wavelength of the radiation emitted, when in a hydrogen atom electron falls from infinity to stationary state 1, would be (Rydberg constant = $1.097 \times 10^{7}\ m^{-1}$) **(2004)**
(a) 91 nm (b) 192 nm
(c) 406 nm (d) 9.1×10^{-8} nm

10. Which of the following sets of quantum numbers is correct for an electron in 4f orbital? **(2004)**
(a) n = 4, l = 3, m = + 4, s = + 1/2
(b) n = 4, l = 4, m = – 4, s = – 1/2
(c) n = 4, l = 3, m = + 1, s = + 1/2
(d) n = 3, l = 2, m = – 2, s = + 1/2

11. Consider the ground state of Cr atom (Z = 24). The numbers of electrons with the azimuthal quantum numbers, l = 1 and 2 are, respectively **(2004)**
(a) 12 and 4 (b) 12 and 5
(c) 16 and 4 (d) 16 and 5

12. Which of the following statements in relation to the hydrogen atom is correct? **(2005)**
(a) 3s, 3p and 3d orbitals all have the same energy
(b) 3s and 3p orbitals are of lower energy than 3d orbital
(c) 3p orbital is lower in energy than 3d orbital
(d) 3s orbital is lower in energy than 3p orbital

13. In a multielectron atom, which of the following orbitals described by the three quantum numbers will have the same energy in the absence of magnetic and electric fields?
(A) n = 1, l = 0 m = 0 (B) n = 2, l = 0, m = 0
(C) n = 2, l = 1, m = 1 (D) n = 3, l = 2, m = 1
(E) n = 3, l = 2, m = 0 **(2005)**
(a) (D) and (E) (b) (C) and (D)
(c) (B) and (C) (d) (A) and (B)

14. Of the following sets which one does not contain isoelectronic species ? **(2005)**

(a) $BO_3^{3-}, CO_3^{2-}, NO_3^-$ (b) $SO_3^{2-}, CO_3^{2-}, NO_3^-$

(c) CN^-, N_2, C_2^{2-} (d) $PO_4^{3-}, SO_4^{2-}, ClO_4^-$

15. Uncertainty in the position of an electron (mass = 9.1×10^{-31} kg) moving with a velocity 300 ms^{-1}, Accurate upto 0.001%, will be ($h = 6.63 \times 10^{-34}$ Js) **(2006)**

(a) 19.2×10^{-2} m (b) 5.76×10^{-2} m

(c) 1.92×10^{-2} m (d) 3.84×10^{-2} m

16. According to Bohr's theory, the angular momentum of an electron in 5th orbit is **(2006)**

(a) $25\frac{h}{\pi}$ (b) $1.0\frac{h}{\pi}$

(c) $10\frac{h}{\pi}$ (d) $2.5\frac{h}{\pi}$

17. The 'spin-only' magnetic moment [in units of Bohr magneton (μ_B)] of Ni^{2+} in aqueous solution would be (Atomic number : Ni = 28) **(2006)**

(a) 2.84 (b) 4.90

(c) 0 (d) 1.73

18. Which of the following sets of quantum numbers represents the highest energy of an atom ? **(2007)**

(a) n = 3, l = 1, m = 1, s = + 1/2

(b) n = 3, l = 2, m = 1, s = + 1/2

(c) n = 4, l = 0, m = 0, s = + 1/2

(d) n = 3, l = 0, m = 0, s = + 1/2

19. Which of the following nuclear reactions will generate an isotope ? **(2007)**

(a) Neutron particle emission

(b) Positron emission

(c) α-particle emission (d) β-particle emission

20. The ionization enthalpy of hydrogen atom is 1.312×10^6 J mol^{-1}. The energy required to excite the electron in the atom from n = 1 to n = 2 is **(2008)**

(a) 8.51×10^5 J mol^{-1} (b) 6.56×10^5 J mol^{-1}

(c) 7.56×10^5 J mol^{-1} (d) 9.84×10^5 J mol^{-1}

21. In an atom, an electron is moving with a speed of 600 m/s with an uncertainity of 0.005%, the position of the electron can be ($h = 6.6 \times 10^{-34}$ kg m^2s^{-1}, mass of electron $e_m = 9.1 \times 10^{-31}$ kg) **(2009)**

(a) 1.52×10^{-4} m (b) 5.01×10^{-3} m

(c) 1.92×10^{-3} m (d) 3.84×10^{-3} m

22. Calculate the wavelength (in nanometer) associated with a proton moving at 1.0×10^3 ms^{-1} (Mass of proton = 1.67×10^{-27} kg and $h = 6.63 \times 10^{-34}$ Js) **(2009)**

(a) 0.032 nm (b) 0.40 nm

(c) 2.5 nm (d) 14.0 nm

23. The energy required to break one mole of Cl–Cl bonds in Cl_2 is 242 kJ mol^{-1}. The longest wavelength of light capable of breaking a single Cl–Cl bond is **(2010)**

(a) 594 nm (b) 640 nm

(c) 700 nm (d) 494 nm

24. Ionisation energy of He^+ is 19.6×10^{-18} J $atom^{-1}$. The energy of the first stationary state (n = 1) of Li^{2+} is **(2010)**

(a) 4.41×10^{-16} J $atom^{-1}$ (b) -4.41×10^{-17} J $atom^{-1}$

(c) -2.2×10^{-15} J $atom^{-1}$ (d) 8.82×10^{-17} J $atom^{-1}$

25. A gas absorbs photon of 355 nm and emits at two wavelengths. If one of the emission is at 680 nm, the other is at **(2011)**

(a) 1035 nm (b) 325 nm

(c) 743 nm (d) 518 nm

26. The frequency of light emitted for the transition n = 4 to n = 2 of He^+ is equal to the transition in H atom corresponding to which of the following ? **(2011)**

(a) n = 3 to n = 1 (b) n = 2 to n = 1

(c) n = 3 to n = 2 (d) n = 4 to n = 3

27. The electrons identified by quantum numbers n and l **(2012)**

(1) n = 4, l = 1 (2) n = 4, l = 0

(3) n = 3, l = 2 (4) n = 3, l = 1

can be placed in order of increasing energy as

(a) (3) < (4) < (2) < (1) (b) (4) < (2) < (3) < (1)

(c) (2) < (4) < (1) < (3) (d) (1) < (3) < (2) < (4)

28. Energy of an electron is given by

$$E = -2.178 \times 10^{-18}\ J\left(\frac{Z^2}{n^2}\right)$$

Wavelength of light required to excite an electron in an hydrogen atom from level n = 1 to n = 2 will be **(2013)**

($h = 6.62 \times 10^{-34}$ Js and $c = 3.0 \times 10^{8}\ ms^{-1}$)

(a) 1.214×10^{-7} m (b) 2.816×10^{-7} m

(c) 6.500×10^{-7} m (d) 8.500×10^{-7} m

29. The correct set of four quantum numbers for the valence electrons of rubidium atom (Z=37) is : **(2014)**

(a) $5, 1, 0, +\frac{1}{2}$ (b) $5, 1, 1, +\frac{1}{2}$

(c) $5, 0, 1, +\frac{1}{2}$ (d) $5, 0, 0, +\frac{1}{2}$

30. Which of the following is the energy of a possible excited state of hydrogen ? **(2015)**

(a) – 3.4 eV (b) + 6.8 eV

(c) + 13.6 eV (d) – 6.8 eV

31. A stream of electrons from a heated filament was passed between two charged plates kept at a potential difference V esu. If e and m are charge and mass of an electron, respectively, then the value of h/λ (where λ is wavelength associated with electron wave) is given by: **(2016)**

(a) 2meV (b) $\sqrt{meV}$

(c) $\sqrt{2meV}$ (d) meV

32. The radius of the second Bohr orbit for hydrogen atom is : **(2017)**

(Planck's Const. $h = 6.6262 \times 10^{-34}$ Js;

Mass of electron $= 9.1091 \times 10^{-31}$ kg;

Charge of electron $e = 1.60210 \times 10^{-19}$ C;

Permittivity of vacuum

$\varepsilon_0 = 8.854185 \times 10^{-12}\ kg^{-1}m^{-3}A^2$)

(a) 4.76 Å (b) 0.529 Å

(c) 2.12 Å (d) 1.65 Å

JEE MAINS ONLINE QUESTION

1. The energy of an electron in first Bohr orbit of H-atom is –13.6 eV. The energy value of electron in the excited state Li^{2+} is: **(Online 2014 SET-1)**

(a) –27.2 eV (b) 30.6 eV

(c) – 30.6 eV (d) 27.2 eV

2. Chloro compound of vanadium has only spin magnetic moment of 1.73 BM. This vanadium chloride has the formula: (at. no. of V = 23) **(Online 2014 SET-1)**

(a) VCl_2 (b) VCl_4

(c) VCl_3 (d) VCl_5

3. Based on the equation:

$\Delta E = -2.0 \times 10^{-18} J\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right)$ the wavelength of the light that must be absorbed to excite hydrogen electron from level n= 1 to level n = 2 will be ($h = 6.625 \times 10^{-34}$ Js, $C = 3 \times 10^{8}\ ms^{-1}$) **(Online 2014 SET-2)**

(a) 2.650×10^{-7} m (b) 1.325×10^{-7} m

(c) 5.300×10^{-10} m (d) 1.325×10^{-10} m

4. If λ_0 and λ be the threshold wavelength and wavelength of incident light, the velocity of photoelectron ejected from the metal surface is: **(Online 2014 SET-2)**

(a) $\sqrt{\frac{2h}{m}\left(\frac{1}{\lambda_0} - \frac{1}{\lambda}\right)}$ (b) $\sqrt{\frac{2h}{m}(\lambda_0 - \lambda)}$

(c) $\sqrt{\frac{2hc}{m}(\lambda_0 - \lambda)}$ (d) $\sqrt{\frac{2hc}{m}\left(\frac{\lambda_0 - \lambda}{\lambda\lambda_0}\right)}$

5. The de-Broglie wavelength of a particle of mass 6.63 g moving with a velocity of 100 ms^{-1} is: **(Online 2014 SET-3)**

(a) 10^{-33} m (b) 10^{-35} m

(c) 10^{-25} m (d) 10^{-31} m

6. Excited hydrogen atom emits light in the ultraviolet region at 2.47×10^{15} Hz. With this frequency, the energy of a single photon is:

($h = 6.63 \times 10^{-34}$ Js) **(Online 2014 SET-3)**

(a) 8.041×10^{-40} J (b) 6.111×10^{-17} J

(c) 2.680×10^{-19} J (d) 1.640×10^{-18} J

7. If m and e are the mass and charge of the revolving electron in the orbit of radius r for hydrogen atom, the total energy of the revolving electron will be:

(Online 2014 SET-3)

(a) $\frac{me^2}{r}$ (b) $-\frac{e^2}{r}$

(c) $\frac{1}{2}\frac{e^2}{r}$ (d) $-\frac{1}{2}\frac{e^2}{r}$

8. Ionization energy of gaseous Na atoms is 495.5 kJ mol^{-1}. The lowest possible frequency of light that ionizes a sodium atom is ($h = 6.626 \times 10^{-34}$ Js, NA $= 6.022 \times 10^{23}$ mol^{-1}) **(Online 2014 SET-4)**

(a) 7.50×10^{4} s^{-1} (b) 4.76×10^{14} s^{-1}

(c) 1.24×10^{15} s^{-1} (d) 3.15×10^{15} s^{-1}

9. At temperature T, the average kinetic energy of any particle $\frac{3}{2}$kT. The de Broglie wavelength follows the order : **(Online 2015 SET-1)**

(a) Visible photon > Thermal neutron > Thermal electron

(b) Thermal proton > Thermal electron > Visible photon

(c) Thermal proton > Visible photon > Thermal electron

(d) Visible photon > Thermal electron > Thermal neutron

10. The total number of orbitals associated with the principal quantum number 5 is **(Online 2016 SET-1)**

(a) 5 (b) 10

(c) 20 (d) 25

11. Aqueous solution of which salt will not contain ions with the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6$?

(Online 2016 SET-2)

(a) NaF (b) NaCl

(c) KBr (d) CaI_2

12. If the shortest wavelength in Lyman series of hydrogen atom is A, then the longest wavelength in Paschen series of He^+ is

(Online 2017 SET-1)

(a) $\frac{5A}{9}$ (b) $\frac{9A}{5}$

(c) $\frac{36A}{5}$ (d) $\frac{36A}{7}$

13. The electron in the hydrogen atom undergoes transition from higher orbitals to orbital of radius 211.6 pm. This transition is associated with : **(Online 2017 SET-2)**

(a) Lyman series (b) Balmer series

(c) Paschen series (d) Brackett series

14. Ejection of the photoelectron from metal in the photoelectric effect experiment can be stopped by applying 0.5 V when the radiation of 250 nm is used. The work function of the metal is :

(Online 2018 SET-1)

(a) 4 eV (b) 4.5 eV

(c) 5 eV (d) 5.5 eV

15. The de-Broglie's wavelength of electron present in first Bohr orbit of 'H' atom is : **(Online 2018 SET-2)**

(a) 0.529 Å (b) $2\pi \times 0.529$ Å

(c) $\frac{0.529}{2\pi}$ Å (d) 4×0.529 Å

16. Which of the following statements is false ?

(Online 2018 SET-3)

(a) Photon has momentum as well as wavelength.

(b) Splitting of spectral lines in electrical field is called Stark effect.

(c) Rydberg constant has unit of energy.

(d) Frequency of emitted radiation from a black body goes from a lower wavelength to higher wavelength as the temperature increases.

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

1. All questions marked **"S"** are single choice questions
2. All questions marked **"M"** are multiple choice questions
3. All questions marked **"C"** are comprehension based questions
4. All questions marked **"A"** are assertion–reason type questions

   (A) If both assertion and reason are correct and reason is the correct explanation of assertion.

   (B) If both assertion and reason are true but reason is not the correct explanation of assertion.

   (C) If assertion is true but reason is false.

   (D) If reason is true but assertion is false.
5. All questions marked **"X"** are matrix–match type questions
6. All questions marked **"I"** are integer type questions

Fundamental Particles

1. (A) **Assertion (A) :** $\frac{\text{Charge}}{\text{Mass}}$ ratio of anode rays is found different for different gases.

Reason : Proton is the fundamental particle present in the gases.

(a) A (b) B
(c) C (d) D

2. (S) The minimum real charge on of any particle, which can exist is :

(a) 1.6×10^{-19} coulomb (b) 1.6×10^{-10} coulomb
(c) 4.8×10^{-10} coulomb (c) zero

3. (S) The ratio of specific charge (e/m) of an electron to that of a hydrogen ion is :

(a) 1 : 1 (b) 1840 : 1
(c) 1 : 1840 (d) 2 : 1

4. (S) The ratio of e/m, i.e., specific charge for a cathode ray :

(a) has the smallest value when the discharge tube is filled with H_2

(b) is constant

(c) varies with the atomic number of gas in the discharge tube

(d) varies with the atomic number of an element forming the cathode

5. (A) **Assertion :** Cathode rays are produced only when the pressure of the gas inside the discharge tube is very low.

Reason : At high pressure, no electric current flows through the tube as gases are poor conductor of electricity.

(a) A (b) B
(c) C (d) D

6. (A) **Assertion :** α – Particles are helium nuclei.

Reason : They are deflected slightly towards the negative plate and hence carry positive change.

(a) A (b) B
(c) C (d) D

7. (X)

Column I	**Column II**
(A) Plum-Pudding model	(P) 1.6022×10^{-19} C
(B) Planetory model of atom	(Q) Thomson's model
(C) Atoms are indivisible	(R) Rutherford's model
(D) Charge on electron	(S) $\pm 1/2$
(E) The spin of electron is	(T) Dalton theory

Rutherford's Model

8. (A) **Assertion :** When α – rays hit a thin foil of gold, only a few α – particles are deflected back.

Reason : Within an atom, there is a very small positively charged heavy body is present.

(a) A (b) B
(c) C (d) D

9. (I) With what velocity should an alpha (α) particle travel towards the nucleus of a copper atom so as to arrive at a distance 10^{-13} m from the nucleus of the copper atom ? (in 10^6 m/s)

10. (S) The nucleus of an atom can be assumed to be spherical. The radius of the nucleus of mass number A is given by $1.25 \times 10^{-13} \times A^{1/3}$ cm Radius of atom is one Å. If the mass number is 64, then the fraction of the atomic volume that is occupied by the nucleus is

(a) 1.0×10^{-3} (b) 5.0×10^{-5}

(c) 2.5×10^{-2} (d) 1.25×10^{-13}

Atomic Number & Mass Number

11. (S) Which are isoelectronic with each other ?

(a) Na^+ and Ne (b) K^+ and O

(c) Ne and O (d) Na^+ and K^+

12. (S) Naturally occurring elements are mixtures of :

(a) isotone (b) isobars

(c) isotopes (d) isomers

13. (S) Li^{2+} and Be^{3+} are :

(a) isotopes (b) isomers

(c) isobars (d) isoelectronic

14. (I) How many of the following atoms/ions are isoelectronic with Ca^{2+} :

Ar, Na^+, Mg^{2+}, K^+, Cl^-, F^-, S^{2-}, N^{3-}

15. (S) The mass number of three isotopes of an element are 11, 12, and 13 units. Their percentage abundances 80, 15, and 5, respectively. What is the atomic weight of the element ?

(a) 11.25 (b) 20

(c) 16 (d) 10

Maxwell's EM Wave Theory

16. (A) **Assertion :** Wave number of visible light with wave-length of 5000 Å is 2×10^6 m^{-1}.

Reason : Wave number is defined as the number of waves present in 1 unit length.

(a) A (b) B
(c) C (d) D

17. (A) **Assertion :** According to the wave theory, the radiation emitted by the body being heated should have the same colour but its intensity varies as the heating is continued.

Reason : Energy of any electromagnetic radiation depends upon its frequency.

(a) A (b) B
(c) C (d) D

Comprehension

Electromagnetic wave theory was proposed by James Clark Maxwell in 1864. Acc. to this theory, the energy is emitted from any energy source contineously in the form of radiations & is called radiant energy.

Some important chanracteristics of a wave are :

1. **Wavelength (λ)** :- Distance between any two consecutive crests or troughs

2. **Frequency (ν) :-** Number of waves passing through a point in one second.

3. **Amplitude :-** Height of crest or depth of through

4. **Wave Number** $(\bar{\nu}) = \dfrac{1}{\lambda}$

Relation between velocity, wavelength & frequency

$c = \nu \times \lambda$

18. (C) How long would it take a radiowave of frequency, 6×10^3 sec^{-1} to travel from Mars to Earth, a distance of 8×10^7 km ?

(a) 266 sec (b) 246 sec

(c) 280 sec (d) None of these

19. (C) Which of the following is correct relation of wavelength among various radiations :

(a) Cosmic < X-rays < Micro-waves < γ-rays

(b) UV rays < Radio waves < visible < IR rays

(c) Cosmic < UV rays < IR rays < Radio wave

(d) γ-rays < Cosmic rays < IR rays < Micro waves

20. (C) Which of the following is incorrect w.r.t Maxwell wave theory.

(a) EM radiations travell with the speed of light.

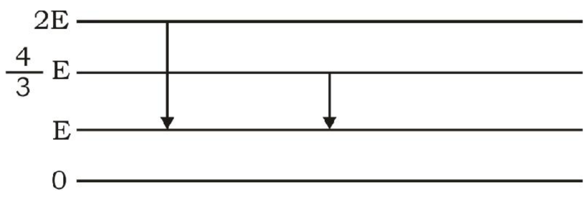
(b) The wave number of radiations having frequency of 4×10^{14} Hz is 1.33×10^4 cm^{-1}

(c) Electric field of radiations is perpendicular to magnetic field but parallel to direction of propagation.

(d) Radiant energy is a contineous form of energy.

Planck's Quantum Theory

21. (S) The given diagram indicates the energy levels of a certain atom. When the system moves from 2E level to E level, a photon of wavelength λ is emitted. The wavelength of photon produced during the transition from $\frac{4E}{3}$ to E level is :



(a) $\frac{\lambda}{3}$ (b) $\frac{3\lambda}{4}$

(c) $\frac{4\lambda}{3}$ (d) 3λ

22. (S) Suppose 10^{-17}J of light energy is needed by the interior of human eye to see an object. The photons of green light (λ = 550 nm) needed to see the object are :

(a) 27 (b) 28

(c) 29 (d) 30

23. (S) A photon of 300 nm is absorbed by a gas and then re-emits two photons. One re-emitted photon has wavelength 496 nm, the wavelength of second re-emitted photon is :

(a) 757 (b) 857

(c) 957 (d) 657

24. (S) 4000 Å photon is used to break the iodine molecule, then the % of energy converted to the K.E. of iodine atoms if bond dissociation energy of I_2 molecule is 246.5 kJ/mol

(a) 8% (b) 12%

(c) 17% (d) 25%

25. (S) Consider a 20 W light source that emits monochromatic light of wavelength 600 nm. The number of photons ejected per second in the form of Avogadro's constant N_{AV} is approximately :

(a) N_{AV} (b) $10^{-2}N_{AV}$

(c) $10^{-4} N_{AV}$ (d) $10^{-6} N_{AV}$

26. (I) Infrared lamps are used in restaurants to keep the food warm. The infrared radiation is strongly absorbed by water, raising its temperature and that of the food. If the wavelength of infrared radiation is assumed to be 1500 nm, then the number of photons per second of infrared radiation produced by an infrared lamp that consumes energy at the rate of 100 W and is 12% efficient only is $x \times 10^{19}$. Here, x is

Blackbody Radiation and Photoelectric Effect

27. (A) **Assertion (A) :** The photoelectrons produced by a monochromatic light beam incident on a metal surface have a spread in their kinetic energy.

Reason : The work function the metal varies as the function of depth from surface.

(a) A (b) B

(c) C (d) D

28. (S) The work function of a metal is 4.2 eV. If radiations of 2000 Å fall on the metal. then the kinetic energy of the fastest photoelectron is

(a) 1.6×10^{-19} J (b) 16×10^{10} J

(c) 3.2×10^{-19} J (d) 6.4×10^{-10} J

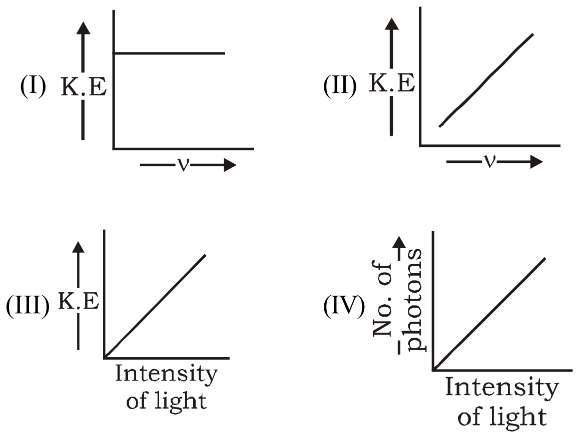
29. (S) If a certain metal was irradiated by using two different light radiations of frequency 'x' and '2x', the maximum kinetic energy of the ejected electrons are 'y' and '3y' respectively. The threshold frequency of the metal will be :

(a) x/3 (b) x/2

(c) 3x/2 (d) 2x/3

30. (I) Calculate the velocity of electron (in 10^5 m/s) ejected from platinum surface when radiation of 200 nm falls on it. Work function of platinum is 5 eV. (1 eV = 1.6×10^{-19})

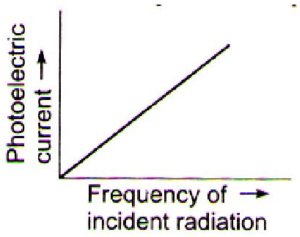
31. (S) Which of the correct graphical representation based on photoelectric effect (assuming $n > n_0$)

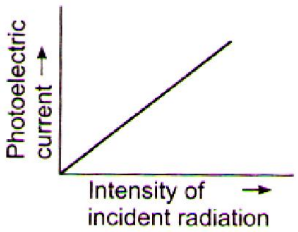


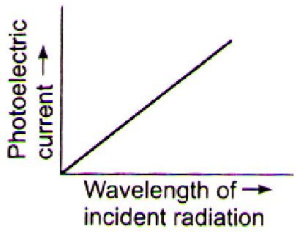
(a) I and II (b) II and III

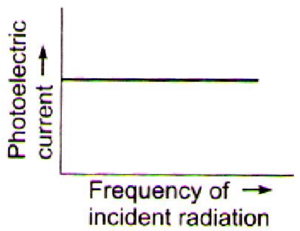
(c) III and IV (d) II and IV

32. (M) Select the correct plots for the photoelectric current.

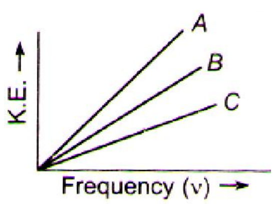
(a) 

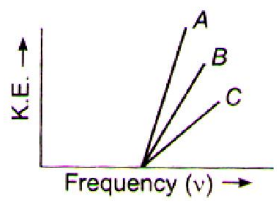
(b) 

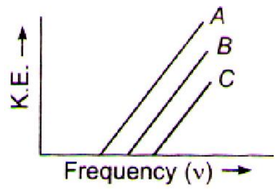
(c) 

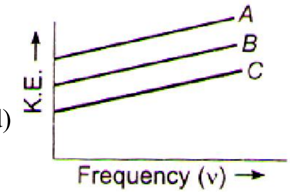
(d) 

33. (S) Photoelectron emission is observed for three different metals A, B and C. The kinetic energy of the fastest photoelectrons versus frequency 'ν' is plotted for each metal. Which of the following graph shows the phenomenon correctly ?

(a) 

(b) 

(c) 

(d) 

Spectra

34. (A) **Assertion :** Emission spectrum of a pure atom is line spectrum, not the continuous.

Reason : Energy of the atoms are quantized.

(a) A (b) B

(c) C (d) D

35. (A) **Assertion :** In the emission spectrum of hydrogen atom, lines are closely spaced in the region of large wavelengths.

Reason : In the region of large wavelengths, electronic transitions occur more frequently.

(a) A (b) B

(c) C (d) D

Bohr's Atomic Model

36. (S) The approximate quantum number of a circular orbit of diameter, 20.6 nm of the hydrogen atom according to Bohr's theory is :

(a) 10 (b) 14

(c) 12 (d) 16

37. (M) The ionisation energy of hydrogen atom is 13.6 eV. Hydrogen atoms in the ground state are excited by monochromatic light of energy 12.1 eV. The spectral lines emitted by hydrogen atoms according to Bohr's theory will be

(a) $n = 3$ to $n = 1$ (b) $n = 3$ to $n = 2$

(c) $n = 2$ to $n = 1$ (d) $n = 4$ to $n = 1$

38. (M) Which of the following parameters are not same for all hydrogen like atoms and ions in their ground state ?

(a) Radius of orbit (b) Speed of electron

(c) Energy of the atom

(d) Orbital angular momentum of electron

39. (X) P_n = potential energy, E_n = total energy
f = frequency, Z = atomic number

v_n = velocity of n^{th} orbit T_n = time period in n^{th} orbit

	Column–I	Column–II	
(A)	$E_n \propto r^y, y = ?$	(P)	1/2
(B)	E_n/P_n	(Q)	1
(C)	$\frac{1}{f_n^{-x}} \propto z,\ x = ?$	(R)	2
(D)	$(v_n \times T_n)^t \propto r_n,\ t = ?$	(S)	–1

40. (X) Match the entries in Column I with the correctly related quantities in Column II.

Column - I	Column - II
(A) Angular momentum	(P) Increase by increasing n
(B) Kinetic energy	(Q) Decreases by decreasing Z
(C) Potential energy	(R) Increases by decreasing Z
(D) Velocity	(S) Decreases by decreasing n

41. (S) As electron moves away from the nucleus, its potential energy

(a) Increases (b) Decreases
(c) Remains constant (d) None of these

42. (S) Which of the following electronic transition in a hydrogen atom will require the largest amount of energy

(a) From n = 1 to n = 2 (b) From n = 2 to n = 3

(c) From n = ∞ to n = 1 (d) From n = 3 to n = 5

43. (S) If the total energy of an electron in the 1^{st} shell of H atom = 0.0 eV then its potential energy in the 1^{st} excited state would be

(a) + 6.8 eV (b) + 20.4 eV
(c) – 6.8 eV (d) + 3.4 eV

44. (S) Ratio of frequency of revolution of electron in the 2^{nd} excited state of He^+ and 2^{nd} state of hydrogen is

(a) 32/27 (b) 27/32
(c) 1/54 (d) 27/2

Comprehension

In a mixture of H – He^+ gas (He^+ is singly ionized He atom), H atoms and He^+ ions are excited to their respective first excited states. Subsequently, H atoms transfer their total excitation energy to He^+ ions (by collisions). Assuming that the Bohr model of atom is applicable, answer the following questions.

45. (C) The quantum number n of the state finally populated in He^+ ions is :

(a) 2 (b) 3
(c) 4 (d) 5

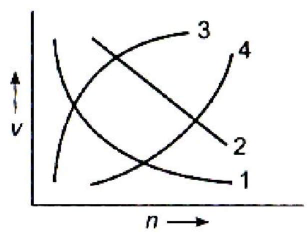
46. (C) The wavelength of light emitted in the visible region by He^+ ions after collisions with H atoms is : (in Å)

(a) 6.5×10^{-7} m (b) 5.6×10^{-7} m
(c) 4.8×10^{-7} m (d) 4.0×10^{-7} m

47. (C) The ratio of the potential energy of the n = 2 electron for the H atom to that of He^+ ion is :

(a) 1/4 (b) 1/2
(c) 1 (d) 2

48. (S) Which of the following curves represent the speed of the electron of hydrogen atom as a function of principal quantum number 'n' ?



(a) 1 (b) 2

(c) 3 (d) 4

49. (M) The angular momentum of electron can have the value (s) :

(a) $0.5\frac{h}{\pi}$ (b) $\frac{h}{\pi}$

(c) $\frac{h}{0.5\pi}$ (d) $2.5\frac{h}{2\pi}$

50. (M) Which of the following are the limitations of Bohr's model ?

(a) It could not explain the intensities or the fine spectrum of the spectral lines.

(b) No justification was given for the principle of the quantization of angular momentum.

(c) It could not explain why atoms should combine to form bond.

(d) It could not be applied to multi-electron atoms.

51. (M) If the radius of first Bohr's orbit of H-atom is x, which of the following is the correct conclusion ?

(a) The de-Broglie wavelength in the third Bohr orbit of H-atom = $6\pi x$.

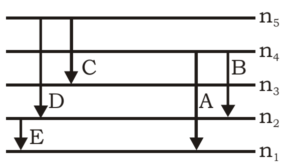
(b) The fourth Bohr's radius of He^+ ion = 8x.

(c) The de-Broglie wavelength in third Bohr's orbit of $Li^{2+} = 2x$

(d) The second Bohr's radius of Be^{2+} = x

Energy Levels of Hydrogen Atom

52. (S) For a hypothetical H like atom which follows Bohr's model, some spectral lines were observed as shown. If it is known that line 'E' belongs to the visible region, then the lines possibly belonging to ultra violet region will be (n_1 is necessarily ground state)



[Assume for this atom, no spectral series shows overlaps with other series in the emmission spectrum]

(a) B and D (b) D only
(c) C only (d) A only

53. (S) If each hydrogen atom in the ground state of 1.0 mole of H-atoms is excited by absorbing photons of energy 8.4eV, 12.09 eV and 15.0 eV of energy, then the number of spectral lines emitted is equal to :

(a) None (b) Two
(c) Three (d) Four

54. (S) A certain transition in H-spectrum from an excited state to ground state in one or more steps gives rise to a total of ten lines. How many of these belong to visible spectrum ?

(a) 3 (b) 4
(c) 5 (d) 6

55. (I) A hydrogen atom in the ground state is hit by a photon exciting the electron to 3rd excited state. The electron then drops to 2nd Bohr orbit. What is the frequency of radiation emitted in the process ? (in 10^{14} hz)

56. (S) The series limit for Balmer series of H-spectra is

(a) 3800 Å (b) 4200 Å
(c) 3646 Å (d) 4000 Å

57. (S) In a sample of H-atoms, electrons make transitions from n = 5 to n = 1. If all the spectral lines are observed, then the line having the 3rd highest energy will correspond to

(a) $5 \rightarrow 3$ (b) $4 \rightarrow 1$
(c) $3 \rightarrow 1$ (d) $5 \rightarrow 4$

Comperhension

One of the fundamental laws of physics is that matter is most stable with the lowest possible energy. Thus, the electron in a hydrogen atom usually moves in the n = 1 orbit, the orbit in which it has the lowest energy. When the electron is in this lowest energy orbit, the atom is said to be in its ground electronic state. If the atom receives energy from an outside source, it is possible for the electron to move to an orbit with a higher n value, in which case the atoms is in an excited with a higher energy.

The law of conservation of energy says that we cannot create or destroy energy. Thus, if a certain amount of external energy is required to excite an electron from one energy level to another, then that same amount of energy will be liberated when the electron returns to its initial state.

Lyman series is formed when the electron returns to the lowest orbit while Balmer series is formed when the electron returns to second orbit. Similarly Paschen, Brackett and Pfund series are formed when electrons returns to the third, fourth and fifth orbits from higher energy orbits respectively.

When an electron returns from n_2 to n_1 state, the number of lines in the spectrum will equal to

$$\frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

If the electron comes back from energy level having energy E_2 to energy level having energy, E_1, then the difference may be expressed in terms of energy of photon as :

$$E_2 - E_1 = \Delta E, \Delta E \Rightarrow \frac{hc}{\lambda}$$

Since, h and c are constants, ΔE corresponds to definite energy ; thus, each transition from one energy level to another with produce a radiation of definite wavelength. This is actually observed as a line in the spectrum of hydrogen atom.

Wave number of a spectral line is given by the formula

$$\bar{v} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$$

where R is a Rydberg's constant ($R = 1.1 \times 10^7\ m^{-1}$)

58. (C) If the wavelength of series limit of Lyman series for He^+ ion is x Å, then what will be the wavelength of series limit of Balmer series for Li^{2+} ion ?

(a) $\frac{9x}{4}$Å (b) $\frac{16x}{9}$Å

(c) $\frac{5x}{4}$Å (d) $\frac{4x}{7}$Å

59. (C) The emission spectra is observed by the consequence of transition of electron from higher energy state to ground state of He^+ ion. Six different photons are observed during the emission spectra, then what will be the minimum wavelength during the transition ?

(a) $\frac{4}{27R_H}$ (b) $\frac{4}{15R_H}$

(c) $\frac{15}{16R_H}$ (d) $\frac{16}{15R_H}$

60. (C) What transition in the hydrogen spectrum would have the same wavelength as Balmer transition, n = 4 to n = 2 in the He^+ spectrum ?

(a) n = 3 to n = 1 (b) n = 3 to n = 2

(c) n = 4 to n = 1 (d) n = 2 to n = 1

61. (S) What is the total number of pairs of electrons atleast same quantum numbers for Be ?

(a) 2 (b) 4

(c) 3 (d) 8

62. (M) If the shortest wavelength of transition of H-atom in Lyman series is x, then the correct conclusion is (are)

(a) the longest wavelength in Balmer series of He^+ is $\frac{x}{4}$

(b) the shortest wavelength in Balmer series of He^+ is x

(c) the longest wavelength in Lyman series of H atom is $\frac{4}{3}x$

(d) the longest wavelength in Paschen series of Li^{2+} is $\frac{16x}{7}$

de-Broglie Concept

63. (M) The ratio of the de Broglie wavelength of a proton and a-particles will be 1 : 2 if their :

(a) velocity are in the ratio 1 : 8

(b) velocity are in the ratio 8 : 1

(c) kinetic energy are in the ratio 16 : 1

(d) kinetic energy are in the ratio 1 : 16

64. (S) The accelerating potential that must be imparted to proton beam to give an wavelength 5 pm.

(a) 32.8 V (b) 3.28 V

(c) 328 V (d) 0.328 V

65. (S) A proton accelerated from rest through a potential difference of 'V' volts has a wavelength l associated with it. An alpha particle in order to have the same wavelength must be accelerated from rest through a potential difference of

(a) V volt (b) 4V volt

(c) 2V volt (d) V/8 volt

66. (S) What is the de–Broglie wavelength associated with the hydrogen electron in its third orbit

(a) 9.96×10^{-10} cm (b) 9.96×10^{-8} cm

(c) 9.96×10^{4} cm (d) 9.96×10^{8} cm

67. (S) De-Broglie wavelength for an electron is related to applied voltage as

(a) $\lambda = \frac{12.3}{\sqrt{h}}$ Å (b) $\lambda = \frac{12.3}{\sqrt{V}}$ Å

(c) $\lambda = \frac{12.3}{\sqrt{E}}$ Å (d) $\lambda = \frac{12.3}{\sqrt{m}}$ Å

68. (S) If E_1, E_2 and E_3 represent respectively the kinetic energies of an electron, an alpha particle and a proton each having same de Broglie wavelength then :

(a) $E_1 > E_3 > E_2$ (b) $E_2 > E_3 > E_1$
(c) $E_1 < E_3 < E_2$ (d) $E_1 = E_2 = E_3$

69. (I) The energy of separation of an electron is 30.6 eV moving in an orbit of Li^{+2}. Find out the number of waves made by the electron in one complete revolution in the orbit.

Heisenberg's Principle

70. (M) The uncertainties in measurement of position and momentum of an electron are equal. Choose the correct statement.

(a) The uncertainty in measurement of speed = 8×10^{12}.

(b) The uncertainty in measurement of kinetic energy = $\frac{h}{8\pi m}$.

(c) The uncertainty in measurement of time = 9.1×10^{-31}.

(d) Increasing the wavelength of light used in the experiment will decrease uncertainty in position and increase the uncertainty in momentum.

71. (A) **Assertion :** Accurate measurement of both positions and momentum can be done simultaneously for a macroparticle.

Reason : Heisenberg's uncertainty principle is more significant to microparticles only.

Comprehension

Werner Heisenberg considered the limits of how precisely we can measure the properties of an electron or other microscopic particle. He determined that there is a fundamental limit to how closely we can measure both position and momentum. The more accurately we measure the momentum of a particle, the less accurately we can determine its position. The converse is also true. This is summed up in what we now call the Heisenberg uncertainty principle.

The equation is $\Delta x.(mv) \geq \frac{h}{4\pi}$

The uncertainty in the position or in the momentum of a macroscopic object like a baseball is too small to observe. However, the mass of microscopic object such as an electron is small enough for the uncertainty to be relatively large and significant.

72. (C) If the uncertainties in position and momentum are equal, the uncertainty in the velocity is :

(a) $\sqrt{\frac{h}{\pi}}$ (b) $\sqrt{\frac{h}{2\pi}}$

(c) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$ (d) None of these

73. (C) If the uncertainty in velocity and position is same, then the uncertainty in momentum will be :

(a) $\sqrt{\frac{hm}{4\pi}}$ (b) $m\sqrt{\frac{h}{4\pi}}$

(c) $\sqrt{\frac{h}{4\pi m}}$ (d) $\frac{1}{m}\sqrt{\frac{h}{4\pi}}$

74. (C) What would be the minimum uncertainty in de-Broglie wavelength of a moving electron accelerated by potential difference of 6 volt and whose uncertainty in position is $\frac{7}{22}$ nm?

(a) 6.25 Å (b) 6 Å
(c) 0.625 Å (d) 0.3125 Å

Quantum Numbers

75. (M) The number of orbitals in n^{th} Bohrs orbit of an atom equal to

(a) n^2 (b) $2n^2$
(c) $(2l+1)$ (d) possible values of 'm'

76. (M) 'g' orbital is possible if

(a) $n = 5, l = 4$ (b) it will have 18 electrons
(c) it will have 9 types of orbitals
(d) it will have 22 electrons.

77. (X) Matching Column Types

Quantum number	Orbitals
(A) $n = 2, l = 1, m = -1$	(P) $2p_x$, or $2p_y$
(B) $n = 4, l = 2, m = 0$	(Q) $4\,dz^2$
(C) $n = 3, l = 1, m = \pm 1$	(R) $3p_x$ or $3p_y$
(D) $n = 4, l = 0, m = 0$	(S) 4s
(E) $n = 3, l = 2, m = \pm 2$	(T) $3d_{x^2-y^2}, 3d_{xy}$

78. (X) Matching Column Types

Column-I	Column-II
(A) Orbit angular momentum	(P) $\sqrt{n(n+2)}$
(B) Orbital angular momentum	(Q) $nh/2\pi$
(C) Spin angular momentum	(R) $\sqrt{s(s+1)}\ h$
(D) Magnetic moment	(S) $\sqrt{l\,(l+1)}\ h$
	(T) $\sqrt{n\,(n+1)}\ h$

79. (M) Which of the following statement about quantum number is correct ?

(a) If the value of l = 0, the electron distribution is spherical.

(b) The shape of orbital is given by subsidiary quantum number.

(c) The Zeeman's effect is explained by magnetic quantum number.

(d) The spin quantum number gives the orientations of electron cloud.

80. (S) If the value of $(n + l)$ is more than 3 and less than 6, then what will be the possible number of orbitals ?

(a) 6 (b) 9

(c) 10 (d) 13

Electronic Configuration

81. (M) The number of electrons in Na(11) having l = 0 are equal to

(a) there are two electrons in 1*s* orbital.
(b) there are two electrons in 2*s* orbital.

(c) there are two electrons in 2*p* orbital.
(d) there are one electrons in 3*s* orbital.

82. (M) Which of the following configuration is/are correct in the first excited state ?

(a) Cr $[Ar]\,3d^5\,4s^1$ (b) Fe^{2+} $[Ar]\,3d^5\,4s^1$

(c) Mn^{2+} $[Ar]\,4s^0\,3d^5$ (d) Co^{3+} $[Ar]\,3d^5\,4s^1$

Comprehension

The sequence of filling electrons in sub-shells of elements with few exceptions in d-block and f-block elements is governed by Aufbau principle followed by Hund's rule and Pauli's exclusion principle.

(a) The electron prefers to enter into sub-shells with lower $(n + l)$ value.

The energy for any sub-shell of an element other than hydrogen is proportional to the sum of principal quantum number (n) and angular momentum quantum number

(b) If $(n + l)$ value is same for many sub-shells, priority of electron filling is given to the sub-shell with lowest n value.

(c) (i) Fulfilled sub shell is most stable.

(ii) Half filled sub-shell is more stable less than half filled.

83. (C) Which pair of sub-shell has same energy for above described exceptional element under rule (a) ?

(a) 1s, 2s (b) 2s, 2p

(c) 3d, 4p (d) 5p, 4d

84. (C) If Hunds rule is not obeyed by some elements given below then which atom has maximum magnetic moment.

(a) Fe (b) Cu

(c) Cr (d) Mn

85. (C) Which pair of element follow rule (c) (ii) ?

(a) Cr, Mo (b) Mn, Fe

(c) Cu, Ag (d) N, P

86. (S) In any subshell, the maximum number of electrons having same value of spin quantum number is

(a) $\sqrt{\ell(\ell+1)}$ (b) $\ell + 2$

(c) $2\ell + 1$ (d) $4\ell + 2$

87. (X) Matching Column Types

Column - I	Column - II
(A) Configuration of Cr is	(P) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}$
(B) Configuration of Cu is	(Q) 5
(C) Number of unpaired electrons in Fe^{3+}	(R) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^5\,4s^1$
(D) Electronic configuration of Zn^{2+}	(S) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^{10}\,4s^1$
	(T) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^9\,4s^2$
	(U) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^4\,4s^2$

88. (M) Which of the following is/are correct ?

(a) The orbital angular momentum for a d-electron is $\sqrt{6}\frac{h}{2\pi}$.

(b) The number of orbitals in a shell with principal quantum number n is $2n^2$.

(c) The correct set of quantum numbers for the last unpaired electron of Cl atoms is $\left\{3,1,1,-\frac{1}{2}\right\}$.

(d) The ratio of energy in the first Bohr orbit of H-atom to the electron in the first excited state of Be^{3+} is 1 : 4.

89. (S) Which of the following pairs of ions have the same electronic configuration ?

(a) Cr^{3+}, Fe^{3+} (b) Fe^{3+}, Mn^{2+}

(c) Fe^{3+}, Co^{3+} (d) Se^{3+}, Cr^{3+}

90. (A) **Assertion :** Half-filled and fully-filled degenerate orbitals are more stable.

Reason : Extra stability is due to the symmetrical distribution of electrons and exchange energy.

(a) A (b) B

(c) C (d) D

Magnetic Moment

91. (M) The magnetic moment of X^{n+} is $\sqrt{24}$ BM. Hence, the species can be

(a) Fe^{2+} (b) Cr^{2+}

(c) Mn^{3+} (d) Co^{3+}

92. (S) How many of the following ions have the same magnetic moments ?

Fe^{2+} Mn^{2+} Cr^{2+} Ni^{2+}

(a) 1 (b) 2

(c) 3 (d) 4

93. (A) **Assertion** : The spin only magnetic moment of Zn^{2+} is zero.

Reason : Zn^{2+} had $3d^{10}$ $4s^0$ configuration. If has no unpaired electron.

(a) A (b) B

(c) C (d) D

94. (M) The spin only magnetic moment of V (Z = 23), Cr (Z = 24) and Mn (Z = 25) are x, y and z respectively. Which of the following is are correct relationships ?

(a) x > y (b) x > z

(c) y > z (d) x < z

Probability Distribution Curves & Nodes

95. (M) For radial probability distribution curves, which of the following is/are correct ?

(a) The number of maxima in 2*s* orbital are two

(b) The number of spherical or radial nodes is equal to $n - l - 1$

(c) The number of angular nodes are 'l'

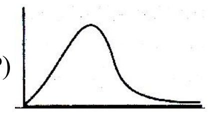
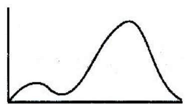
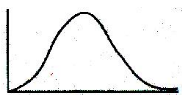
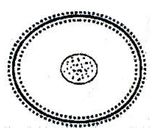
(d) $3d_z^2$ has two angular nodes.

96. (M) Pick out the orbitals with the maximum number of nodal planes ?

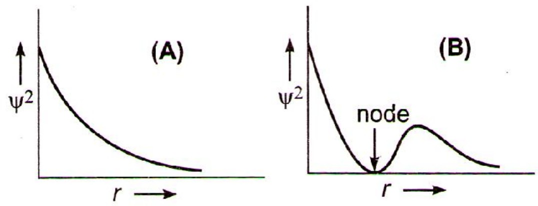
(a) $3d_{xy}$ (b) $4d_{z^2}$

(c) $4d_{xy}$ (d) $2p_x$

97. (X) Matching Column Types

Column I	Column II
(A) Radial probability distribution graphs for 1s orbital	(P) 
(B) Radial probability distribution graph for 2s orbital	(Q) 
(C) Radial probability distribution graph for 2p orbital	(R) 
(D) Electron cloud picture of 2s orbital	(S) 

98. (S) In following two plots, ψ^2 is plotted against the distance 'r' from nucleus.



Select the correct statement :

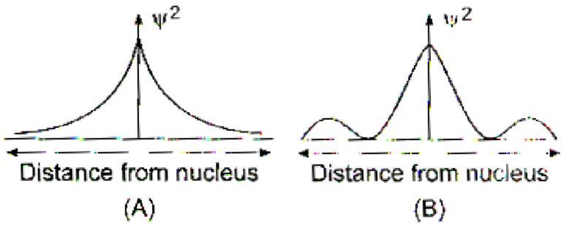
(a) 'A' is for 1s and 'B' for 2s

(b) 'A' is for 2s and 'B' for 1s

(c) 'A' is for 2s and 'B' for 2p

(d) 'A' is for 2p and 'B' for 2s

99. (S) In the following two figures, (ψ^2) is plotted against (r) the distance from nucleus :



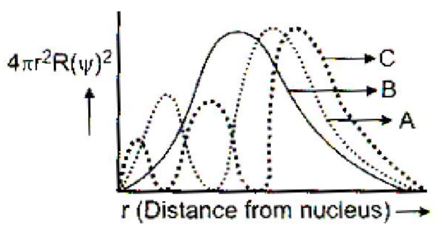
(a) Both (A) and (B) are for 1s

(b) Both (A) and (B) are for 2s

(c) (A) is for 1s and (B) is for 2s

(d) (A) is for 2s and (B) is for 1s

100. (S) Let us consider following graph for radial distribution function. Which of the following has correct matching of curve and orbital ?



(a) A (3p) B (3d) C (3s) (b) A (3s) B (3p) C (3d)

(c) A (3d) B (3p) C (3s) (d) A (3s) B (3d) C (3p)

101. (S) The Schrodinger wave equation for hydrogen atom is

$$\psi\,(\text{radial}) = \frac{1}{16\sqrt{4}}\left(\frac{Z}{a_0}\right)^{3/2}$$

$$[(\sigma-1)(\sigma^2-8\sigma+12)]e^{-\sigma/2}$$

where a_0 and Z are the constant in which answer can be expressed and $\sigma = \frac{2Zr}{a_0}$ minimum and maximum position of radial nodes from nucleus are .... respectively.

(a) $\frac{a_0}{Z}, \frac{3a_0}{Z}$ (b) $\frac{a_0}{2Z}, \frac{a_0}{Z}$

(c) $\frac{a_0}{2Z}, \frac{3a_0}{Z}$ (d) $\frac{a_0}{2Z}, \frac{4a_0}{Z}$

102. (M) Select the correct statement(s) :

(a) Radial distribution function indicates that there is a higher probability of finding the 3s electron close to the nucleus than in case of 3p and 3d electrons

(b) Energy of 3s orbital is less than for the 3p and 3d orbitals

(c) At the node, the value of the radial function changes from positive to negative

(d) The radial function depends upon the quantum numbers n and l

103. (M) The probability of finding the electron in p_x – orbitals is :

(a) maximum on two opposite side of the nucleus along x-axis

(b) zero at the nucleus

(c) same on all the sides around the nucleus

(d) zero on the z-axis

104. (A) **Assertion :** The $3d_{x^2-y^2}$ orbital has zero probability of finding an electron along x and y axes.

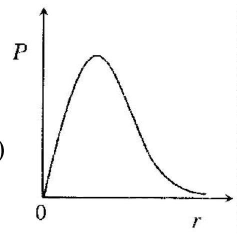
Reason : The orbitals $d_{x^2-y^2}$ lies in x-y plane.

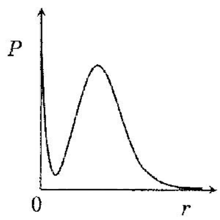
(a) A (b) (B)

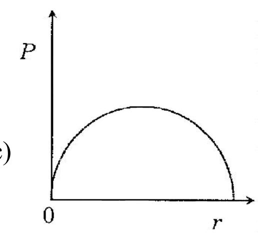
(c) C (d) D

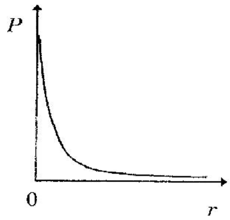
EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

1. P is the probability of finding the 1s electron of hydrogen atom in a spherical shell of infinitesimal thickness, dr, at a distance r from the nucleus. The volume of this shell is $4\pi r^2 dr$. The qualitative sketch of the dependence of P on r is **(2016)**

(a) 

(b) 

(c) 

(d) 

2. The kinetic energy of an electron in the second Bohr orbit of a hydrogen atom is [a_0 is Bohr radius] **(2013)**

(a) $\dfrac{h^2}{4\pi^2 m a_0^2}$ (b) $\dfrac{h^2}{16\pi^2 m a_0^2}$

(c) $\dfrac{h^2}{32\pi^2 m a_0^2}$ (d) $\dfrac{h^2}{64\pi^2 m a_0^2}$

PARAGRAPH FOR QUESTIONS 3 TO 5

The hydrogen-like species Li^{2+} is in a spherically symmetric state S_1 with one radial node. Upon absorbing light the ion undergoes transition to a state S_2. The state S_2 has one radial node and its energy is equal to the ground state energy of the hydrogen atom

3. The state S_1 is **(2010)**

(a) 1s (b) 2s

(c) 2p (d) 3s

4. Energy of the state S_1 in units of the hydrogen atom ground state energy is **(2010)**

(a) 0.75 (b) 1.50

(c) 2.25 (d) 4.50

5. The orbital angular momentum quantum number of the state S_2 is **(2010)**

(a) 0 (b) 1

(c) 2 (d) 3

6. The number of radial nodes in 3s and 2p respectively are **(2005)**

(a) 2 and 0 (b) 0 and 2

(c) 1 and 2 (d) 2 and 1

7. The number of nodal planes in a p_x orbital is **(2005)**

(a) One (b) Two

(c) Three (d) Zero

8. Which hydrogen like species will have same radius as that of Bohr orbit of hydrogen atom ? **(2004)**

(a) n = 2, Li^{2+} (b) n = 2, Be^{3+}

(c) n = 2, He^{+} (d) n = 3, Li^{2+}

9. Rurtherford's experiment, which established the nuclear model of the atom, used a beam of **(2002)**

(a) β-particles, which impinged on a metal foil and got absorbed

(b) γ-rays, which impinged on a metal foil and ejected electrons

(c) Helium atoms, which impinged on a metal foil and got scattered

(d) Helium nuclei, which impinged on a metal foil and got scattered

10. If the nitrogen atom had electronic configuration $1s^7$, it would have energy lower than that of the normal ground state configuration $1s^2\ 2s^2\ 2p^3$, because the electrons would be closer to the nucleus, yet $1s^7$ is not observed because it violates **(2002)**

(a) Heisenberg uncertainty principle

(b) Hund's rule

(c) Pauli exclusion principle

(d) Bohr postulate of stationary orbits

11. The quantum numbers $+\frac{1}{2}$ and $-\frac{1}{2}$ for the electron spin represent **(2001)**

(a) rotation of the electron in clockwise and anticlockwise direction respectively

(b) rotation of the electron in anticlockwise and clockwise direction respectively

(c) magnetic moment of the electron pointing up and down respectively

(d) two quantum mechanical spin states which have no classical analogue

12. The wavelength associated with a golf ball weighing 200g and moving at a speed of 5 m/h is of the order **(2001)**

(a) 10^{-10} m (b) 10^{-20} m

(c) 10^{-30} m (d) 10^{-40} m

13. The number of nodal planes in a p_x orbital is **(2001)**

(a) one (b) two

(c) three (d) zero

14. The electronic configuration of an element is $1s^2$, $2s^2 2p^6$, $3s^2 3p^6 3d^5$, $4s^1$. This represents its **(2000)**

(a) excited state (b) ground state

(c) cationic form (d) anionic form

15. Which of the following relates to photons both as wave motion and as a stream of particles ? **(1992)**

(a) Inference (b) $E = mc^2$

(c) Diffraction (d) $E = h\nu$

16. The electrons, identified by numbers n and l,

(i) $n = 4, l = 1$, (ii) $n = 4, l = 0$, (iii) $n = 3, 1 = 2$, and (iv) $n = 3$, $l = 1$ can be placed in order of increasing energy, from the lowest to highest, as **(1999)**

(a) (iv) < (ii) < (iii) < (i) (b) (ii) < (iv) < (i) < (iii)

(c) (i) < (iii) < (ii) < (iv) (d) (iii) < (i) < (iv) < (ii)

17. The energy of an electron in the first Bohr orbit of H-atom is –13.6 eV. The possible energy value(s) of the excited state(s) for electrons in Bohr orbits of hydrogen is (are) **(1998)**

(a) –3.4 eV (b) – 4.2 eV

(c) – 6.8 eV (d) + 6.8 eV

18. For a d-electron, the orbital angular momentum is **(1997)**

(a) $\sqrt{6}\left(\frac{h}{2\pi}\right)$ (b) $\sqrt{2}\left(\frac{h}{2\pi}\right)$

(c) $\left(\frac{h}{2\pi}\right)$ (d) $2\left(\frac{h}{2\pi}\right)$

19. The first use of quantum theory to explain the structure of atom was made by **(1997)**

(a) Heisenberg (b) Bohr

(c) Planck (d) Einstein

20. Which of the following has the maximum number of unpaired electrons ? **(1996)**

(a) Mg^{2+} (b) Ti^{3+}

(c) V^{3+} (d) Fe^{2+}

21. The orbital angular momentum of an electron in 2s orbital is

(a) $+\frac{1}{2}.\frac{h}{2\pi}$ (b) zero

(c) $\frac{h}{2\pi}$ (d) $\sqrt{2}.\frac{h}{2\pi}$

22. Which of the following does not characterise X-rays ? **(1992)**

(a) The radiation can ionise gases

(b) It causes ZnS to fluoresce

(c) Deflected by electric and magnetic fields

(d) Have wavelengths shorter than ultraviolet rays

23. The correct set of quantum numbers for the unpaired electron of chlorine atom is **(1989)**

	n	l	m		n	l	m
(a)	2	1	0	(b)	2	1	1
(c)	3	1	1	(d)	3	0	0

24. The correct ground state electronic configuration of chromium atom is **(1989)**

(a) $[Ar]\ 3d^5\ 4s^1$ (b) $[Ar]\ 3d^4\ 4s^2$

(c) $[Ar]\ 3d^6\ 4s^0$ (d) $[Ar]\ 4d^5\ 4s^1$

25. The orbital diagram in which the aufbau principle is violated **(1988)**

(a)  (b) 

(c)  (d) 

26. The wavelength of a spectral line for an electronic transition is inversely related to **(1988)**

(a) the number of electrons undergoing the transition

(b) the nuclear charge of the atom

(c) the difference in the energy of the energy levels involved in the transition

(d) the velocity of the electron undergoing the transition

27. The ratio of the energy of a photon of 200 Å wavelength radiation to that of 4000 Å radiation is **(1986)**

(a) $\frac{1}{4}$ (b) 4

(c) $\frac{1}{2}$ (d) 20

28. Which one of the following sets of quantum numbers represents an impossible arrangement ? **(1986)**

	n	l	m	s
(a)	3	2	–2	$\frac{1}{2}$
(b)	4	0	0	$\frac{1}{2}$
(c)	3	2	–3	$\frac{1}{2}$
(d)	5	3	0	$-\frac{1}{2}$

29. Rutherford's alpha particle scattering experiment eventually led to the conclusion that **(1986)**

(a) mass and energy are related

(b) electrons occupy space arount the nucleus

(c) neutrons are burried deep in the nucleus

(d) the point of impact with matter can be precisely determined

30. Electromagnetic radiation with maximum wavelength is **(1985)**

(a) ultraviolet (b) radio wave

(c) X-ray (d) infrared

31. The radius of an atomic nucleus is of the order of **(1985)**

(a) 10^{-10} cm (b) 10^{-13} cm

(c) 10^{-15} cm (d) 10^{-8} cm

32. Bohr's model can explain **(1985)**

(a) the spectrum of hydrogen atom only

(b) spectrum of an atom or ion containing one electron only

(c) the spectrum of hydrogen molecule

(d) the solar spectrum

33. Which electronic level would allow the hydrogen atom to absorb a photon but not be emit a photon ? **(1984)**

(a) 3s (b) 2p

(c) 2s (d) 1s

34. Correct set of four quantum numbers for the valence (outermost) electron of rubidium (Z = 37) is **(1984)**

(a) $5, 0, 0, +\frac{1}{2}$ (b) $5, 1, 0, +\frac{1}{2}$

(c) $5, 1, 1, +\frac{1}{2}$ (d) $6, 0, 0, +\frac{1}{2}$

35. The increasing order (lowest first) for the values of e/m (charge/mass) for electron (e), proton (p), neutron (n) and alpha particle (α) is **(1984)**

(a) e, p, n, α (b) n, p, e, α

(c) n, p, α, e (d) n, α, p, e

36. Retherford's scattering experiment is related to the size of the **(1983)**

(a) nucleus (b) atom

(c) electron (d) neutron

37. The principal quantum number of an atom is related to the **(1983)**

(a) size of the orbital

(b) spin angular momentum

(c) orientation of the orbital in space

(d) orbital angular momentum

38. Any p-orbital can accommodate up to **(1983)**

(a) four electrons (b) six electrons

(c) two electrons with parallel spins

(d) two electrons with opposite spins

39. Rutherford's experiment on scattering of α-particles showed for the first time that the atom has **(1981)**

(a) electrons (b) protons

(c) nucleus (d) neutrons

(One or more than one correct option)

40. Ground state electronic configuration of nitrogen atom can be represented by **(1999)**

(a) $\boxed{\uparrow\downarrow}\ \boxed{\uparrow\downarrow}\ \ \boxed{\uparrow}\boxed{\uparrow}\boxed{\uparrow}$

(b) $\boxed{\uparrow\downarrow}\ \boxed{\uparrow\downarrow}\ \ \boxed{\uparrow}\boxed{\downarrow}\boxed{\uparrow}$

(c) $\boxed{\uparrow\downarrow}\ \boxed{\uparrow\downarrow}\ \ \boxed{\uparrow}\boxed{\downarrow}\boxed{\downarrow}$

(d) $\boxed{\uparrow\downarrow}\ \boxed{\uparrow\downarrow}\ \ \boxed{\downarrow}\boxed{\downarrow}\boxed{\downarrow}$

41. Which of the following statement (s) is (are) correct ? **(1998)**

(a) The electronic configuration of Cr is [Ar] $3d^5\ 4s^1$. (Atomic number of Cr = 24).

(b) The magnetic quantum number may have a negative value.

(c) In silver atom, 23 electrons have a spin of one type and 24 of the opposite type. (Atomic number of Ag = 47).

(d) The oxidation state of nitrogen in HN_3 is –3.

42. The energy of an electron in the first Bohr orbit of H-atom is –13.6 eV. The possible energy value (s) of the excited state (s) for electrons in Bohr orbits of hydrogen is (are) **(1988)**

(a) – 3.4 eV (b) – 4.2 eV

(c) – 6.8 eV (d) + 6.8 eV

43. The atomic nucleus contains

(a) protons (b) neutrons

(c) electrons (d) photons

44. The sum of the number of neutrons and proton in the isotope of hydrogen is **(1986)**

(a) 6 (b) 5

(c) 4 (d) 3

45. When alpha particles are sent through a thin metal foil, most of them go straight through the foil because **(1984)**

(a) alpha particles are much heavier than electrons

(b) alpha particles are positively charged

(c) most part of the atom is empty space

(d) alpha particles move with high velocity

46. Many elements have non-integral masses because **(1984)**

(a) they have isotopes

(b) their isotopes have non-integral masses

(c) their isotopes have different masses

(d) the constiuents, neutrons, protons and electrons, combine to give fractional masses

47. An isotone of $^{76}_{32}Ge$ is **(1984)**

(a) $^{76}_{32}Ge$ (b) $^{77}_{33}As$

(c) $^{77}_{34}Se$ (d) $^{78}_{34}Se$

48. When alpha particles are sent through a thin metal foil, most of them go straight through the foil because **(1982)**

(a) Alpha particles are much heavier than electrons

(b) Alpha particles are positively charged

(c) Most part of the atom is empty space

(d) Alpha particle move with high velocity

Fill in the Blanks

49. The outermost electronic configuration of Cr is .... **(1994)**

50. The $2p_x$, $2p_y$ and $2p_z$ orbitals of atom have identical shapes but differ in their .... **(1993)**

51. Wave functions of electrons in atoms and molecules are called .... **(1993)**

52. The light radiations with discrete quantities of energy are called .... **(1993)**

53. The uncertainty principle and the concept of wave nature of matter were proposed by .... and .... respectively. **(1988)**

54. Elements of the same mass number but of different atomic numbers are known as .... **(1983)**

55. When there are two electrons in the same orbital, they have .... spins. **(1983)**

56. Isotopes of an element differ in the number of .... in their nuclei. **(1982)**

True/False

57. The electron density in the XY plane in $3d_{x^2-y^2}$ orbital is zero. **(1986)**

58. The energy of the electron in the 3d-orbital is less than that in the 4s-orbital in the hydrogen atom. **(1983)**

59. Gamma rays are electromagnetic radiations of wavelengths of 10^{-6} cm to 10^{-5} cm. **(1983)**

60. The outer electronic configuration of the ground state chromium atom is $3d^4 4s^2$. **(1982)**

Subjective Questions

61. (a) Calculate velocity of electron in first Bohr orbit of hydrogen atom (Givern, $r = a_0$)

(b) Find de-Broglie wavelength of the electron in first Bohr orbit.

(c) Find the orbital angular momentum of 2p-orbital in terms of $h/2\pi$ units. **(2005)**

62. (a) The Schrodinger wave equation for hydrogen atom is **(2004)**

$$\Psi_{2s} = \frac{1}{4\sqrt{2\pi}}\left(\frac{1}{a_0}\right)^{3/2}\left(2-\frac{r_0}{a_0}\right)e^{-r_0/a_0}$$

where a_0 is Bohr's radius. If the radial node in 2s be at r_0, then find r_0 in terms of a_0.

(b) A base ball having mass 100 g moves with velocity 100 m/s. Find out the value of wavelength of base ball.

63. The wavelength corresponding to maximum energy for hydrogen is 91.2 nm. Find the corresponding wavelength for He^+ ion. **(2003)**

64. Calculate the energy required to excite 1 L of hydrogen gas at 1 atm and 298 K to the first excited state of atomic hydrogen. The energy for the dissociation of H—H bond is 436 kJ mol^{-1}. **(2000)**

65. With what velocity should an α-particle travel towards the nucleus of a copper atom so as to arrive at a distance 10^{-13} m from the nucleus of the copper atom ? **(1997)**

66. Consider the hydrogen atom to be proton embedded in a cavity of radius a_0 (Bohr's radius) whose charge is neutralised by the addition of an electron to the cavity in vacuum, infinitely slowly. Estimate the average total energy of an electron in its ground state in a hydrogen atom as the work done in the above neutralisation process. Also, if the magnitude of the average kinetic energy is half of average potential energy, find the average potential energy. **(1996)**

67. Calculate the wave number for the shortest wavelength transition in the Balmer series of atomic hydrogen. **(1996)**

68. Iodine molecule dissociates into atoms after absorbing light to 4500Å. If one quantum of radiation is absorbed by each molecule, calculate the kinetic energy of iodine atoms.

(Bond energy of I_2 = 240 kJ mol^{-1}) **(1995)**

69. A bulb emits light of λ = 4500Å. The bulb is rated as 150 watt and 8% of this energy is emitted as light. How many photons are emitted by the bulb per second ? **(1995)**

70. Find out the number of waves made by a Bohr's electron in one complete revolution in its 3rd orbit. **(1994)**

71. What transition in th e hydrogen spectrum would have the same wavelength as the Balmer transition n = 4 to n = 2 of He^+ spectrum ? **(1993)**

72. Estimate the difference in energy between 1st and 2nd Bohr's orbit for a hydrogen atom. At what minimum atomic number, a transition from n = 2 to n = 1 energy level would result in the emission of X-rays with $l = 3.0 \times 10^{-8}$ m ? Which hydrogen atom-like species does this atomic number correspond to ? **(1993)**

73. According to Bohr's theory, the electronic energy of hydrogen atom in the nth Bohr's orbit is given by $E_n = \frac{-21.76\times10^{-19}}{n^2}$J. Calculate the longest wavelength of light that will be needed to remove an electron from the third Bohr orbit of the He^+ ion. **(1990)**

74. The electron energy in hydrogen atom is given by $E_n = \frac{21.7\times10^{-12}}{n^2}$ erg. Calculate the energy required to remove an electron completely from the n = 2 orbit. What is the longest wavelength (in cm) of light that can be used to cause this transition ? **(1984)**

75. Give reason why the ground state outermost electronic configuration of silicon is : **(1985)**

and not


76. What is the maximum number of electrons that may be present in all the atomic orbitals with principal quantum number 3 and azimuthal quantum number 2 ? **(1985)**

77. Calculate the wavelength in Angstroms of the photon that is emitted when an electron in the Bohr's orbit, n = 2 returns to the orbit, n = 1 in the hydrogen atom. The ionisation potential of the ground state hydrogen atom is 2.17×10^{-11} erg per atom. **(1982)**

78. The energy of the electron in the second and third Bohr's orbits of the hydrogen atom is -5.42×10^{-12} erg and -2.41×10^{-12} erg respectively. Calculate the wavelength of the emitted light when the electron drops from the third to the second orbit. **(1981)**

79. No considering the electronic spin, the degeneracy of the second excited state (n = 3) of H atom is 9, while the degeneracy of the second excited state of H^- is **(2014)**

80. In an atom, the total number of electrons having quantum numbers n = 4, $|m_l| = 1$ and $m_s = -1/2$ is **(2015)**

Match the Columns

81. Match the entries in Column I with the correctly related quantum number(s) in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS **(2008)**

Column - I	Column - II
(A) Orbital angular momentum of the electron in a hydrogen -like atomic orbital	(P) Principal quantum number
(B) A hydrogen-like one-electron wave function obeying Pauli principle	(Q) Azimuthal quantum number
(C) Shape, size and orientation of hydrogen like atomic orbitals	(R) Magnetic quantum number
(D) Probability density of electron at the nucleus in hydrogen -like atom	(S) Electron spin quantum number

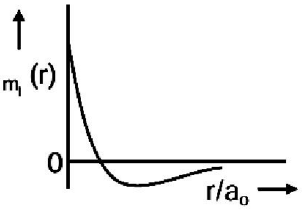
82. In the following, V_n, K_n and E_n represent potential energy, kinetic energy and total energy of an electron in the nth Bohr orbit (radius : r) of hydrogen like species (nuclear charge : Z). Match the entries on the left with those given on right. **(2006)**

(A) $V_n/K_n = ?$	(P) –1
(B) $V_n/E_n = ?$	(Q) – 2
(C) $K_n/E_n = ?$	(R) 1
(D) $1/r \propto (Z)^x : x = ?$	(S) 2

PARAGRAPH

Answer Q. 83, Q. 84 and Q. 85 appropriately matching the information given in the three columns of the following table.

The wave function, ψ_{n,l,m_l} is a mathematical function whose value depends upon spherical polar coordinates (r, θ, ϕ) of the electron and characterized by the quantum numbers n, l and m_l. Here r is distance from nucleus, θ is colatitude and ϕ is azimuth. In the mathematical functions given in the Table, Z is atomic number and a_0 is Bohr radius. **(2018)**

Column 1	**Column 2**	**Column 3**
(I) 1s-orbital	(i) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{3}{2}} e^{-\left(\frac{Zr}{a_0}\right)}$	(P) 
(II) 2s-orbital	(ii) One radial node	(Q) Probability density at nucleus $\propto = \frac{1}{a_0^3}$
(III) $2p_z$-orbital	(iii) $\psi_{n,l,m_l} \propto \left(\frac{Z}{a_0}\right)^{\frac{5}{2}} re^{-\left(\frac{Zr}{a_0}\right)} \cos\theta$	(R) Probability density is maximum at nucleus
(IV) $3d_z^2$ – orbital	(iv) xy-plane is a nodal plane	(S) Energy needed to excite electron from n = 2 state to n = 4 state is $\frac{27}{32}$ times the energy needed to excite electron from n = 2 state to n = 6 state

83. For He^+ ion, the only **INCORRECT** combination is

(a) (I) (i) (S) (b) (II) (ii) (Q) (c) (I) (iii) (R) (d) (I) (i) (R)

84. For the given orbital in Column 1, the only **CORRECT** combination of any hydrogen-like species is

(a) (II) (ii) (P) (b) (I) (ii) (S) (c) (IV) (iv) (R) (d) (III) (iii) (P)

85. For hydrogen atom, the only **CORRECT** combination is

(a) (I) (i) (P) (b) (I) (iv) (R) (c) (II) (i) (Q) (d) (I) (i) (S)

ANSWER KEY

EXERCISE - 1 : (Basic Objective Questions)

1. (c)	**2.** (b)	**3.** (a)	**4.** (b)	**5.** (d)	**6.** (a)	**7.** (b)	**8.** (a)	**9.** (a)	**10.** (c)
11. (a)	**12.** (b)	**13.** (c)	**14.** (a)	**15.** (d)	**16.** (d)	**17.** (d)	**18.** (a)	**19.** (d)	**20.** (a)
21. (a)	**22.** (a)	**23.** (a)	**24.** (c)	**25.** (b)	**26.** (a)	**27.** (d)	**28.** (a)	**29.** (c)	**30.** (c)
31. (a)	**32.** (b)	**33.** (a)	**34.** (a)	**35.** (d)	**36.** (c)	**37.** (c)	**38.** (d)	**39.** (c)	**40.** (a)
41. (a)	**42.** (a)	**43.** (a)	**44.** (b)	**45.** (b)	**46.** (d)	**47.** (d)	**48.** (d)	**49.** (b)	**50.** (c)
51. (a)	**52.** (c)	**53.** (b)	**54.** (c)	**55.** (d)	**56.** (a)	**57.** (b)	**58.** (b)	**59.** (a)	**60.** (d)
61. (b)	**62.** (d)	**63.** (b)	**64.** (d)	**65.** (b)	**66.** (c)	**67.** (c)	**68.** (c)	**69.** (b)	**70.** (d)
71. (c)	**72.** (d)	**73.** (c)	**74.** (d)	**75.** (a)	**76.** (b)	**77.** (c)	**78.** (a)	**79.** (c)	**80.** (a)
81. (b)	**82.** (a)	**83.** (a)	**84.** (b)	**85.** (a)	**86.** (a)	**87.** (c)	**88.** (d)	**89.** (d)	**90.** (b)
91. (d)	**92.** (a)	**93.** (b)	**94.** (c)	**95.** (d)	**96.** (d)	**97.** (c)	**98.** (c)	**99.** (a)	**100.** (c)
101. (c)	**102.** (c)	**103.** (d)	**104.** (c)	**105.** (c)	**106.** (a)	**107.** (c)	**108.** (d)		

EXERCISE - 2 : (Previous year JEE Mains Questions)

1. (c)	**2.** (a)	**3.** (a)	**4.** (a)	**5.** (a)	**6.** (c)	**7.** (b)	**8.** (b)	**9.** (a)	**10.** (c)
11. (b)	**12.** (a)	**13.** (a)	**14.** (b)	**15.** (c)	**16.** (d)	**17.** (a)	**18.** (b)	**19.** (a)	**20.** (d)
21. (c)	**22.** (b)	**23.** (d)	**24.** (b)	**25.** (c)	**26.** (b)	**27.** (b)	**28.** (a)	**29.** (d)	**30.** (a)
31. (c)	**32.** (c)								

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1. (c)	**2.** (b)	**3.** (b)	**4.** (d)	**5.** (a)	**6.** (d)	**7.** (d)	**8.** (c)	**9.** (d)	**10.** (d)
11. (a)	**12.** (d)	**13.** (b)	**14.** (b)	**15.** (b)	**16.** (d)				

EXERCISE - 3 : (Advanced Objective Questions)

1. (c)	**2.** (a)	**3.** (b)	**4.** (b)	**5.** (a)	**6.** (b)	**7.** ($A \rightarrow Q$; $B \rightarrow R$; $C \rightarrow R$; $D \rightarrow P$; $E \rightarrow S$)			
8. (a)	**9.** (0006)	**10.** (d)	**11.** (a)	**12.** (c)	**13.** (d)	**14.** (0004)	**15.** (a)	**16.** (b)	**17.** (a)
18. (a)	**19.** (c)	**20.** (c)	**21.** (d)	**22.** (b)	**23.** (a)	**24.** (c)	**25.** (c)	**26.** (0009)	**27.** (a)
28. (c)	**29.** (b)	**30.** (0006)	**31.** (d)	**32.** (b,d)	**33.** (c)	**34.** (b)	**35.** (c)	**36.** (b)	**37.** (a,b,c)

38. (a,b,c) **39.** ($A \rightarrow S$; $B \rightarrow P$; $C \rightarrow P$; $D \rightarrow Q$) **40.** ($A \rightarrow P,S$; $B \rightarrow Q$; $C \rightarrow P,R,S$; $D \rightarrow Q$) **41.** (a)

42. (a) **43.** (a) **44.** (a) **45.** (c) **46.** (c) **47.** (a) **48.** (a) **49.** (abc) **50.** (a), (b), (c), (d)

51. (ab) **52.** (d) **53.** (c) **54.** (a) **55.** (0006) **56.** (c) **57.** (c) **58.** (b) **59.** (b) **60.** (d)

61. (b) **62.** (bcd) **63.** (b, c) **64.** (a) **65.** (d) **66.** (b) **67.** (b) **68.** (a) **69.** (0002) **70.** (abc)

71. (a) **72.** (c) **73.** (a) **74.** (c) **75.** (a, d) **76.** (a,b,c) **77.** ($A \rightarrow P$; $B \rightarrow Q$; $C \rightarrow R$; $D \rightarrow S$; $E \rightarrow T$)

78. ($A \rightarrow Q$; $B \rightarrow S$; $C \rightarrow R$; $D \rightarrow P$) **79.** (abc) **80.** (d) **81.** (a,b,d) **82.** (b,d) **83.** (b) **84.** (c) **85.** (a)

86. (c) **87.** ($A \rightarrow R$; $B \rightarrow S$; $C \rightarrow Q$; $D \rightarrow R$) **88.** (a,c,d) **89.** (b) **90.** (a) **91.** (a,b,c,d) **92.** (b)

93. (a) **94.** (cd) **95.** (a,b,c,d) **96.** (a,c) **97.** ($A \rightarrow P$; $B \rightarrow Q$; $C \rightarrow R$; $D \rightarrow S$) **98.** (a)

99. (c) **100.** (a) **101.** (c) **102.** (a,b,c,d) **103.** (a,b,d) **104.** (d)

EXERCISE - 4 : (Previous year JEE Advaced Questions)

1. (a) **2.** (c) **3.** (b) **4.** (c) **5.** (b) **6.** (a) **7.** (a) **8.** (b) **9.** (d) **10.** (c)

11. (d) **12.** (c) **13.** (a) **14.** (b) **15.** (d) **16.** (a) **17.** (a) **18.** (a) **19.** (b) **20.** (d)

21. (b) **22.** (c) **23.** (c) **24.** (a) **25.** (b) **26.** (c) **27.** (d) **28.** (c) **29.** (b) **30.** (b)

31. (b) **32.** (b) **33.** (d) **34.** (a) **35.** (d) **36.** (a) **37.** (a) **38.** (d) **39.** (c) **40.** (ad)

41. (abc) **42.** (a) **43.** (ab) **44.** (d) **45.** (ac) **46.** (ac) **47.** (bd) **48.** (ac) **49.** $Cr = [Ar]\ 3d^5, 4s^1$

50. Orientation in space **51.** orbital **52.** photons **53.** Heisenberg, de-Broglie **54.** isobars

56. opposite **56.** neutrons **57. (F)** **58. (T)** **59. (F)** **60. (F)**

61. (a) 2.16×10^6 m/s **(b)** 3.3Å **(c)** $\sqrt{2} \cdot \frac{h}{2\pi}$ **62.** $r_0 = 2a_0$ **63.** 22.8 nm **69.** 20. 2.72×10^{20}

73. 2.055×10^{-7}m **81.** A – Q; B – S; C – P, Q, R; D – P, Q, R **82.** A – Q; B – S; C– P; D – R

83. (c) **84.** (a) **85.** (d)

Dream on !!